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FINAL DISSERTATION

INTERLEAVED VISION FUSION FOR TIME-SERIES FOUNDATION MODELS

Measuring What Vision Contributes to Photovoltaic Ramp Forecasting

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...thanks to...

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Abstract

Photovoltaic power forecasting is usually posed as a per-plant supervised problem: one model is fitted to the history of one installation. That formulation does not survive contact with deployment, where a newly connected plant arrives with a couple of weeks of measurements and no training set of its own. This thesis studies the alternative formulation — *cross-plant zero-shot forecasting*, in which a model trained on one fleet must forecast plants it has never observed — and asks whether frozen foundation models, fused across modalities, are the right instrument for it.

The work makes three contributions. First, it constructs a dataset of record for the problem: 1.34 million rows over 110 photovoltaic sites in two regions, pairing measured power and fourteen weather and solar-geometry covariates with co-registered satellite imagery on a single timestamp grid, together with a plant-disjoint split and a tensor contract that every model in the study consumes unchanged. Second, it defines a fairness protocol — identical physical-time windows, identical splits, identical capacity-normalised metrics, no domain physics inside models — and runs 28 forecasters under it, spanning statistical references, classical machine learning, supervised deep networks, four families of time-series foundation model, retrieval and adapter-based adaptation of frozen backbones, and both endogenous and exogenous multimodal systems. Third, it proposes MMTSFM, an architecture that fuses a mostly-fixed Chronos-2 time-series backbone with a frozen V-JEPA 2.1 video encoder through two mechanisms: *selective temporal interleaving*, which weaves visual tokens into the time-series sequence only inside a short refinement window at roughly two percent token overhead, and *causal Grassmann mixing*, an $O(L)$ temporal operator that encodes the transition between consecutive hidden states as a subspace on a Grassmann manifold rather than as a dot-product similarity.

The benchmark yields four findings that reframe the problem. A well-tuned supervised transformer (iTransformer, skill score 0.552) remains the model to beat, ahead of every foundation model evaluated. Zero-shot time-series foundation models underdeliver on photovoltaic dynamics, one of them falling below the naive reference. Retrieval over a frozen backbone reaches skill 0.478 against 0.331 for fine-tuning the same backbone, making it the most effective and cheapest adaptation tested. And the best multimodal model in the suite synthesises its images *from the series itself*, while every system consuming genuine satellite frames sits mid-pack — evidence that visual inductive bias, not visual information, is what currently converts into forecast skill. MMTSFM is presented in full, with its training curriculum, its controls and its evaluation plan; the experimental campaign for the model was still running at the time of writing, and the chapter that reports it states the measurement protocol and the placeholders rather than provisional numbers.

1 Introduction

1.1 Forecasting a plant you have never seen

A photovoltaic installation converts a stochastic, weather-driven input into an electrical output that a grid operator has to schedule around. The forecasting problem that follows is a short-horizon regression: given the recent power history of a plant and whatever exogenous information is available, predict the next few hours of generation. Framed that way it looks like a solved family of problems, and for a *single, instrumented, long-lived* plant it very nearly is — a gradient-boosted model over lagged power and weather covariates is a strong forecaster, and a supervised transformer trained on years of that plant’s own history is stronger still.

The formulation breaks down at the point where it matters operationally. Photovoltaic capacity is added continuously and in small increments: rooftop systems, community installations, new inverters

on an existing site. A plant that was energised last month has no multi-year history to fit a model to. If every installation requires its own training run, the forecasting stack scales linearly in the number of assets and degrades exactly where the fleet is growing fastest. The deployment-realistic question is therefore not “how well can we forecast a plant we have studied” but

how well can we forecast a plant we have never seen, given two weeks of its history and nothing else?

This is a generalisation question, not a photovoltaic-engineering question, and it is the question this thesis is built around. The distinction matters for how the work is evaluated. Under the usual protocol a model is trained and tested on the same plants, separated only in time, and reported accuracy conflates two very different abilities: learning the idiosyncrasies of a specific installation, and learning something transferable about how irradiance, cloud cover and time of day map onto power. Splitting by *plant identity* rather than by time isolates the second ability, and it is the only split under which the deployment question can be answered honestly. Throughout this thesis, train, validation and test sets are disjoint sets of *plants*; no test plant contributes a single row, a single frame or a single normalisation statistic to anything that is fitted.

1.1.1 What makes the signal hard

It is worth being precise about the difficulty, because “predict solar power” sounds easier than it is. The signal decomposes into two parts with completely different character.

The first is deterministic: the position of the sun, and therefore the clear-sky irradiance reaching a fixed panel, is known analytically for all future time. Any model with a calendar and a latitude reproduces it. It also accounts for the majority of the raw variance, which means that a forecast evaluated against a reference that does not know it will look excellent while having learned nothing.

The second is the modulation of that envelope by the atmosphere — overwhelmingly by cloud. It is intermittent, spatially structured, advects at wind speed, and changes the output of an array by most of its range within minutes. It is not well captured by hourly reanalysis weather, which is what numerical covariates provide: an hourly areal cloud fraction says little about whether the sun is occluded at one rooftop at 14:20. This is the part a forecaster is actually paid to predict, and it is the part for which the numerical track is an impoverished description.

That gap is the opening for a visual modality. Cloud fields are directly observable in satellite imagery, at a spatial and temporal resolution that reanalysis does not approach, and they advect predictably enough over a few hours that what is visible upwind now is overhead later. Chapter 3 quantifies the residual information available: conditioning on cloud state changes median output by roughly 0.09 in normalised power *after* conditioning on irradiance. Whether any current architecture converts that availability into forecast skill is, in the end, the question this thesis is about.

1.2 Why foundation models, and why multimodal

Two developments make the cross-plant question tractable in a way it was not a few years ago.

The first is the arrival of *time-series foundation models*: transformers pretrained on large, heterogeneous corpora of time series that forecast unseen series zero-shot, with no per-series fitting [11, 16, 17]. Their promise is precisely the promise this problem needs — transfer to a series that was not in the training set. The second is the maturation of *video foundation models* trained by self-supervised prediction rather than reconstruction [25]. This matters because the dominant source of short-horizon photovoltaic uncertainty is cloud advection, a spatiotemporal process that is visible in satellite imagery hours before it registers in the power signal. A model that reads only the power history is, in the ramp regime, structurally blind: the information that would let it anticipate a cloud edge exists, but not in the modality it consumes.

Combining the two is not novel in itself; a growing literature attaches visual encoders to time-series forecasters [15, 19, 23, 26]. What is contested is *how deeply* the two modalities should be coupled. The prevailing designs are shallow: a visual encoder produces a summary vector which is concatenated to the series representation (late fusion), or injected as a soft prompt [19], or used to condition an adapter on an otherwise frozen backbone [12]. In each case the two modalities meet once, after each

has been processed independently. The alternative investigated here is to let them meet *inside* the temporal operator, as neighbouring tokens in one sequence, so that the model computes cross-modal interactions at every layer rather than a single cross-modal readout at the end.

The claim under test is therefore:

A frozen multimodal foundation model stack (Chronos-2 and V-JEPA 2.1) with deep token-level fusion achieves cross-plant photovoltaic power forecasting on disjoint test plants, beating late fusion, unimodal foundation models, and domain-specific architectures.

Every baseline in Chapter 5 exists to falsify one part of that sentence. The design principle behind the benchmark is that a reader must not be able to say “the gain could come from X , and X was never tested”.

1.3 Research question and hypotheses

The research question, stated in the terms the thesis is evaluated in:

Can a frozen multimodal foundation model stack (a time-series foundation model and a vision foundation model) achieve cross-plant photovoltaic power forecasting on disjoint test plants by deep token-level fusion, rather than by late fusion or by domain-specific architectures?

It decomposes into a ladder of hypotheses, each of which is answered by a specific experiment, and each of which is falsifiable independently of the ones above it.

- **H0 — Foundation-model anchor.** A time-series foundation model applied zero-shot is a meaningful reference point for this task. Tested in Chapter 5 against classical and supervised alternatives.
- **H1 — Vision helps.** Adding a visual stream through a late-fusion adapter improves over an otherwise identical time-series-only model. Tested by the modality ablation of Chapter 8.
- **H2 — Fusion depth matters.** Selective temporal interleaving of visual tokens improves over late fusion, with the same backbones and the same data. This is the architectural contribution of Chapter 7.
- **H3 — Cross-plant transfer is achievable.** A model can forecast completely disjoint plants from a short history, relying on structure learned from other plants. This is the evaluation protocol of Chapter 4, not an ablation.
- **H4 — Retrieval is a competitive alternative.** Retrieval augmentation on a frozen backbone closes the gap to full fine-tuning. Tested in Chapter 5.

1.4 Contributions

A dataset of record for multimodal cross-plant photovoltaic forecasting. Four public sources — UK domestic generation [7], EUMETSAT SEVIRI high-resolution visible imagery [2], NREL PVDAQ systems with GOES-16 crops [4], and Open-Meteo reanalysis covariates [3] — are fused into one flat table of 1,337,654 rows over 110 sites, plus a packed image archive addressed by a single timestamp-exact pointer. The construction, the quality flags and the exact plant partition are given in Chapter 3.

A 28-model benchmark under one fairness contract. All models consume the same tensor contract, the same 14-day history and 6-hour horizon defined in *physical* time, the same plant-disjoint split, and the same capacity-normalised metrics. The suite spans seven tiers, from naive persistence to photovoltaic-specialised multimodal networks. To the best of this work’s knowledge it is the largest single-protocol comparison of time-series foundation models, frozen-backbone adaptation methods and multimodal forecasters on photovoltaic data.

MMTSFM, an architecture for deep multimodal fusion on frozen backbones. Two mechanisms are contributed. *Selective temporal interleaving* inserts visual summary tokens into the time-series sequence only inside a short refinement window, leaving the macro-context purely numerical; the token overhead is proportional to the number of visual tokens rather than to context length, on the order of two percent. *Causal Grassmann mixing* replaces $O(L^2)$ temporal attention with an $O(L)$ operator that encodes the transition between hidden states as a two-dimensional subspace via a Plücker embedding, aggregated over multiple temporal offsets. A matched diagnostic twin using ordinary causal self-attention is retained so that any measured difference is attributable to the inductive bias and not to the training schedule.

A diagnosis of where the headroom actually is. Two measurements from the benchmark reorient the problem. The gap between a privileged-information oracle variant of Chronos-2 and its honest fine-tuned counterpart is 0.173 skill — larger than any architectural difference observed in the entire suite, and evidence that *conditioning*, not architecture, is the binding constraint. Separately, the strongest multimodal model in the suite uses images synthesised from the series itself, while every model consuming real satellite frames ranks mid-pack; the distinction between visual *inductive bias* and visual *information* is developed in Chapters 5 and 9.

1.5 Scope and non-goals

The framing is deliberately that of a machine-learning contribution with photovoltaics as the testbed, not a photovoltaic-engineering contribution. Several things are consequently out of scope, and are listed here so the reader does not go looking for them.

- **No domain physics inside models.** Clear-sky-index conversions and irradiance-to-power physical models are excluded from every learned model. The single exception is smart persistence, which *is* the physics reference and serves as the denominator of the skill score.
- **No few-shot in-context adaptation curves.** Accuracy as a function of K support days per test plant is a different protocol — context matching rather than zero-shot transfer — and was excluded by an explicit design decision. The evaluation harness supports it as an appendix experiment should a reader require it.
- **No grid-operations or electricity-market analysis.** The output of this work is a forecast and its error, not a dispatch decision or an economic valuation.
- **No pre-2025 methods as primary comparators.** Older architectures appear only where they are still the community reference (persistence, gradient boosting, DLinear as the simplicity check).
- **No dataset construction beyond what is required.** The consolidated dataset is treated as read-only once built; no result in this thesis depends on re-processing it.

1.6 A note on what this thesis reports

Two editorial decisions shape how the results are presented, and both are worth declaring at the outset because they are unusual enough to be mistaken for omissions.

Negative results are reported as results. Several of the most informative measurements in this work are things that did not happen: foundation models did not transfer zero-shot, real satellite imagery did not outperform synthesised imagery, and pretraining did not beat a well-matched supervised architecture. Each is stated in the chapter where it was measured, with the control that establishes it, and none is relegated to a limitations paragraph.

Provisional numbers are not reported at all. The experimental campaign for the proposed architecture was still running at the time of writing. Chapter 8 therefore states the arms, the controls, the interpretation rules and the reporting template in full, and leaves the result cells empty. A number that later moves is worse than no number, because it propagates silently into every comparison that

cites it — and because a design whose interpretation is fixed only after the result is known is not a test of anything.

The second decision has a consequence for how the thesis should be read: its empirical weight sits in Chapters 3 through 6, which are complete, rather than in Chapter 8, which is a specification.

1.7 Structure of the thesis

Chapter 2 positions the work against the 2025–2026 literature on time-series foundation models, frozen-backbone adaptation, video foundation models and multimodal forecasting, and states the gap that motivates the benchmark. Chapter 3 describes the construction of the dataset of record, its schema, its quality flags and its plant-disjoint splits. Chapter 4 defines the evaluation protocol: the fairness rules, the physical-time windows, the metrics and the scenario battery. Chapter 5 presents the baseline suite, its implementation, the full leaderboard and the four findings that follow from it. Chapter 7 presents MMTSFM in full: both branches, the fusion mechanisms, the Grassmann operator, the output head and the four-stage training curriculum. Chapter 8 gives the experimental design for the model, its ablation and control battery, and the reporting protocol under which its numbers will be recorded; the campaign was still running at the time of writing and no provisional figures are stated. Chapter 9 answers the research question as far as the current evidence permits, states the limitations, and lays out the next experiments in priority order.

2 Background and Related Work

This chapter fixes the notation used throughout the thesis, then surveys the four literatures the work sits between: time-series foundation models, methods for adapting frozen backbones, video foundation models, and multimodal time-series forecasting. It closes with the photovoltaic-specific literature, a comparison against the two other public multimodal photovoltaic corpora, and the gap statement that motivates Chapters 3–5.

2.1 Problem formalisation and notation

Let a *plant* (equivalently, an *entity*) be indexed by i , and let time be discretised on the plant’s native cadence. At forecast origin t_0 a model observes a history window of length T and must produce a horizon of length H .

- $\mathbf{Y} \in \mathbb{R}^{N \times T \times 1}$ — normalised historical power for N entities. The target is *capacity-normalised* power, $y = P/P_{\text{installed}} \in [0, 1]$, so that plants of different nameplate size are directly comparable.
- $\mathbf{X}_{\text{cov}} \in \mathbb{R}^{N \times (T+H) \times C}$ — covariates over history *and* horizon. The horizon portion contains only quantities that are genuinely known ahead of time (solar geometry, calendar encodings) unless a variant is explicitly labelled as using privileged information.
- $\mathbf{V} \in \mathbb{R}^{N \times T_v \times C_{\text{img}} \times H_{\text{img}} \times W_{\text{img}}}$ — T_v satellite frames over a short window ending at t_0 .
- $\hat{\mathbf{Y}}_{\text{fut}} \in \mathbb{R}^{N \times H \times Q}$ — the prediction, as Q quantiles per horizon step rather than a point estimate.

The *cross-plant* constraint is a property of the index set, not of the tensors: if $\mathcal{P}_{\text{train}}$, $\mathcal{P}_{\text{val}}$ and $\mathcal{P}_{\text{test}}$ are the plant sets, then $\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$, and no statistic computed on $\mathcal{P}_{\text{test}}$ — including normalisation constants — may enter any fitted component. This rules out the standard practice of per-series standardisation using test-series statistics, which is why the target is normalised by nameplate capacity, a quantity known at installation time and independent of the measured signal.

2.1.1 A taxonomy of the forecasting task

Solar forecasting is not one problem but a family indexed by lead time, and the dominant information source changes with it. Table 2.1 states the regimes and where this thesis sits.

Regime	Lead time	Dominant information source
Nowcasting	0–30 min	Sky-camera imagery; cloud advection within the local field of view
Intra-hour	30 min – 2 h	Satellite imagery; advection at mesoscale
Intra-day	2–6 h	Satellite imagery plus numerical weather; this thesis
Day-ahead	6–48 h	Numerical weather prediction; imagery is stale
Seasonal	weeks–months	Climatology; individual weather is irrelevant

Table 2.1: Forecasting regimes by lead time. The six-hour horizon studied here spans the boundary at which satellite imagery ceases to dominate and numerical weather takes over, which is why the visual window is short and the numerical context long.

The six-hour horizon is a deliberate choice of the hardest interesting regime. Below two hours the problem is largely advection and a good image model wins; beyond twelve hours the observed cloud field has left the domain and only numerical weather matters. Between them both sources are partially informative and neither is sufficient, which is exactly the setting in which a fusion architecture should earn its complexity. It is also the regime that matters operationally for intra-day market participation and reserve scheduling.

2.1.2 Why cross-plant differs from cross-time

The distinction between splitting by time and splitting by entity is worth stating precisely, because it determines what a reported number means.

Under a *temporal* split, a model is fitted on $\{(x_t, y_t) : t < t_0\}$ for plant i and evaluated on later observations of the same plant. The model may — and in practice does — absorb plant-specific constants into its parameters: panel orientation, local shading geometry, inverter clipping thresholds, the systematic bias of the nearest weather grid cell. These are genuine, stable, exploitable regularities, and learning them is not cheating. But they are not *transferable*, and a benchmark that rewards them measures memorisation of a plant alongside understanding of the physics.

Under a *cross-plant* split, plants in the test set contribute nothing to fitting. The model receives, at inference, a short history from which everything plant-specific must be inferred on the fly. Formally the model must learn a mapping conditioned on context rather than a per-entity parameter vector: $\hat{y}_{i,t+h} = f(\mathbf{Y}_{i,<t}, \mathbf{X}_i, \mathbf{V}_{i,<t})$ with f shared across all i and no i -indexed parameters anywhere. This is the same requirement that in-context learning imposes on language models, and it is why foundation models are a natural hypothesis for the problem.

Figure 3.12 quantifies what has to be inferred: after capacity normalisation, median daytime output still ranges from below 0.10 to above 0.55 across plants. That residual spread is exactly the per-plant information a temporal split would hand the model for free.

2.2 Time-series foundation models

A time-series foundation model is a sequence model pretrained on a large corpus of heterogeneous series and applied to unseen series without per-series fitting. The design space has converged on a few recurring choices: patch-based tokenisation, in which contiguous blocks of timesteps become tokens; a normalisation scheme that makes series of arbitrary scale comparable; and a probabilistic output head, either quantile-based or sampling-based.

Chronos-2 [11] is the backbone used in this thesis. It is an encoder-only transformer operating on patches of size 16, with an arcsinh-based normalisation, a nine-quantile output head, and — the property that matters most here — a *group attention* mechanism that attends across the batch axis over rows sharing a group identifier. This is what allows covariates, and in principle other entities, to be supplied as additional token rows rather than as extra channels, and it is the hook that the multimodal design of Chapter 7 exploits.

TimesFM 2.5 [16] is a decoder-only model from an independent lineage. **TiRex** [17] is built on the xLSTM family rather than on attention and is among the strongest zero-shot performers on general-purpose leaderboards. **Tiny Time Mixers** [9] occupy the opposite end of the scale axis: a

few million parameters, mixer-based rather than attention-based, intended to test whether size is what matters.

2.2.1 Shared design elements

Four design choices recur across the families and are worth stating once, because they determine what these models can and cannot do on photovoltaic data.

Patch tokenisation. Rather than one token per timestep, contiguous blocks of p steps become one token. This buys a factor of p in sequence length — decisive when the context is 672 steps — and imposes a mild smoothing prior, since sub-patch structure must survive a linear projection to be represented at all. For photovoltaic data with $p = 16$ at a 30-minute cadence, one token spans eight hours: an entire daylight period is roughly one and a half tokens. Fine-grained ramp structure is therefore compressed aggressively, which is one plausible mechanism behind the zero-shot results of Chapter 6.

Scale-free normalisation. A model pretrained across corpora cannot assume a scale. Solutions range from instance normalisation with stored statistics to inverse-hyperbolic-sine transforms that compress large magnitudes while remaining approximately linear near zero. The choice interacts badly with a target that is exactly zero for half of every day: a bounded, heavily zero-inflated signal is not what these transforms were designed for.

Probabilistic heads. Output is a predictive distribution, most commonly a fixed quantile grid trained with pinball loss, occasionally a sampling-based head. Quantile heads are cheap and directly scoreable but cannot represent multi-modality — and a forecast under broken cloud is genuinely bimodal (the sun is either occluded at the array or it is not).

Covariate handling. This is where the families differ most, and it matters for this task more than any other design axis. Chronos-2 admits covariates as additional token rows fused by attention across the batch axis; TimesFM and TiRex in the configurations evaluated here are target-only; TTM accepts exogenous channels. Given that the covariate track carries most of the exploitable signal (Figure 3.8), a model that cannot ingest it is competing with one hand tied — a caveat that must be attached to the zero-shot comparison.

Model	Family	Backbone	Covariates	Output
Chronos-2	encoder transformer	attention + group attention	yes	9 quantiles
TimesFM 2.5	decoder transformer	causal attention	no	point / quantile
TiRex	xLSTM	recurrent state	no	quantile
TTM-R2	mixer	MLP-mixer blocks	yes	point

Table 2.2: The four time-series foundation model families evaluated, in the configurations used here. Covering three distinct architectural paradigms is what makes a negative zero-shot result a statement about the task rather than about a model.

Including at least three distinct families is a methodological requirement rather than a completeness gesture. A result obtained with one backbone can always be attributed to that backbone’s pretraining corpus; a result that replicates across an encoder transformer, a decoder transformer and an xLSTM is a property of the *task*. Chapter 5 reports all four families under one protocol, and the conclusion — that zero-shot time-series foundation models underdeliver on photovoltaic dynamics — is consistent across them.

2.3 Adapting frozen backbones

If a pretrained backbone is not good enough zero-shot, three strategies are available that do not require training a new model from scratch.

Fine-tuning. Update some or all backbone weights on in-domain data. Effective, but it produces a task-specific copy of a large model and forfeits the generality that motivated pretraining.

Retrieval augmentation. Keep the backbone frozen and condition it on similar windows retrieved from a datastore. **TS-RAG** [18] retrieves analogous historical windows and fuses them into the forecast; **Cross-RAG** [21] extends this across series with clear-sky-aware keys and per-step mixing weights. The appeal is that adaptation cost is a nearest-neighbour lookup rather than a gradient step. The fairness question it raises is central to this thesis and is answered in Section 4.6: the datastore may contain training plants only, since populating it with test-plant history is transduction, not zero-shot forecasting.

Adapters and covariate injection. **CoRA** [12] injects exogenous covariates — of any modality, in principle including image features — into a frozen time-series backbone through a zero-initialised residual adapter. CoRA is the closest published competitor to the architecture proposed here: it occupies the same design space (frozen backbone, external information injected through a learned interface) but couples the modalities once, at a single insertion point, rather than throughout the temporal operator. If deep token-level fusion cannot beat CoRA supplied with image features, the fusion-depth claim of this thesis does not survive. Parametric-memory alternatives [13] occupy the same niche and are positioned but not run.

2.4 Video foundation models

The visual branch of the proposed model uses **V-JEPA 2.1** [25], a ViT-L/16 video encoder trained by joint-embedding predictive self-supervision: it learns to predict the *representation* of masked spatiotemporal regions rather than their pixels. Three properties motivate the choice over reconstruction-based alternatives such as VQ-VAE video tokenisers.

First, the objective is predictive rather than reconstructive. A reconstruction loss spends capacity on pixel-level detail that is irrelevant — and, for satellite imagery, largely sensor noise. A predictive objective in representation space yields spatially structured, semantically coherent features, which is what cloud-structure detection requires. Second, temporal modelling is native: V-JEPA consumes an entire clip and encodes motion internally through tubelet embedding with temporal stride, so a downstream summariser only needs to perform *spatial* compression. A per-frame image encoder would force the temporal modelling back onto the fusion layer. Third, the model is built for frozen use, remaining strong under linear probing and adapter tuning — the operating regime of the staged curriculum in Section 7.11.

2.4.1 The joint-embedding predictive objective

The distinction between predictive and reconstructive self-supervision is worth stating concretely, because it is the reason this branch of the architecture uses a video model rather than an image tokeniser.

A reconstructive objective — a masked autoencoder, or a vector-quantised autoencoder — asks the network to output the missing *pixels*. The loss is therefore computed in pixel space, and every source of pixel variance contributes to it, including sensor noise, sub-pixel registration error and radiometric calibration drift. For satellite crops these are substantial and carry no meteorological content whatsoever.

A joint-embedding predictive objective asks the network to output the missing *representation*, as computed by a slowly updated copy of itself. The loss lives in representation space, so any variation the encoder chooses not to represent costs nothing. The network is free to discard exactly the nuisance variance a reconstruction loss forces it to model. What survives is structure that is predictable across space and time — which, in a sequence of satellite crops, is cloud morphology and its motion.

Two properties follow that matter for this application. The representation is spatially structured rather than globally pooled, so downstream modules can attend to regions rather than to a single vector. And the model is robust under frozen use: because the objective never required the encoder to serve a decoder, its features are not entangled with a reconstruction head, and linear probes and

light adapters recover most of their utility. That is precisely the operating regime of the curriculum in Section 7.11.

2.4.2 What retrieval augmentation actually does

Since retrieval turns out to be the strongest frozen-backbone strategy in Chapter 6, its mechanism deserves more than a citation.

Given a query context $\mathbf{Y}_{i,<t}$, a retrieval-augmented forecaster embeds it, finds the k nearest neighbours in a datastore of historical windows under some key function, and combines the neighbours’ *continuations* with the backbone’s own forecast — typically as a convex combination whose weight α is tuned on held-out data, sometimes per horizon step. The backbone is untouched.

Two design choices distinguish the variants. The *key* determines what counts as similar: raw values, a learned embedding, or — in the clear-sky-aware variant — a representation that has already divided out the deterministic diurnal component, so that neighbours match on cloud regime rather than on time of day. The *scope* determines where neighbours may come from: within the same series, or across series. Cross-series retrieval is the more powerful option and is exactly what a cross-plant protocol makes legitimate — a test plant with two weeks of history has no useful within-series precedent for an unusual regime, but the training fleet does.

This also makes the fairness rule of Section 4.6 non-negotiable. If the datastore contained test-plant history, retrieval would be doing in-context memorisation of the target plant, and the resulting number would not be a zero-shot result at all.

2.5 Multimodal time-series forecasting

The multimodal forecasting literature divides along an axis that turns out to be empirically decisive, and this thesis adopts the distinction as a reporting category.

Endogenous multimodal. The second modality is *synthesised from the numerical series itself*. **Time-VLM** [15] renders the series as an image and feeds it, with a text description, to pretrained vision-language backbones. **VisionTS++** [20] maps series to images to exploit a frozen masked autoencoder. **Aurora** [10] templates weather text from covariates. No external sensor is involved; what these methods exploit is the *inductive bias* of a visual backbone — its priors over locality, periodicity and texture — applied to a re-rendering of data the model already had.

Exogenous multimodal. The second modality is a genuine external observation. **UniCast** [19] prompts a frozen time-series foundation model with vision and text embeddings. **CrossViVit** [26] is the reference deep satellite-plus-series model, cross-attending video tokens into a temporal transformer, and remains the strongest non-foundation-model multimodal competitor. **Solar-VLM** [23] and **SUNSET** [5] are the photovoltaic-domain instances.

The proposed architecture is exogenous by construction, and the benchmark of Chapter 5 evaluates both classes side by side. The result — that the best endogenous model outperforms every exogenous one — is the sharpest empirical finding in this thesis and is what Chapter 9 argues should reorient the design.

2.6 Photovoltaic forecasting and its conventions

The solar forecasting community has its own reference model and its own headline metric, and both are adopted here to keep the results legible to that community.

Smart persistence carries the last observed *clearness index* — the ratio of measured power to a clear-sky model [30] — forward across the horizon, rather than carrying power itself forward. It therefore respects the deterministic diurnal shape and only assumes that cloud conditions persist. It is a much harder reference than naive persistence, and it is the denominator of the *skill score* $SS = 1 - \text{NRMSE}_{\text{model}} / \text{NRMSE}_{\text{smart persistence}}$, the number by which the field reports progress.

Domain-specific architectures follow a common pattern: a convolutional or transformer encoder over sky or satellite imagery, fused with a recurrent or attention-based encoder over the power series. SUNSET [5] is the canonical convolutional instance; CrossViVit [26] the deep attention-based one;

Solar-VLM [23] and PV-VLM [14] add a language backbone; FusionSF [8] and M3S-Net [22] pursue vector-quantised and multi-scale fusion respectively.

2.6.1 Four generations of method

The domain literature has moved through four broad phases, and the assumptions of each are still visible in the baselines that survive from it.

Physical and statistical models came first: clear-sky models with empirical cloud attenuation, persistence of the clearness index, and autoregressive corrections. They are interpretable, require no training data, and remain the reference against which everything else is measured — smart persistence in this benchmark is a direct descendant.

Feature-engineered machine learning followed: gradient boosting and support-vector regression over lagged power and numerical weather. This generation established that exogenous weather is worth more than a longer power history, a finding this benchmark reproduces in the gap between DLinear (target only) and every covariate-consuming model.

Deep sequence and image models came next: convolutional encoders over sky or satellite imagery combined with recurrent power encoders, of which SUNSET is the canonical example, and later attention-based spatiotemporal architectures such as CrossViVit. This is the generation that established the multimodal premise — that imagery should help — largely on nowcasting horizons where it demonstrably does.

Foundation-model adaptation is the current phase: frozen pretrained backbones adapted by fine-tuning, retrieval, adapters or prompting, with vision-language models entering the domain through Solar-VLM and PV-VLM. The premise inherited from the previous generation — that real imagery is the payload — has not been re-tested under the new one, which is part of what this thesis does.

What these works do not share is a protocol. Horizons range from minutes to a day ahead, history windows from one hour to a year, splits are usually temporal rather than plant-disjoint, and normalisation conventions differ enough that reported errors are not comparable across papers. This is the practical reason a new benchmark was necessary: without one, the question of whether foundation-model fusion beats domain-specific fusion cannot be answered from the published numbers.

2.7 Public multimodal photovoltaic datasets

Two public corpora pair photovoltaic generation with imagery, and both were considered before the dataset of Chapter 3 was built.

ClimateHack.AI 2023 [1] covers Great Britain at 5-minute resolution with generation as a fraction of installed capacity, two EUMETSAT satellite streams (high-resolution visible and an 11-channel non-HRV product) as [12, 128, 128] crops, DWD ICON-EU numerical weather forecasts over 38 variables, and ECMWF CAMS air-quality forecasts. It is organised as a *nowcasting* task: one hour of history in, four hours out at 5-minute granularity. The layout is per-window tensors rather than a tall table, which targets minutes-to-hours nowcasting and would require complete re-windowing to serve a 14-day-history cross-plant protocol. It is also a competition release with no accompanying peer-reviewed publication, so there is no citable account of its construction.

MMSP, released with FusionSF [8], covers 88 plants across a Chinese province at hourly resolution from January 2021 to June 2022, fusing power, Himawari-8/9 crops at 64×64 pixels (one channel in the 10-plant subset, four in the full set) and 17 ECMWF weather variables. The task is day-ahead: 24 hours in, 24 hours out. Again the structure — hourly cadence, fixed day-ahead windows, per-window forecast tensors — is a poor fit for a longer-history intra-day cross-plant protocol.

Neither corpus is deficient; they are built for different questions. But neither supports the combination this thesis requires: a long numerical history, a short high-cadence visual window, native per-plant cadences, plant-disjoint splits across two regions, and real satellite frames rather than nowcast tensors.

2.8 Fusion depth as a design axis

The multimodal methods above differ along an axis that the literature rarely names explicitly: *where*, in the computation, the two modalities first interact. Table 2.3 orders the design space by that criterion.

Depth	Mechanism	Interactions	Examples
Post-hoc	Average or stack two independent forecasts	0	Ensembles
Late / readout	Concatenate encoder outputs before the head	1, at the end	SUNSET, adapter-style fusion
Prompt	Project visual features into the input embedding space	1, at the input	UniCast
Adapter	Residual injection at one or more insertion points	few	CoRA
Cross-attention	Visual tokens attended from the temporal stream at every layer	L	CrossViVit, Solar-VLM
Token interleaving	Visual tokens placed <i>inside</i> the temporal sequence	L , and inside the temporal operator itself	this work

Table 2.3: Fusion depth, ordered by where the modalities first interact and how many times. The proposed mechanism differs from cross-attention in that visual tokens are not a separate stream being attended *to*; they occupy positions in the sequence the temporal operator runs over, so cross-modal pairs are formed by the same mechanism that forms temporal pairs.

The implicit assumption in most of this literature is that deeper is better — more interaction points give the model more opportunity to condition one modality on the other. That assumption is rarely tested, because comparisons across the rows of Table 2.3 are almost always confounded: different papers, different backbones, different data, different horizons. Isolating the axis requires holding everything else fixed, which is what the late-fusion and interleaved arms of Chapter 8 are constructed to do.

There is also a cost argument that cuts the other way. Cross-attention between a temporal sequence of length L_t and a visual sequence of length L_v costs $O(L_t L_v)$ per layer. For a long numerical context this becomes the dominant term, which is why cross-attention multimodal forecasters typically use short numerical contexts — and short contexts are precisely what harmed the foundation models in preliminary experiments. Interleaving, by contrast, costs $O(L_t + L_v)$ under a linear temporal operator, which is what makes a fourteen-day context and deep fusion simultaneously affordable.

2.9 Evaluation methodology

A forecasting result is a claim about a metric, and the metrics used here carry assumptions worth making explicit.

The skill score and its reference. Skill scores of the form $1 - \mathcal{L}_{\text{model}}/\mathcal{L}_{\text{ref}}$ are standard in the atmospheric sciences, and their interpretation depends entirely on the reference. A skill score against climatology answers “does the model know anything about today”; against persistence, “does it know anything beyond the present”; against smart persistence, “does it know anything beyond the present that is not implied by solar geometry”. Only the third is a statement about forecasting skill in the sense this thesis cares about, and it is a substantially harder bar — the difference is visible in Chapter 6, where several models that comfortably beat naive persistence sit near zero against the smart variant.

Proper scoring for distributions. The continuous ranked probability score is a strictly proper scoring rule [29]: its expectation is uniquely minimised by the true predictive distribution, so a model cannot improve it by misrepresenting its own uncertainty. Approximating it by pinball loss over a finite quantile grid preserves propriety up to discretisation. Reporting it alongside point metrics is what distinguishes a forecast that is useful for decision-making from one that is merely accurate on average.

Significance. Comparing forecast accuracy on overlapping windows violates the independence assumptions of naive tests, since consecutive forecast origins share most of their context. The standard remedy is the Diebold–Mariano test [27] on the loss differential series with an autocorrelation-consistent variance estimator. This thesis does not report such tests — a limitation stated plainly in Section 9.4 — and consequently treats small differences between models as ties rather than as orderings.

Macro-averaging. Metrics are computed per plant and then averaged, rather than pooled over all observations. Pooling would weight plants by their number of valid steps and by their capacity factor, letting a few data-rich, high-output sites dominate. Since the research question is about generalisation *across* plants, each plant is one unit of evidence.

2.10 Positioning

Table 2.4 places this work against its closest neighbours on the four axes that matter: whether the backbone is frozen, whether real external imagery is consumed, how deeply the modalities are fused, and whether evaluation is cross-plant.

Work	Frozen	Real imagery	Fusion depth	Cross-plant
Chronos-2 [11]	—	no	— (unimodal)	n/a
TS-RAG [18]	yes	no	— (retrieval)	varies
CoRA [12]	yes	optional	adapter	no
UniCast [19]	yes	yes	prompt	no
Time-VLM [15]	yes	no (rendered)	readout	no
CrossViVit [26]	no	yes	cross-attention	partial
Solar-VLM [23]	partly	yes	cross-attention	no
This work	yes	yes	token interleaving	yes

Table 2.4: Positioning against the closest published methods. No prior work occupies all four columns simultaneously, and — more importantly — no prior work evaluates these alternatives against each other under one protocol.

The last column is the one most often left empty. Cross-plant evaluation is expensive: it requires a multi-plant corpus, disjoint splits and a normalisation scheme that does not leak, and it produces worse numbers than a temporal split on the same data. That combination is a strong incentive to report the easier protocol, and it is why the benchmark of Chapter 6 is a contribution independent of the architecture it was built to evaluate.

2.11 Gap statement

Three gaps follow from the survey above.

1. **No single-protocol comparison exists** across time-series foundation models, frozen-backbone adaptation methods and multimodal forecasters on photovoltaic data. Published numbers are not comparable, so the relative merit of the three strategies is unknown.
2. **Fusion depth has not been isolated.** Late fusion, prompting, adapters and deep cross-attention have each been proposed, but rarely against each other with the same backbones, the same data and the same schedule — so “deep fusion is better” is an assumption rather than a measurement.
3. **The endogenous/exogenous distinction is not controlled for.** Methods that render the series as an image and methods that consume real imagery are reported in the same tables, although they exploit entirely different resources.

Chapters 3 and 4 build the instrument that closes the first gap, Chapter 5 runs it, and Chapters 7 and 8 address the second and third.

2.11.1 Why these gaps have persisted

It is worth asking why, given how active this literature is, the gaps above are still open. Three structural reasons, none of them a criticism of any individual paper.

Protocols are expensive and unrewarded. Building a cross-plant benchmark requires a multi-plant corpus, a leak-free normalisation scheme, disjoint splits and a fairness contract that every model must be ported to. The work is substantial, it produces worse headline numbers than an easier protocol on the same data, and it is not what a methods paper is reviewed on. The incentives favour reporting a new architecture under a convenient protocol.

Comparisons cross paper boundaries. Fusion depth, adaptation strategy and modality type are each explored within papers, but the comparison that matters — late against deep, retrieval against fine-tuning, endogenous against exogenous — falls between them. Nobody owns the cross-cutting question, and citation-based comparison cannot answer it because the protocols differ.

Negative controls are not publication-shaped. A modality-off arm run under an identical recipe is the single most informative experiment a multimodal paper can report, and it is the one most likely to weaken the paper’s headline. That asymmetry is enough to explain its rarity, and it is the reason this thesis treats that arm as the primary result rather than as an appendix.

3 Dataset Construction

The benchmark of this thesis rests on a dataset that did not previously exist in the form required: a single table pairing measured photovoltaic power, weather and solar-geometry covariates, and co-registered satellite imagery, across two regions, at native cadence, with plant identity as the split key. This chapter states the requirements that shaped it, the sources fused, the harmonisation performed, the resulting schema, the quality flags, and the plant-disjoint partition used everywhere in Chapters 5 and 8.

3.1 Design requirements

Five requirements were fixed before construction, each of which excludes a common shortcut.

1. **One row per (plant, timestamp).** Power, covariates and a frame pointer live on the same row. This makes windowing a slicing operation and removes any possibility of misalignment between modalities.
2. **Native cadence, no resampling.** The two source datasets sample at 30 and 15 minutes. Resampling to a common grid would either discard information or fabricate it; instead, windows are defined in *physical time* and resolved to a different number of steps per dataset (Section 4.2).
3. **No gap interpolation beyond a strict threshold.** Long gaps are dropped rather than filled, so that a model is never scored against invented targets.
4. **Plant identity as the split key.** Every row carries `site_id`, and splits partition plants, never timestamps.
5. **Frames addressable in constant time.** Millions of individual image files are impractical on a shared cluster filesystem; frames are packed into one HDF5 archive with a direct integer pointer from each table row.

3.2 Sources

Four public sources are fused. Table 3.1 summarises what each contributes.

Track	Source	Contribution
PV power, UK	openclimatefix/uk_pv [7]	30-minutely domestic generation, 2019–2020, with per-site metadata: rounded latitude/longitude and nameplate capacity in kWp.
Imagery, UK	EUMETSAT SEVIRI RSS, high-resolution visible channel [2]	Approximately 1 km per pixel, reprojected to each site and cropped to 128×128 pixels (≈ 128 km), sized for cloud-advection context.
PV power and imagery, US	NREL PVDAQ via the OEDI data lake [4], paired with GOES-16 crops	10 sites with site metadata (coordinates, nameplate power) and 256×256 RGB satellite crops.
Weather	Open-Meteo Archive API [3]	Eight hourly variables per site: 2 m temperature, shortwave / direct / diffuse radiation, direct normal irradiance, cloud cover, 10 m wind speed, precipitation.

Table 3.1: The four public sources fused into the dataset of record.

3.3 Harmonisation

Power. UK generation is published as energy over a 30-minute interval (`generation_Wh`) and is converted to instantaneous power in watts by a factor of two. Gaps of at most three steps are linearly interpolated; longer gaps are dropped rather than filled. Nameplate capacity is audited per site and stored as `installed_power_w`; the modelling target is

$$\text{norm_power} = \frac{\text{power_w}}{\text{installed_power_w}} \in [0, 1],$$

set to NaN on rows flagged as outage or stuck. Capacity normalisation is what makes a 1.5 kW rooftop and a 408 kW utility array commensurable, and — critically for the cross-plant protocol — it uses a quantity known at installation rather than a statistic estimated from the measured signal.

Weather. The eight Open-Meteo variables are hourly, while the power tracks are sub-hourly; they are joined to each row by nearest timestamp using an as-of merge, so a 30-minute row inherits the enclosing hour’s reanalysis values.

Solar geometry and clear-sky quantities. Deterministic quantities are derived per row from coordinates and timestamp: solar zenith and azimuth, a Haurwitz clear-sky global horizontal irradiance model [30], the clearness index k_t , the clear-sky index (undefined and left as NaN below 50 W/m^2 of clear-sky irradiance), sine and cosine day-of-year encodings, and local apparent solar time. These are the only covariates that are legitimately *known in advance* over the forecast horizon; the distinction is what separates the honest and oracle variants of the foundation-model baselines in Section 4.6.

Frames. Per frame, missing values are set to zero, values are clipped to $[0, 1]$, and empty crops are discarded. Frames are written as a single `uint8` HDF5 archive rather than as individual files.

3.4 Storage layout and the frame pointer

The result is two artefacts:

- `dataset_all.parquet` — 1,337,654 rows $\times$ 35 columns, covering both datasets;
- `images_all.h5` — 27 GB, 110 per-site HDF5 groups named `<dataset>_<site>`, each holding an `images` array and a `timestamps` array of ISO-8601 byte strings.

Alignment between the two is carried by a single canonical column, `image_h5_index`. It is a *group-local* index into `images_all.h5[<dataset>_<site>]["images"]` — not a global row number — and

it is timestamp-exact, verified row by row for both datasets. This deserves emphasis because the table retains two legacy pointer columns: `image_index`, which approximates the canonical pointer, and `image_uk128_index`, which is dead and points into a file that no longer exists. Any re-use of this dataset should read frames exclusively through `image_h5_index`.

A visual step is considered valid — the mask `mask_visual = 1` — if and only if the row has a matching pointer and timestamp. Night and outage steps have no frame, which means the visual mask is structurally correlated with the diurnal cycle; models must handle this rather than assume a dense visual stream.

3.5 Schema

Table 3.2 groups the 35 columns by role.

Group	Columns
Identity	<code>dataset</code> , <code>site_id</code> , <code>station_id</code> , <code>camera_id</code> , <code>timestamp_utc</code> , <code>latitude</code> , <code>longitude</code>
Target	<code>power_w</code> , <code>installed_power_w</code> , <code>norm_power</code>
Weather covariates	<code>temperature_2m</code> , <code>shortwave_radiation</code> , <code>direct_radiation</code> , <code>diffuse_radiation</code> , <code>direct_normal_irradiance</code> , <code>cloudcover</code> , <code>windspeed_10m</code> , <code>precipitation</code>
Solar geometry and clear-sky	<code>solar_zenith</code> , <code>solar_azimuth</code> , <code>clearsky_ghi</code> , <code>kt</code> , <code>csi</code> , <code>doy_sin</code> , <code>doy_cos</code> , <code>solar_time</code>
Quality flags	<code>capacity_fixed</code> , <code>outage_flag</code> , <code>stuck_flag</code> , <code>night_clamped</code> , <code>bad_site_flag</code>
Frame pointers	<code>image_h5_index</code> (canonical), <code>image_index</code> , <code>image_path</code>

Table 3.2: Column groups of `dataset_all.parquet`.

3.6 Quality flags and curation

Curation is recorded as flags rather than applied destructively, so that any downstream decision to include or exclude a row is visible and reversible.

- `bad_site_flag` — whole sites whose series is unusable: `uk_pv` sites 7239 and 8587, `goes_pvdaq` sites 1283 and 51.
- `outage_flag` — 15,486 rows with zero or anomalous output during daylight.
- `stuck_flag` — 1,318 rows where the sensor reports a constant value across a period during which it should have varied.
- `night_clamped` — 1,535 rows where night-time output was clamped to zero.

One inconsistency is recorded here rather than concealed: the committed `goes_pvdaq` split pre-dates the introduction of `bad_site_flag` and still assigns sites 1283 and 51 to train and validation respectively. That split must be regenerated, excluding the two flagged sites and leaving eight usable plants, before the `goes_pvdaq` track is run. No result reported in this thesis is affected, because every reported run is restricted to `uk_pv`; but the limitation is real and is restated in Section 9.4.

3.7 Statistics

Dataset	Sites	Rows	Cadence	Span (UTC)	Frames
<code>uk_pv</code>	100	1,232,862	30 min	2019-01-01 – 2020-12-31	(N , 128, 128) uint8 grey
<code>goes_pvdaq</code>	10	104,792	15 min	2019-01-01 – 2019-09-30	(N , 256, 256, 3) uint8 RGB

Table 3.3: Per-dataset specification. Valid power steps: 1,217,399 and 103,451 respectively.

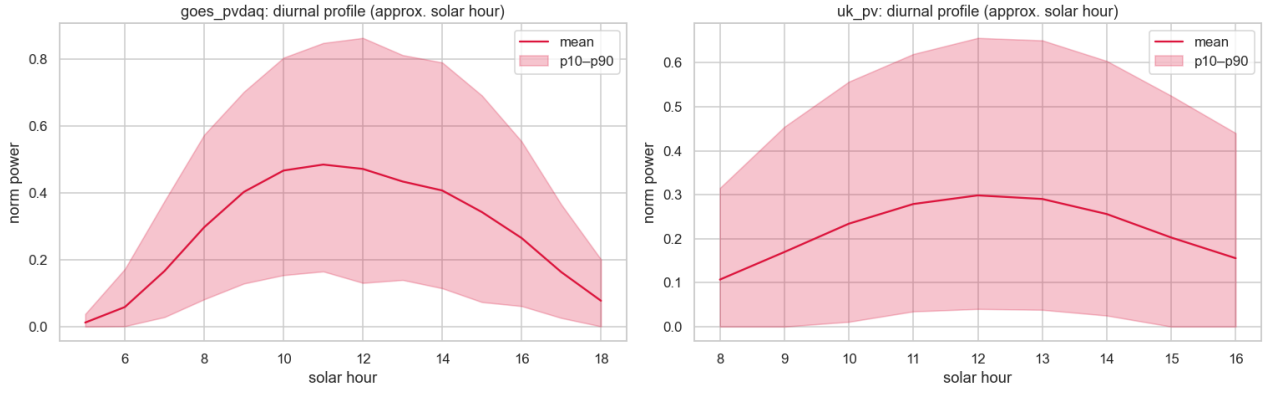


Figure 3.1: Mean capacity-normalised power against solar hour, with the 10th–90th percentile band, for both tracks. The mean is a smooth deterministic arc — which a clear-sky model captures for free — while the band is enormous: at solar noon the tenth percentile of `uk_pv` output is near zero and the ninetieth is above 0.65. Essentially all of the forecasting difficulty lives in that band, not in the mean, which is precisely why smart persistence rather than climatology is the reference the skill score is defined against.

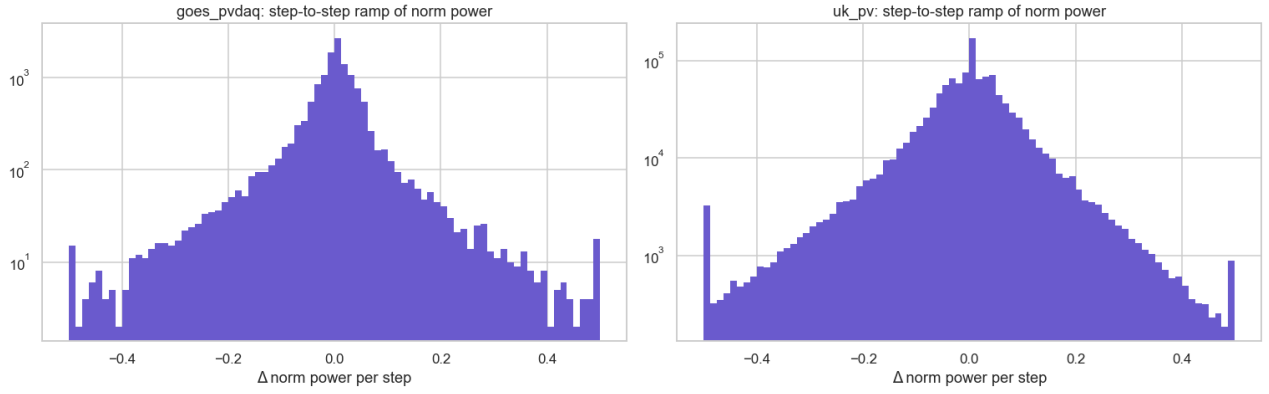


Figure 3.2: Distribution of step-to-step changes in normalised power (logarithmic counts). The distribution is sharply peaked at zero with heavy, near-exponential tails — the signature of a signal that is usually smooth and occasionally discontinuous. The ramp subset of Section 4.3 isolates the tails, which is where a visual modality is expected to contribute and where persistence-like models fail.

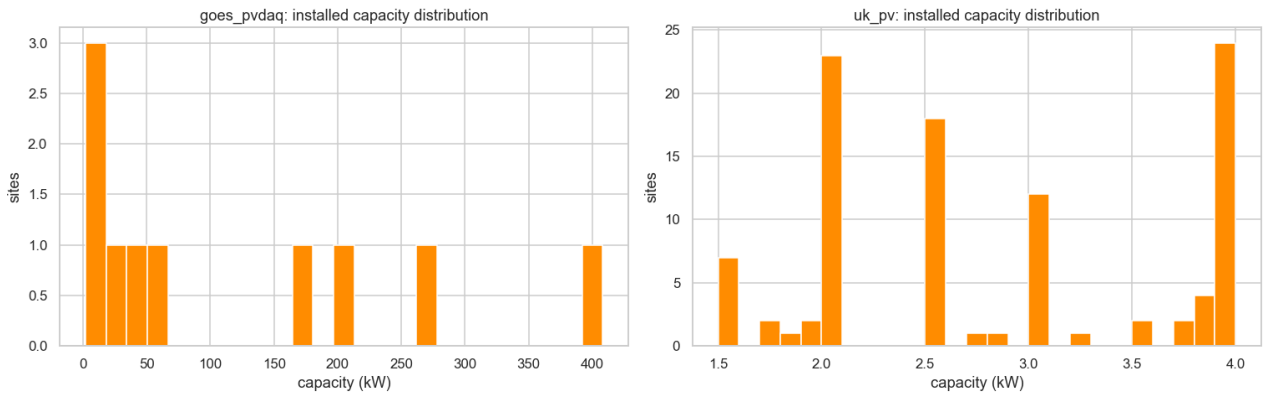


Figure 3.3: Installed capacity per site. `uk_pv` is a narrow residential band clustered at round values between 1.5 and 4.0 kW; `goes_pvdaq` spans more than two orders of magnitude, from small residential systems to a 408 kW array. Capacity normalisation is what makes the two commensurable.

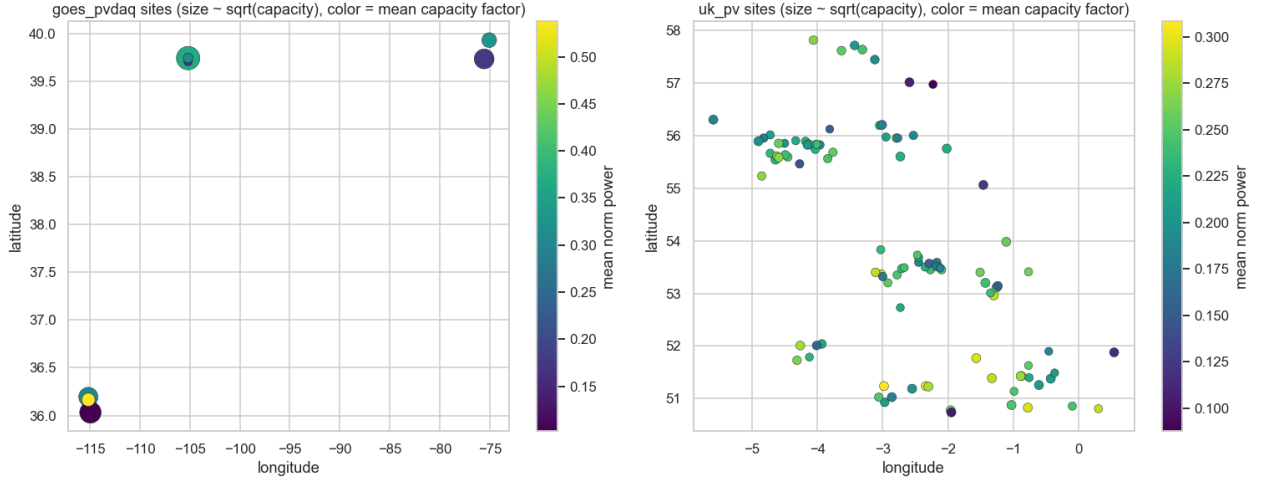


Figure 3.4: Site locations, sized by capacity and coloured by mean capacity factor. The `uk_pv` fleet is spatially dense: many sites are within tens of kilometres of each other and therefore share a cloud field. This is the geometry behind the spatial-leakage caveat of Section 3.9 — a random plant split does not guarantee spatial separation. Note also the systematic north–south gradient in capacity factor, a generalisation axis a cross-plant model must absorb.

The two tracks differ in almost every respect that matters for generalisation, which is what makes their combination valuable. `uk_pv` is a dense fleet of 100 residential rooftop systems between 1.5 and 4.0 kW, confined to a small geographic envelope (latitude 50.7–57.8, longitude –5.6–0.5) and observed through a single-channel high-resolution visible product. `goes_pvdaq` spans capacities from 1.8 to 408 kW — residential through utility scale — across a continental spread of the United States (latitude 36.0–39.9, longitude –115.2––75.0), observed in three-channel RGB at twice the temporal cadence but over nine months rather than two years. A model that transfers between them is transferring across sensor, scale, climate and cadence simultaneously.

3.8 Exploratory analysis

This section characterises the signal a forecaster has to model. It is not a formality: three of the design decisions defended elsewhere in this thesis — the choice of smart persistence as the reference, the decision to keep a short visual window, and the expectation that vision should matter in the ramp regime — rest on properties visible here and nowhere else.

3.8.1 The signal is a deterministic envelope plus a stochastic modulation

Figure 3.5 makes the structure of the problem concrete over a single week. Shortwave radiation (blue) traces smooth, nearly symmetric daily arcs whose amplitude varies with season and synoptic conditions. Measured power (red) follows the same envelope but is persistently rougher: on 05 and 07 January the power curve collapses to a fraction of what the radiation curve implies; on 10 and 19 January the two agree closely; and on several days the power series oscillates at high frequency within a single afternoon while the radiation series — which is hourly reanalysis interpolated to the power grid — stays smooth.

That mismatch is the forecasting problem in one picture. The deterministic part of the signal is nearly free: solar geometry gives it analytically. The residual is what the model must predict, it lives at a temporal scale finer than the reanalysis resolves, and it is driven by cloud structure that is not represented in any numerical channel at that resolution.

3.8.2 The mean is trivial; the spread is the problem

Figure 3.1 already showed that the conditional mean is a smooth arc with an enormous percentile band around it. Figure 3.6 extends this to the seasonal axis. For `uk_pv` the mean normalised power is an almost perfectly separable function of month and solar hour: a bright core around April to July at

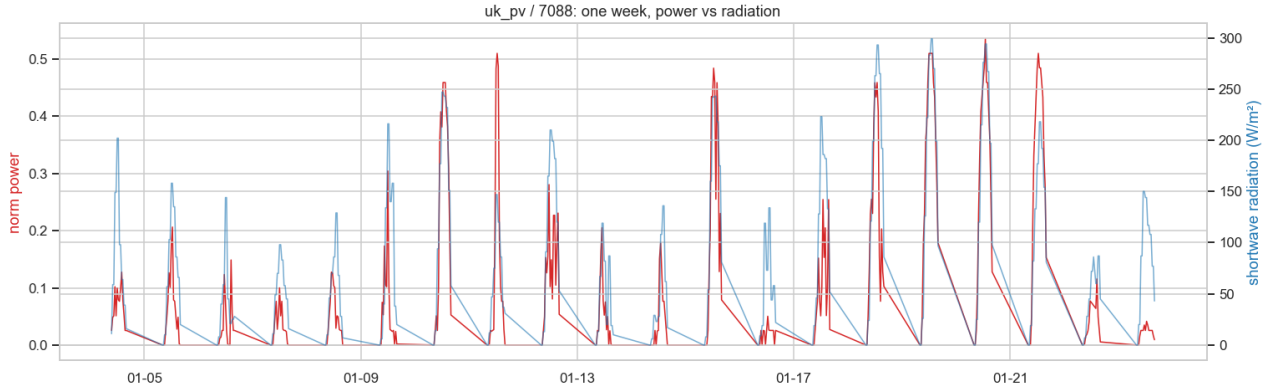


Figure 3.5: One week at a single `uk_pv` site: normalised power (red, left axis) against shortwave radiation (blue, right axis). The radiation channel is hourly reanalysis and is consequently smooth; power varies at the native 30-minute cadence and frequently departs from the envelope the radiation implies. The gap between the two curves is what a forecaster must model, and it is finer-grained than the covariate track can express.

solar noon, decaying symmetrically in both directions, with the winter months (November to January) contributing almost nothing at any hour. The pattern is so regular that a lookup table indexed by month and hour — the hourly climatology baseline — reaches a skill score of 0.234 without observing the current day at all.

The consequence for evaluation is direct. A metric computed against a reference that does *not* know the diurnal and seasonal structure rewards a model mostly for rediscovering astronomy. This is precisely why the skill score in Section 4.3 is defined against smart persistence, which is given the clear-sky curve, rather than against naive persistence or against zero.

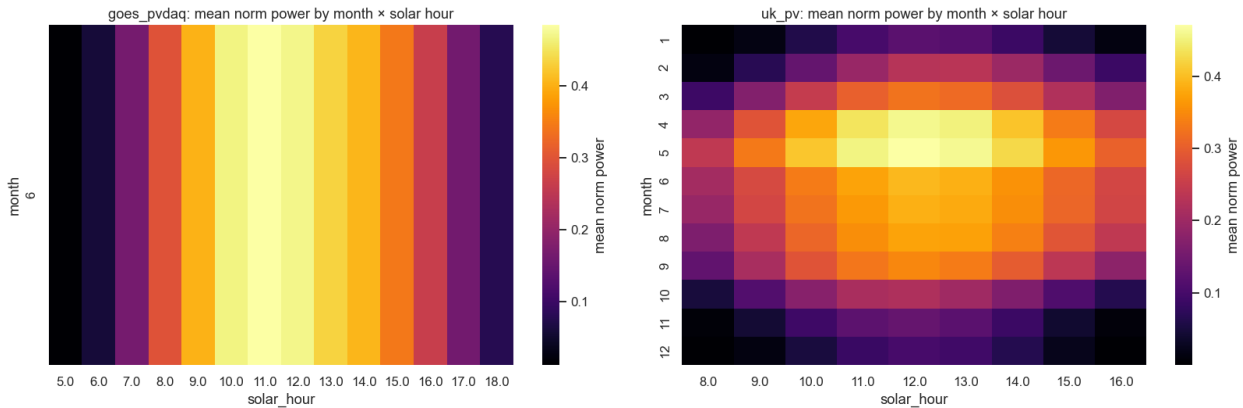


Figure 3.6: Mean normalised power by month and solar hour. The `uk_pv` panel (right) is close to separable in its two arguments — a strong, entirely deterministic seasonal and diurnal structure that any competent model captures and that therefore carries no discriminative information between models. The `goes_pvdaq` panel (left) covers a single month and shows the diurnal axis only.

3.8.3 Predictability decays within the forecast horizon

Figure 3.7 plots the autocorrelation of normalised power against lag in hours. Two readings matter. First, correlation falls below 0.5 within roughly three hours on `uk_pv` and within two on `goes_pvdaq` — comfortably inside the six-hour horizon this thesis forecasts. The task is genuinely hard at the horizon chosen: half the linear predictability is gone before the horizon is half over. Second, the two datasets behave very differently at longer lags. `uk_pv` autocorrelation flattens near 0.35 and then *rises* again towards 0.45 around eight hours, the signature of the diurnal cycle folding back; `goes_pvdaq` plunges to -0.75 at six hours, which is what a single-peaked daily arc produces when the lag reaches

the half-period. The difference is a cadence and latitude effect, and it is a concrete illustration of why metrics are never pooled across the two tracks (Section 4.2).

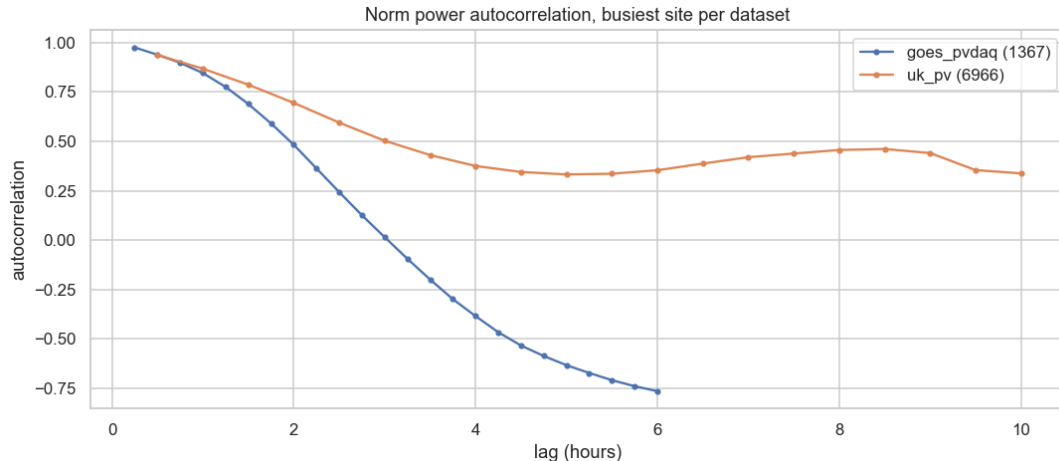


Figure 3.7: Autocorrelation of normalised power against lag in hours, busiest site per dataset. Linear predictability halves within two to three hours — well inside the six-hour forecast horizon. The negative excursion on `goes_pvdaq` is the half-period of the diurnal arc, not a modelling artefact.

3.8.4 Ramps are rare, large, and where the difficulty concentrates

Figure 3.2 showed the step-to-step change distribution: sharply peaked at zero with heavy, near-exponential tails on a logarithmic count axis. Most steps are nearly constant; a small minority move by a third of nameplate capacity in a single interval. A model optimised for average error can ignore the tails almost entirely and still score well, which is exactly why the ramp subset of Section 4.3 exists as a separate reporting column. The spikes at the extremes of both panels are the clipping bounds of the histogram, not a physical accumulation.

3.8.5 Covariate structure: strong, redundant, and incomplete

Figure 3.8 gives the Pearson and Spearman correlation structure over daytime rows. Three observations shape the modelling.

The covariates are *strong*: normalised power correlates at 0.74 with shortwave radiation, 0.69 with direct radiation and 0.60 with direct normal irradiance. A model with access to the covariate track starts from a substantial advantage over one restricted to the power history — which is visible in the leaderboard, where every covariate-consuming model outranks DLinear.

The covariates are *redundant*: shortwave and direct radiation correlate at 0.93 with each other, and direct normal irradiance with cloud cover at -0.70 . There are not eight independent weather channels here; there are perhaps three degrees of freedom presented eight ways. This is the structure that variate-attention architectures exploit and that channel-independent ones cannot, and it is the most plausible explanation for the ordering between iTransformer and PatchTST in Chapter 6.

The covariates are *incomplete*, which is the point that motivates the entire visual branch of this thesis. Figure 3.9 shows the joint density of power against each driver. The relationship with shortwave radiation is broad rather than tight: at 400 W/m^2 observed power ranges over most of the unit interval. Cloud cover, remarkably, shows almost no marginal relationship at all — the density is nearly uniform in the horizontal direction — because cloud cover as an hourly areal fraction says little about whether the sun is occluded at a specific site at a specific minute.

3.8.6 There is residual cloud information beyond irradiance

Figure 3.10 isolates the effect that motivates a visual modality. Daytime observations are binned by shortwave radiation, and within each bin further split by cloud cover. If the reanalysis irradiance already captured the cloud state, the three boxes within a bin would coincide. They do not. In the $250\text{--}450 \text{ W/m}^2$ bin the median falls from 0.33 under clear skies to 0.27 under mixed conditions to 0.24

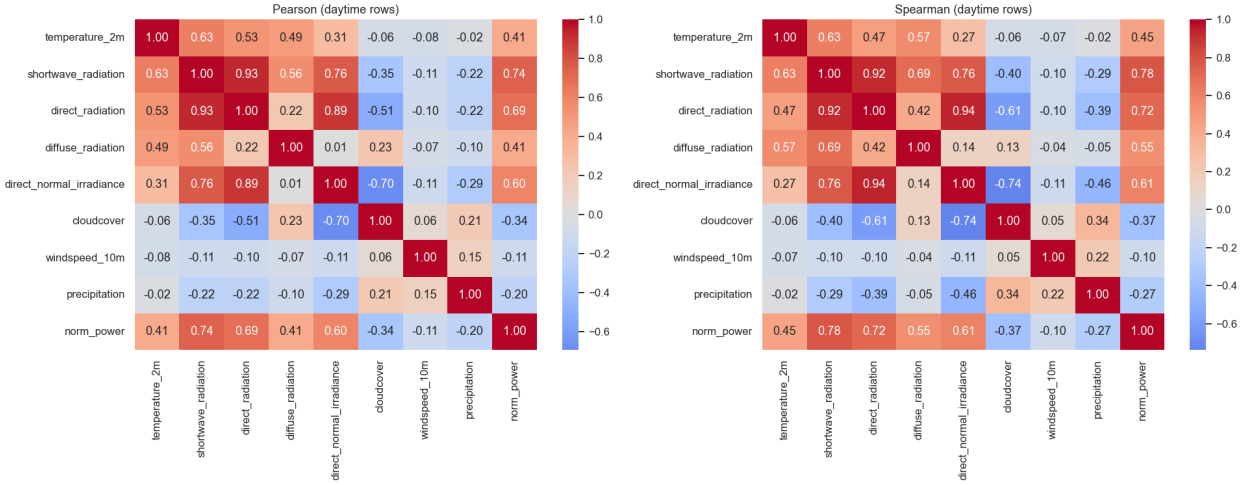


Figure 3.8: Pearson (left) and Spearman (right) correlation over daytime rows. The covariate block is strongly inter-correlated — 0.93 between shortwave and direct radiation — so the effective dimensionality of the weather track is far below its eight channels.

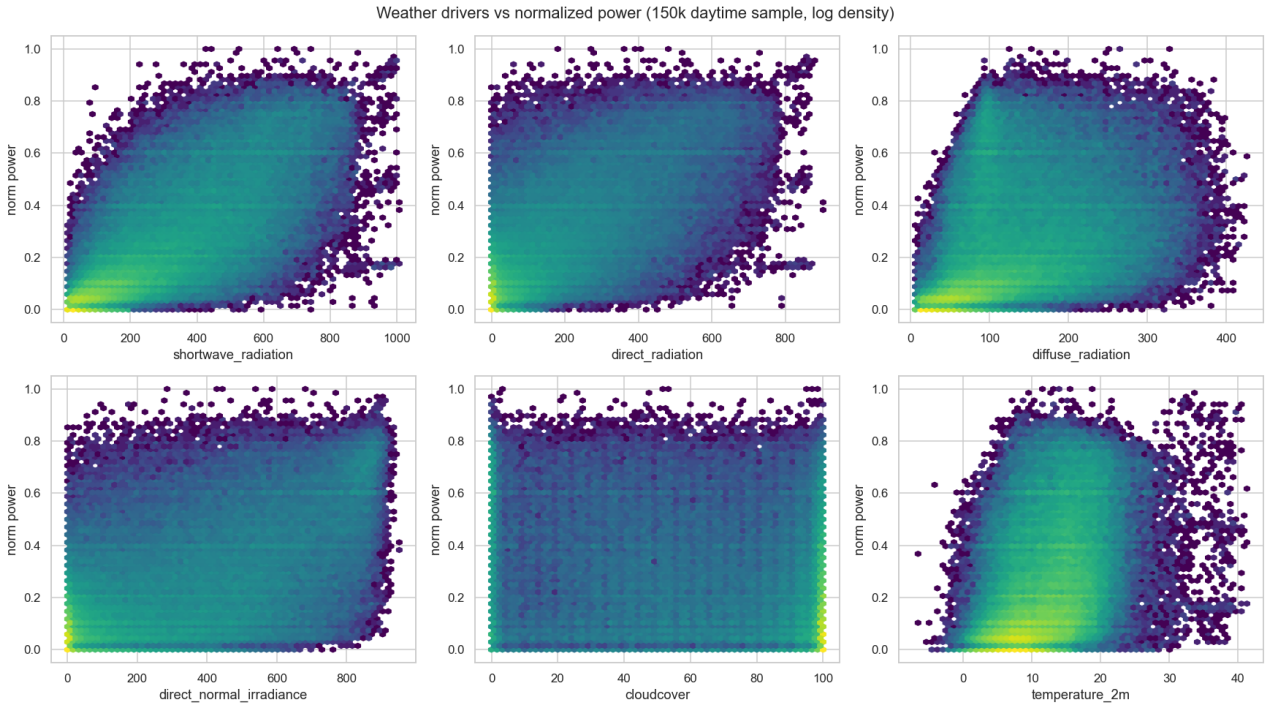


Figure 3.9: Joint density of normalised power against each weather driver (logarithmic density, 150k daytime sample). The irradiance panels show broad, cone-shaped relationships rather than tight ones; the cloud-cover panel is nearly uninformative marginally, because an hourly areal cloud fraction does not determine occlusion at a point.

under overcast, and the same monotone ordering holds in every bin. Conditioning on cloud state adds information *after* conditioning on irradiance.

This is the empirical case for the thesis. There exists cloud-state information not carried by the numerical channels, it is exactly the kind of information a satellite crop observes directly, and the size of the effect — roughly 0.09 in normalised power at fixed irradiance — is of the same order as the differences separating models in the leaderboard. Chapter 6 reports that no model consuming real frames converts this into skill, which makes the figure the sharpest statement of the gap between what is available and what is exploited.

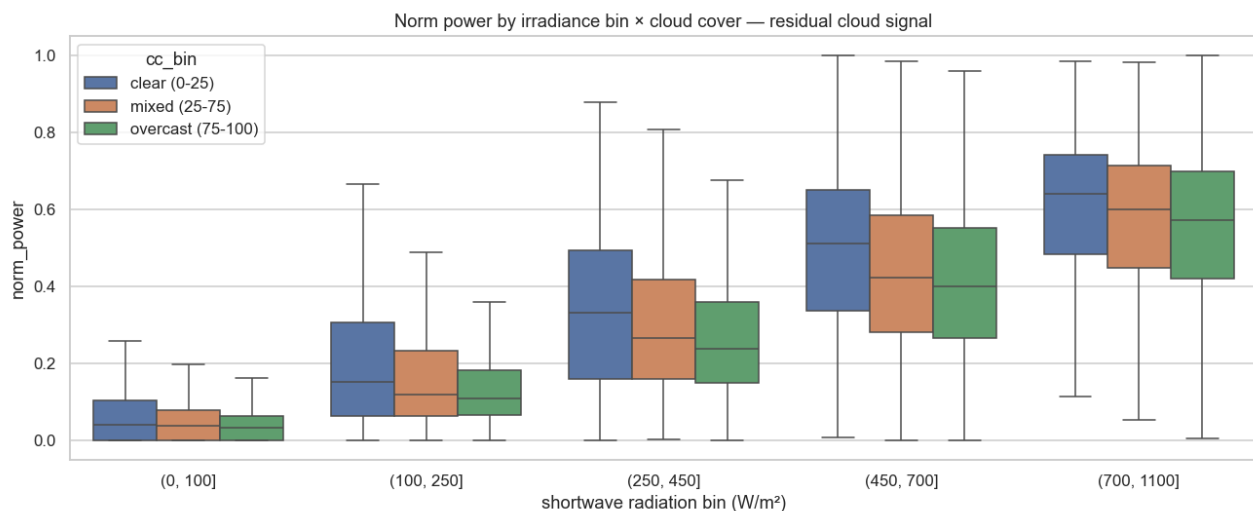


Figure 3.10: Normalised power by irradiance bin, split by cloud cover within each bin. The monotone separation of the three boxes at fixed irradiance is residual cloud-state information that the numerical covariate track does not carry — and that a satellite frame observes directly.

3.8.7 The second track behaves differently

Figure 3.11 shows the same week-long view for a `goes_pvdaq` site. The contrast with Figure 3.5 is the reason the two tracks are kept separate in every table. The cadence is twice as fast, the capacity factor is markedly higher, the daily arcs are broader — a lower-latitude summer — and the power curve tracks the radiation envelope more closely, with fewer of the deep intra-day collapses that characterise the UK fleet.

A model that transfers between these tracks is transferring across sensor (greyscale high-resolution visible against three-channel GOES), spatial scale (a 128 km crop at 1 km resolution against a 256-pixel RGB frame), capacity class (kilowatts against hundreds of kilowatts), climate, latitude and sampling cadence simultaneously. That is a genuine distribution shift and the reason scenario S3 is the strongest available test of generalisation — and the reason its absence from the executed scenarios is the principal limitation of this work.

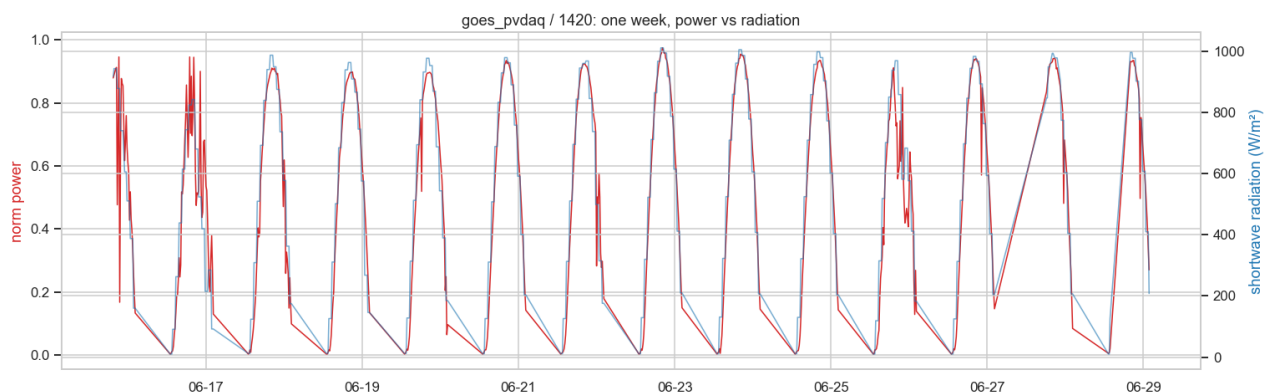


Figure 3.11: One week at a `goes_pvdaq` site, at 15-minute cadence. Compare Figure 3.5: higher capacity factor, broader arcs, closer tracking of the radiation envelope. The two tracks are different forecasting problems sharing a schema.

3.8.8 Between-plant heterogeneity

Figure 3.12 shows the distribution of daytime normalised power per site. Even after capacity normalisation, medians range from below 0.10 to above 0.55, and the inter-quartile ranges differ by a factor of three. Part of this is geography — the north–south gradient visible in Figure 3.4 — and part is

per-installation factors that the dataset does not record: tilt, azimuth, shading, module technology, inverter clipping behaviour.

This heterogeneity is what makes cross-plant transfer non-trivial and what a temporal split would hide entirely. A model fitted on one plant and tested on the same plant absorbs its orientation and shading into its parameters; a model that must generalise across plants cannot, and has to recover whatever is recoverable from a two-week history at inference time. The `goes_pvdaq` sites (orange) sit systematically higher, reflecting both lower latitude and a different mix of installation types — which is why cross-dataset transfer is a genuine distribution shift and not a formality.

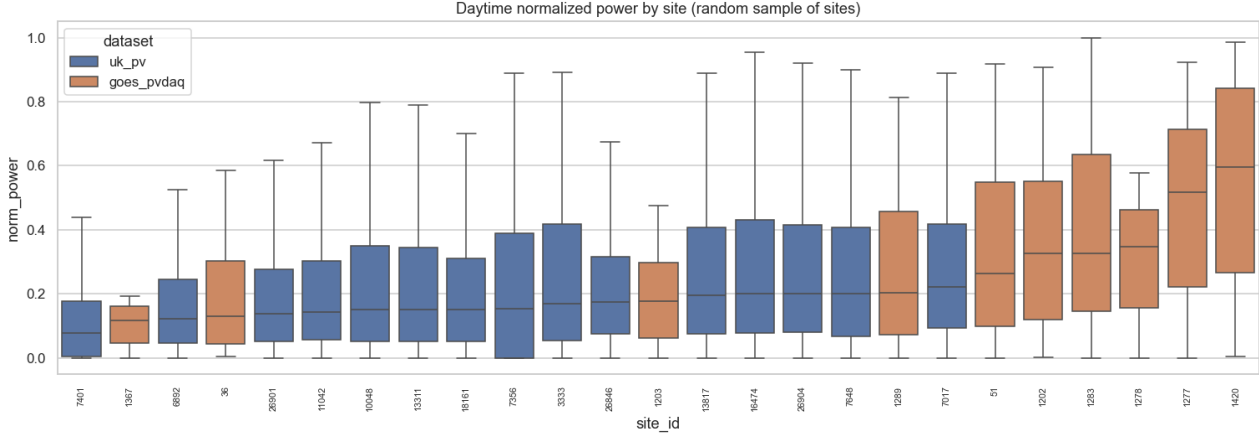


Figure 3.12: Daytime normalised power by site, random sample of sites from both tracks. Capacity normalisation removes the scale difference but not the difference in *capacity factor*; the residual spread is the between-plant variation a cross-plant model must absorb.

3.8.9 The target distribution is zero-inflated and right-skewed

Figure 3.13 shows why the target is awkward for a general-purpose sequence model. Over all rows the distribution has a dominant spike at zero — night, outage and deep-overcast steps — and a long right tail that never approaches the theoretical maximum of one. Restricting to production steps removes the spike but leaves a strongly right-skewed distribution with most mass below 0.3. In absolute units the picture is different again: log-power is roughly bimodal, reflecting the two capacity populations of the two tracks.

Three modelling consequences follow. A model whose output head assumes an unbounded, approximately symmetric predictive distribution — as most general-purpose foundation model heads do — is mismatched at both ends: it can predict negative power, and it wastes probability mass above the observed ceiling. The zero spike means that a large fraction of the loss is contributed by observations that are trivially predictable given solar geometry, which inflates apparent accuracy unless evaluation is restricted to daylight steps — as it is here. And the right skew means mean-squared objectives are dominated by a minority of high-output steps, which is one reason NMAE and NRMSE order models differently in Chapter 6.

3.8.10 Per-plant driver relationships are stable

Figure 3.14 decomposes the between-plant variation of Figure 3.12 into two histograms. Daytime mean capacity factor is tightly concentrated for `uk_pv` — most sites between 0.20 and 0.32 — with the `goes_pvdaq` sites spread much wider and reaching 0.54. But the per-site correlation between normalised power and shortwave radiation is remarkably consistent: nearly every `uk_pv` site falls between 0.70 and 0.87, and the `goes_pvdaq` sites reach 0.94.

This is the empirical basis for believing cross-plant transfer is possible at all. Plants differ substantially in their *level* — how much they produce — but very little in the *structure* of their response to the dominant driver. A model that learns the irradiance-to-power relationship on one fleet has learned something that holds on another; what it cannot carry over is the plant-specific scaling, which is exactly what the short inference history is there to supply.

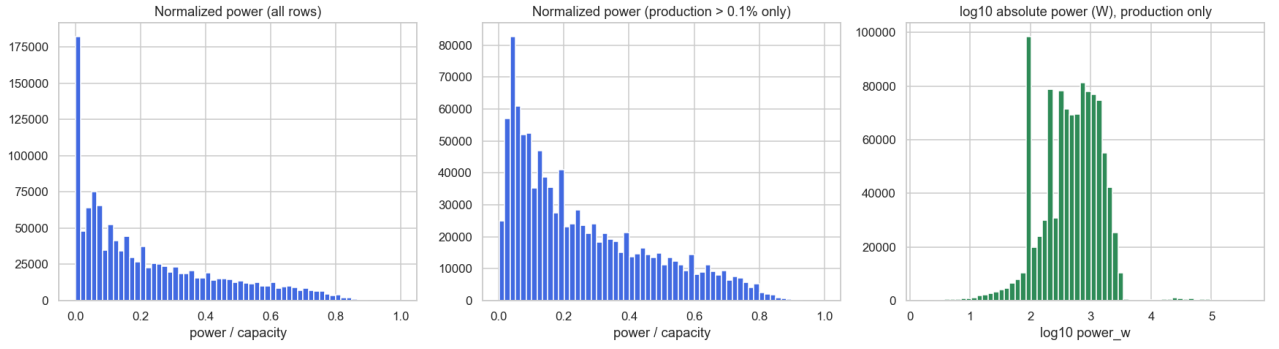


Figure 3.13: Target distribution. Left: all rows, showing the dominant zero spike. Centre: production steps only, right-skewed with most mass below 0.3 and an effective ceiling near 0.85 rather than 1.0. Right: $\log_{10}$ absolute power, whose bimodality reflects the two capacity populations.

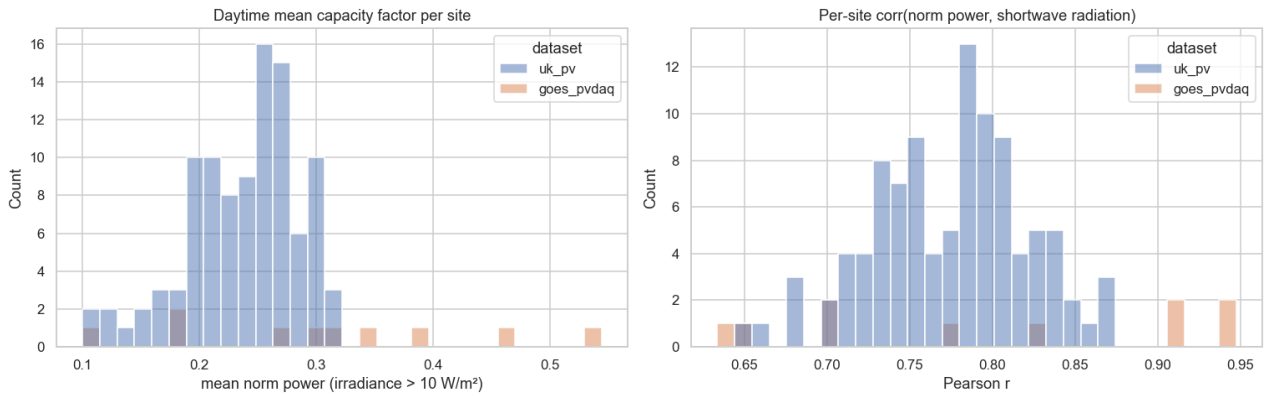


Figure 3.14: Left: daytime mean capacity factor per site. Right: per-site Pearson correlation between normalised power and shortwave radiation. Levels differ substantially across plants; the structure of the driver relationship barely does. Cross-plant transfer is a problem of recovering level, not of relearning structure.

3.8.11 Coverage and missingness

Figure 3.15 shows per-site data availability by month. Coverage is dense but not complete: a minority of sites begin reporting only in mid-2019, and a few have interior gaps. Nothing is imputed across these gaps — windows that would span one are simply not generated — which keeps the evaluation honest at the cost of a slightly uneven sampling of forecast origins across sites.

The structural missingness matters more than the incidental kind. Night steps have no frames at all, and outage and stuck-sensor rows are masked out of the target. The visual mask is therefore correlated with time of day by construction, and any model consuming the visual stream must handle a systematically absent modality rather than a randomly absent one. This is the reason modality dropout is part of the tensor contract of Section 3.10 rather than a training-time detail.

Quantitatively, coverage is better than the timeline suggests. Figure 3.16 shows the fraction of days within each site’s span that carry data: every `uk_pv` site exceeds 99.3 percent, and forty-four of them are complete. The gaps visible in Figure 3.15 are therefore late *starts* rather than interruptions — a set of sites that joined the fleet mid-2019 — which affects how many windows a site contributes but not the continuity of the windows it does contribute.

3.8.12 Seasonality and the two-year span

Figure 3.17 shows mean normalised power by calendar month across the full `uk_pv` span. The seasonal cycle is large — a factor of eight between December (0.05) and May (0.42) — and it repeats closely across the two years, with 2020 running slightly higher in spring than 2019.

Two consequences for evaluation follow. Because the split is by plant rather than by time, every

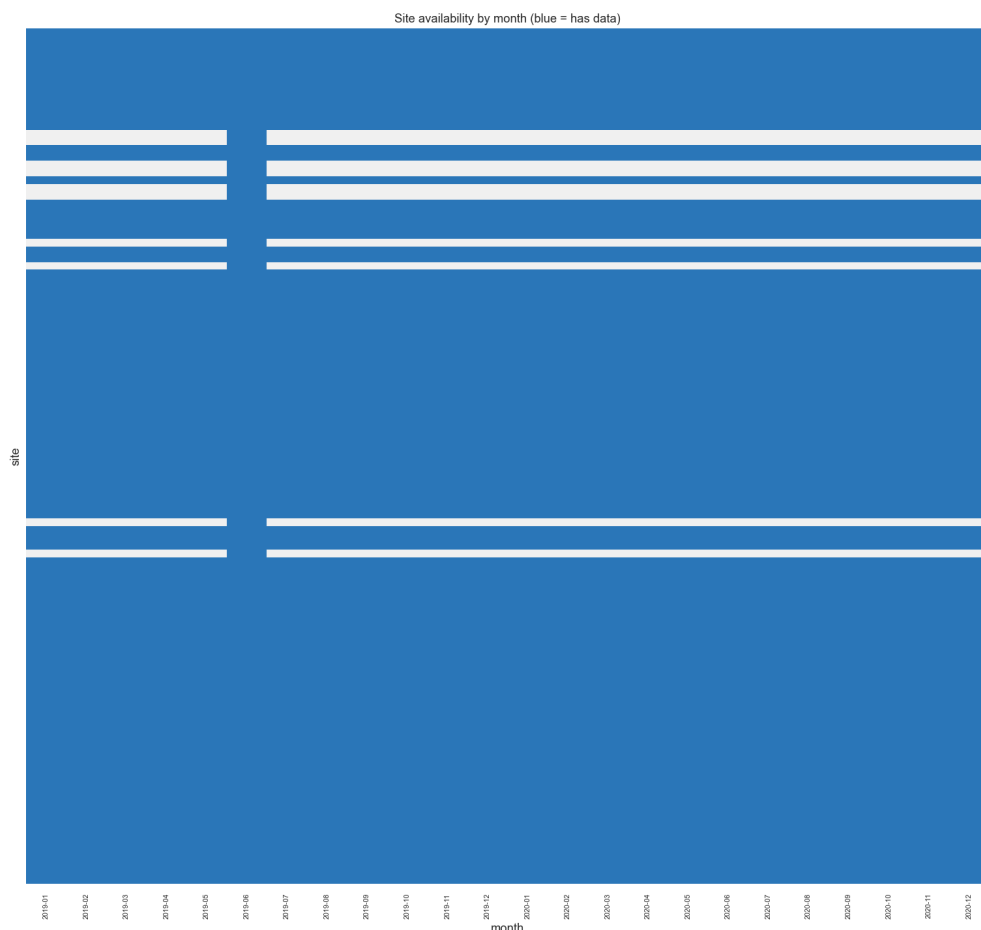


Figure 3.15: Per-site availability by month (blue indicates data present). Coverage is broadly dense; a minority of sites start mid-2019 or carry interior gaps. No values are imputed across gaps — affected windows are not generated.

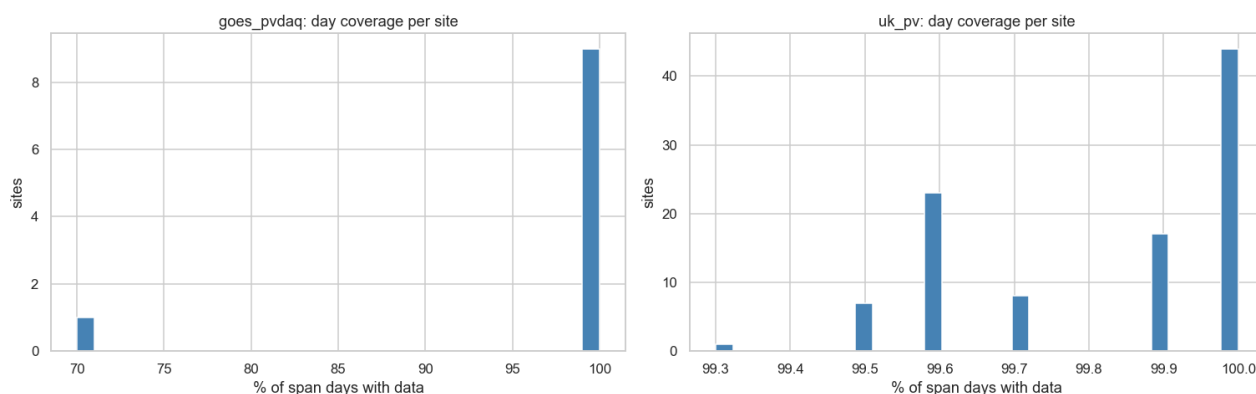


Figure 3.16: Percentage of days within each site's span that carry data. Coverage exceeds 99.3 percent for every `uk_pv` site. The apparent gaps in Figure 3.15 are late starts, not interruptions.

plant contributes windows from every season, so no model is advantaged by a favourable temporal slice; the seasonal cycle is common-mode across train, validation and test. And because the span covers two full years, the seasonal cycle is observed twice, which is the minimum required for a fourteen-day context window to be interpretable: with a single year, a model could not distinguish a seasonal trend from a secular one. The single-month `goes_pvdaq` overlap point is a reminder that the second track covers only nine months and cannot support the same seasonal analysis.

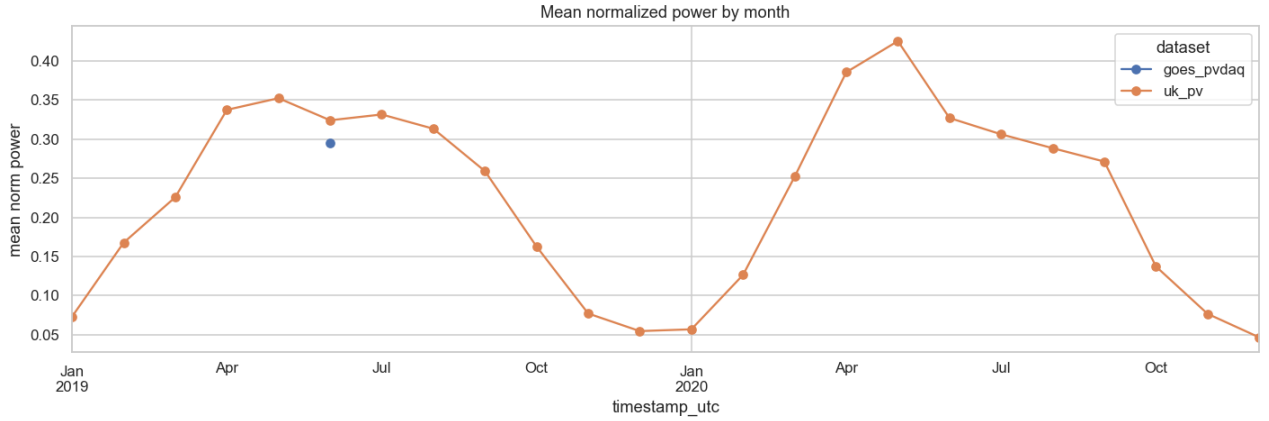


Figure 3.17: Mean normalised power by month over the full span. The seasonal cycle spans a factor of eight and repeats closely across the two years. Plant-based splitting makes this cycle common-mode across train, validation and test.

3.9 Splits

Splits partition *plants*, are generated once with seed 42 at a per-dataset 70/15/15 ratio, exclude sites carrying `bad_site_flag`, and are committed to a version-controlled file. Disjointness is asserted programmatically at every data load rather than trusted; a leak would raise rather than silently improve a score.

Role	Plants	Rows	Valid steps	Purpose
Train	69	850,654	846,633	Fit parameters, fine-tune backbones
Validation	15	184,899	182,809	Early stopping, hyperparameter and mixing-weight tuning
Test	14	172,656	171,543	Final reporting only
Excluded	2	—	—	<code>bad_site_flag</code> : sites 7239, 8587

Table 3.4: The `uk_pv` plant partition. Exact site membership is listed in Appendix B.

The companion `goes_pvdaq` split is 7 train / 2 validation / 1 test. With ten plants, a fixed 15% test share is one or two plants, and per-plant variance would dominate any comparison; that track is therefore additionally evaluated *leave-one-plant-out*, reporting mean and standard deviation over folds.

A limitation of random plant assignment. Assigning plants to splits at random does not guarantee *spatial* disjointness. Two `uk_pv` rooftops a few kilometres apart experience nearly the same cloud field, so a random split can place a test plant’s near-twin in the training set and leak spatial information that a distance-based or region-based split would withhold. The split used here is random because it is the convention the baseline literature follows and because changing it mid-study would invalidate cross-comparison; a region-based variant is identified as future work in Section 9.5. The effect is expected to inflate all models equally rather than to favour any particular one, but it inflates the absolute numbers.

3.10 The tensor contract

Every model in this thesis — baselines and proposed architecture alike — consumes the same dictionary emitted by the dataset adapter. Fixing this contract is what makes the comparison of Chapter 5 a comparison of *models* rather than of data pipelines. With N entities, history T , horizon H and T_v frames:

Key	Shape	Content
Y	(N, T, C_{tgt})	Normalised historical target
Y_future	(N, H, C_{tgt})	Ground-truth horizon
X_cov	$(N, T + H, C_{\text{cov}})$	Historical and future covariates
V	$(N, T_v, C_{\text{img}}, H_{\text{img}}, W_{\text{img}})$	Frames normalised to $[0, 1]$
timestamps	$(T + H,)$	Unix epoch over the full window
timestamps_v	$(T_v,)$	Unix epoch of the frames
entity_ids	$(N,)$	Plant identifiers
mask_target	(N, T, C_{tgt})	Validity of historical targets
mask_future	(N, H, C_{tgt})	Evaluation mask
mask_visual	(N, T_v)	Validity of each frame
mask_modality_dropout	$(N, 2)$	Numeric / visual dropout
adj_matrix	(N, N)	Precomputed spatial adjacency

Table 3.5: The canonical batch dictionary. All models read this and nothing else.

3.10.1 Notes for re-use

Six points would save a future user of this dataset a considerable amount of time, and they are collected here rather than scattered through the chapter.

1. **Read frames only through `image_h5_index`**, and remember it is group-local. `image_uk128_index` is dead; `image_index` is an approximation retained for compatibility.
2. **Respect `bad_site_flag`** before generating splits. The committed `goes_pvdaq` partition predates the flags and must be regenerated.
3. **Do not standardise per series.** Use capacity normalisation; anything estimated from the series leaks under a cross-plant protocol.
4. **Distinguish zero from missing.** Night power is genuinely zero; outage power is absent. The masks carry the distinction and the target column does not.
5. **Do not resample across tracks.** The two cadences are native and physical-time windows resolve to different step counts; pooling raw step-horizon metrics across them is a category error.
6. **Expect a structurally missing visual stream.** There are no night frames, so visual availability is correlated with time of day by construction.

Two design decisions in this contract are worth stating explicitly. Masks are carried alongside the data rather than encoded as sentinel values, so that a model can distinguish “zero power” from “no measurement” — a distinction that matters enormously at dawn and dusk. And `mask_modality_dropout` is part of the contract rather than a training detail, so that robustness to missing frames can be evaluated under exactly the mechanism used during training.

4 Evaluation Protocol

This chapter is short and load-bearing. Every number in Chapters 5 and 8 is only as credible as the protocol that produced it, and the protocol is where a multimodal cross-plant study is most easily compromised — by a normaliser fitted on test plants, by a horizon that differs between models, by a retrieval datastore that quietly contains the answer.

4.1 Three fairness principles

1. **Same horizon and granularity for every model.** All models forecast the same future window from the same history cadence. Context length is itself a fairness variable: a model given more history is not a better model, so no comparison is made across differing T .

2. **Disjoint test plants, with no statistic leakage.** No model may train on, or derive statistics from, held-out plants. This includes normalisers — hence capacity normalisation (Section 3.3) rather than per-series standardisation.
3. **No domain physics inside models.** Clear-sky-index conversions and irradiance-to-power formulas are excluded from every learned model, so that measured improvements are attributable to modelling rather than to a hand-coded solar prior. The sole exemption is smart persistence, which is the physics reference itself.

4.1.1 The failure modes these principles prevent

Each principle exists because of a specific way a cross-plant multimodal benchmark can be compromised, usually without anyone noticing. Naming the failure mode makes the rule checkable.

Normaliser leakage. The most common and least visible failure. Standardising each series by its own mean and standard deviation is routine in forecasting, and under a cross-plant protocol it silently passes test-plant statistics into the model. The symptom is a benchmark on which every method looks unusually strong and no method generalises in deployment. Capacity normalisation eliminates the channel entirely, because nameplate capacity is metadata, not a measurement.

Context-length mismatch. If one model receives fourteen days and another six hours, the comparison measures history length rather than architecture. The failure is easy to introduce accidentally, because foundation models have fixed maximum contexts and quietly truncate. The rule here is that context length is a fairness variable: models that cannot consume the full window are documented as such rather than silently compared.

Transductive retrieval. A retrieval baseline whose datastore contains test-plant history is performing in-context memorisation of the evaluation target. The resulting number can be excellent and means nothing about zero-shot transfer. The datastore rule closes this, and the transductive condition — which is a legitimate thing to study — is reported separately if at all.

Privileged covariates. Supplying observed future weather over the horizon is not forecasting; it is forecasting given a perfect weather forecast. Because covariate arrays naturally span history and horizon, this is a single indexing decision away at all times. The protocol partitions covariates into deterministic and observed classes (Appendix H), and models receiving the observed class over the horizon are labelled oracle and excluded from competitive comparison.

Physics smuggled into a model. A clear-sky conversion inside a learned model converts a modelling result into a statement about how good the clear-sky model is. Since smart persistence — which is that physics — is the reference, a model that internalises it is partly competing against itself. The exemption is granted to exactly one model, and it is the reference.

Metric-window mismatch. If two models are scored on different evaluation windows, their metrics are not comparable even under an identical protocol. This is the defect behind the flagged Time-VLM row of Chapter 5, and it is the reason skill scores are always computed against the reference evaluated on the same windows as the model being scored, rather than against a single global reference value.

4.2 Windows in physical time

Windows are defined in physical time and resolved to steps per dataset, rather than being defined in steps.

- **History $T = 14$ days** — 672 steps at the `uk_pv` 30-minute cadence, 1344 at the `goes_pvdaq` 15-minute cadence. Two weeks is deployment-realistic (a newly connected plant will have that much history) and gives every model multiple diurnal cycles, which shorter contexts denied

the foundation models in preliminary work. Foundation models with a fixed maximum context consume their most recent supported window.

- **Horizon** $H = 6$ **hours** — 12 and 24 steps respectively. Skill-decay curves are additionally reported at 1, 6 and 24 hours, the last covering the day-ahead case.
- **Visual window** $T_v = 8$ **frames** over a short recent window of three to six hours, *decoupled* from the numerical history. Vision carries information only over the cloud-advection horizon; widening the visual window past a few hours adds compute, not signal. This asymmetry — long numerical context, short visual context — is a design commitment of the architecture in Chapter 7, not merely a budget decision.

The cadence rule. Because windows are physical, step counts differ across datasets: 672 against 1344 for the same fourteen days. A step-horizon of 12 therefore means six hours in one dataset and three in the other. Two hard rules follow. Physical lead time is reported beside every per-dataset table; and metrics are never pooled across cadences in raw form — cross-dataset aggregation uses only the scale-free statistics of Section 4.4.

4.3 Metrics

All metrics are computed on the capacity-normalised target over daylight, valid steps, per plant, and then macro-averaged over plants so that large or data-rich plants do not dominate.

Point accuracy. With M evaluation samples, horizon H and plant capacity C_i :

$$\text{NMAE} = \frac{1}{MH} \sum_{i=1}^M \sum_{h=1}^H \frac{|\hat{y}_{i,h} - y_{i,h}|}{C_i}, \quad \text{NRMSE} = \sqrt{\frac{1}{MH} \sum_{i=1}^M \sum_{h=1}^H \left(\frac{\hat{y}_{i,h} - y_{i,h}}{C_i} \right)^2}.$$

Skill score. The headline number, defined against smart persistence and computed from NRMSE:

$$\text{SS} = 1 - \frac{\text{NRMSE}_{\text{model}}}{\text{NRMSE}_{\text{smart persistence}}}.$$

A score of zero means no better than the reference; one means perfect; negative means worse than a model that assumes cloud conditions persist. An NMAE-based variant may be reported as a secondary column but must be labelled as such, since the two rank models differently.

Probabilistic accuracy. For models with a distributional output, the continuous ranked probability score is approximated by the pinball loss averaged over the quantile grid [29]. It rewards forecasts that are both accurate and honestly calibrated, and it is the only metric here that penalises overconfidence. Calibration is additionally reported as empirical coverage of the nominal 80% interval.

Variance tracking. The squared Pearson correlation R^2 between prediction and truth measures how much of the *shape* of the day a forecast tracks, independently of bias or scale.

Reading R^2 and skill together. These two must be read jointly, because their disagreement is diagnostic. A model with high R^2 and low skill tracks the timing of the day correctly but sits off the diagonal — a bias or scale problem, fixable by recalibration. A model with low R^2 and moderate skill is doing the opposite. In Chapter 5 several multimodal baselines exhibit exactly the first pattern, and the distinction changes the interpretation of their ranking entirely.

Ramp regime. Point metrics are additionally reported on a *ramp subset*: the top decile of $|\Delta y|$ per site, that is, the cloud-transition periods. This is where a visual modality should earn its tokens — clear-sky periods are won by persistence — so the ramp columns, not the aggregate ones, are the sharpest test of the multimodal hypothesis.

4.3.1 Metrics considered and rejected

Three alternatives were considered and are recorded here with the reason for rejection, because a metric choice that is not justified is a metric choice a reader has to take on trust.

Mean absolute percentage error. Undefined at zero and unbounded near it. The target is exactly zero for roughly half of every day and arbitrarily close to it at dawn and dusk, so any percentage-of-actual metric is dominated by the least interesting observations in the dataset.

Per-series standardised error. Normalising each plant’s series by its own mean and standard deviation is standard in general-purpose forecasting benchmarks. It is prohibited here: those statistics would have to be computed on test-plant data, which is precisely the leakage the cross-plant protocol exists to prevent. Capacity normalisation uses a quantity known at installation and is leak-free by construction.

Energy-weighted error. Weighting errors by absolute energy would emphasise high-output periods, which is arguably what a grid operator cares about. It was rejected because it would make the metric depend on the capacity mix of the test set, undermining comparability across the two datasets — and because the ramp subset already isolates the high-consequence regime in a way that is transparent rather than baked into the headline number.

4.3.2 Aggregation order matters

Metrics are computed *per plant and then averaged*, not pooled over all observations. The two differ whenever plants contribute unequal numbers of valid steps or have different capacity factors, which they do (Figure 3.12, Figure 3.15). Pooling would weight a data-rich, high-output plant more heavily than a sparse one, and the resulting number would answer “how well does the model do on the average *observation*” rather than “how well does it transfer to the average *plant*”. Since the research question is about generalisation across plants, each plant is one unit of evidence.

The choice has a practical consequence worth noting: macro-averaged metrics are noisier than pooled ones, because a single difficult plant is not diluted. With fourteen test plants the standard error of the mean is not negligible, which is a further reason to treat small differences between models as ties.

4.4 Scale-free aggregation

When results must be summarised across datasets of different cadence, or across scenarios, only rank-preserving and scale-free statistics are used: win rate against the reference, geometric-mean skill, and average rank. Raw step-horizon metrics are never pooled across cadences.

4.5 Scenario battery

Seven scenarios were specified. Table 4.1 states each and, in the last column, whether it was executed for this thesis — an honest accounting matters more than a complete-looking table.

4.5.1 What each scenario is for

S1, in-domain. Train plants, held-out time range. This is the protocol most of the literature reports, included here only as a sanity ceiling: the gap between S1 and S2 measures how much of a model’s apparent accuracy is plant-specific memorisation.

S2, cross-plant. The headline. Disjoint test plants, short inference history, no test-plant statistic anywhere. Everything reported in this thesis is S2.

S3, cross-dataset. Train on one track, test on the other. This is the strongest generalisation test available in the data, because it shifts sensor, scale, climate, latitude and cadence at once (Section 3.8). It also involves a cadence change, so results must be reported with the physical lead time attached

ID	Split	Question	Run
S1	Train plants, held-out time range	Sanity / upper bound	no
S2	Disjoint test plants (primary)	Headline result (H3)	yes
S3	Train <code>uk_pv</code> , test <code>goes_pvdaq</code> and reverse	Distribution-shift generalisation	no
S4	S2 with $H \in \{1, 6, 24\}$ h	Skill-decay curves	partial
S5	S2 with 10/25/50/100% of train plants	Sample efficiency	no
S6	S2 restricted to top-decile $ \Delta y $ windows	Where vision should win	yes
S7	Train on a month subset, test unseen season	Temporal shift	no

Table 4.1: Scenario battery. Everything reported in this thesis is S2 on `uk_pv`, plus the S6 ramp subset and a partial S4 decay profile.

and aggregated only through the scale-free statistics of Section 4.4. A robustness variant resamples the faster track to the slower cadence so that step-horizon and physical horizon coincide.

S4, long horizon. The same protocol at one, six and twenty-four hours. Skill decay curves are how a forecaster’s useful range is communicated, and they separate models that are good at short lead times from models that are good at persistence.

S5, data efficiency. S2 with 10, 25, 50 and 100 percent of training plants. This is the canonical foundation-model claim — that pretraining buys sample efficiency — and it is the scenario in which a pretrained backbone should beat a supervised one even if it does not at full data. Its absence is why the comparison in Chapter 6 between supervised and pretrained models must be read as holding *at 69 training plants* and not in general.

S6, ramp subset. S2 restricted to the top decile of $|\Delta y|$. The regime where a visual modality should win, and therefore the sharpest test of the thesis.

S7, seasonal transfer. Train on a subset of months, test on an unseen season. A temporal analogue of the cross-plant shift, included for completeness and not executed.

The unexecuted scenarios are not decoration. S3 and S5 are the standard evidence a foundation-model claim is expected to provide — transfer under distribution shift, and accuracy as a function of training data — and their absence is the principal limitation of this work, restated in Section 9.4.

4.6 Input parity

Table 4.2 states which inputs each tier may consume. A baseline never receives information the models it is compared against cannot access.

Tier	Y	$\mathbf{X}_{\text{cov}}$	V	Retrieval	Text
T0 Reference	yes	clear-sky only	—	—	—
T1/T2 Supervised	yes	yes	—	—	—
T3 Foundation models	yes	where native	—	—	—
T4 Adaptation	yes	yes (CoRA)	—	yes (RAG)	—
T5/T6 Multimodal	yes	yes	yes	model-specific	model-specific
MMTSFM (ours)	yes	yes	yes	optional (H4)	—

Table 4.2: Input-parity matrix.

Two rules within this matrix are the ones a critical reader attacks first, and both are enforced.

The retrieval datastore rule. Retrieval and memory baselines may populate their datastore with *training-plant data only*. Allowing test-plant history into the datastore converts zero-shot forecasting into transduction, and would make the retrieval tier incomparable with everything else. Transductive retrieval is a legitimate but *separate* condition and never enters the headline table.

The oracle-covariate rule. Two Chronos-2 variants are reported that consume *observed future weather* over the forecast horizon — temperature, cloud cover, irradiance that would not be known at forecast time. They are labelled *oracle* and are upper bounds, not competitors. The honest variants receive only deterministic future covariates: solar geometry and calendar encodings. The gap between the two is one of the most informative measurements in this thesis (Section 6.12), but it must never be read as a competitive result.

Two further parity constraints apply to the multimodal tier. Models that natively want text (Solar-VLM) may generate weather descriptions only from covariates available to every other model — no external weather service. And no model may consume a normaliser, calibration constant or context-selection heuristic derived from the test plants.

4.7 Reproducibility

Reproducibility here means something stronger than “the code is available”. It means that a number in this thesis can be traced to the job that produced it, that the job can be re-specified exactly, and that the data it consumed cannot have changed underneath it. Three mechanisms enforce that.

The data is immutable. The dataset of record is treated as read-only, and no result depends on re-processing it. Splits are committed to version control rather than regenerated at run time, so a later run cannot silently receive a different partition; their disjointness is asserted at load, so a leak raises rather than quietly improving a score.

The configuration is declarative and complete. Every hyperparameter is composed by Hydra from configuration groups, and none is hardcoded in model code. Each baseline’s configuration is self-contained in its own directory, so no model’s behaviour depends on a global setting that another model also reads — a failure mode that is invisible until two models disagree for reasons unrelated to their architectures.

The experiment is registered before it runs. Each experiment declares a one-sentence hypothesis, a configuration diff, a branch and a nominated comparison baseline before the job is submitted. This is a guard against a specific and common failure: reframing an experiment after the fact as a test of whatever it happened to show.

A fixed global seed of 42 governs data loaders, weight initialisation and dropout. Configuration is Hydra-only [37], with each baseline’s hyperparameters self-contained in its own directory, so that no model’s behaviour depends on a global setting another model also reads. Dependencies are managed with a single lockfile. Training runs on the Leonardo cluster under SLURM, with standard output and error retained per job, so that any reported number can be traced to the job that produced it. Splits are committed to version control rather than regenerated, and their disjointness is asserted at load time.

5 The Baseline Suite

A benchmark is only as informative as the alternatives it contains. This chapter describes the twenty-eight forecasters evaluated under the protocol of Chapter 4: why each is present, what it consumes, how it was implemented, and where the implementation departs from the published method. Chapter 6 reports what they scored.

The organising principle is adversarial rather than encyclopaedic. Each model exists to close one escape route — one alternative explanation of a multimodal gain that a sceptical reader would otherwise be entitled to propose. A suite assembled this way is smaller than a survey and more useful

than one, because every row has a job.

5.1 Suite design

Tier	Models	Question it answers
T0	Persistence, smart persistence , hourly climatology, seasonal naive	Is the task trivial? What is the reference floor?
T1	LightGBM (one model per quantile), TabPFN-3	Does a tabular model already suffice?
T2	MLP, DLinear, PatchTST, iTransformer, TFT	Is a simpler or a purely supervised model better?
T3	Chronos-2 zero-shot and fine-tuned (plus oracle variants), TimesFM 2.5, TiRex, TTM-R2 zero-shot and fine-tuned	Do foundation models transfer? Is any finding backbone-specific?
T4	TS-RAG, Cross-RAG, CoRA	Would retrieval or a covariate adapter close the gap without vision?
T5	Time-VLM, Aurora, VisionTS++, UniCast	Do pseudo-images or prompting suffice?
T6	Solar-VLM, CrossViVit, SUNSET	Does the domain state of the art already win?

Table 5.1: The baseline suite and the question each tier answers.

The tiers span the three adaptation strategies of Section 2.3 — full supervision (T1–T2), zero-shot transfer (T3), and frozen-backbone adaptation (T4) — which is what allows Chapter 6 to compare them directly rather than by citation. They also span the endogenous/exogenous distinction of Section 2.5: T5 contains models whose second modality is synthesised from the numerical series, T6 models that consume genuine satellite frames. Holding the protocol fixed across that boundary is what makes the comparison of the two classes meaningful, and it is the comparison that produces the sharpest result in this thesis.

The rebuttal matrix. Table 5.2 states the mapping from objection to evidence explicitly. It is reproduced here because it is the actual specification the suite was built against.

Objection	Baseline that answers it
“A simpler model would do as well”	DLinear, MLP, persistence family
“The gain is just from covariates”	CoRA, TFT, LightGBM (all covariate-rich, no vision)
“Retrieval would give you the same thing”	TS-RAG, Cross-RAG
“The result is specific to your backbone”	TimesFM 2.5, TiRex, TTM-R2 (three other families)
“You never showed the model reads the images”	Shuffled-frames and mismatched-plant controls (Chapter 8)
“Domain-specific models already do this”	Solar-VLM, CrossViVit, SUNSET
“Fusion depth does not matter”	Late-fusion arm; UniCast as published shallow prompting
“Pseudo-images would be cheaper”	Time-VLM, VisionTS++

Table 5.2: Objection-to-evidence mapping. Every cell must be occupied before the multimodal claim can be defended.

5.2 Tier 0 — reference models

Reference models establish the floor and calibrate the reader’s sense of how much of the signal is trivially predictable. On solar data this matters more than in most domains, because the diurnal

cycle alone accounts for a very large fraction of the variance (Figure 3.1), and a metric that rewards predicting the sun’s position is uninformative about forecasting skill.

Persistence. The last observed value is carried flat across the horizon. This is the absolute floor. At very short lead times it is surprisingly hard to beat — power changes slowly under stable conditions — and its failure mode is exactly the diurnal ramp: at 08:00 it predicts that 08:00 output will persist until 14:00.

Smart persistence. The clearness index — the ratio of observed power to a clear-sky model — is carried forward instead of the power itself, and the forecast is reconstructed by multiplying that index by the clear-sky curve over the horizon. This respects the deterministic solar geometry and assumes only that *cloud conditions* persist. It is dramatically stronger than naive persistence and is the community’s standard reference; it is the denominator of the skill score in Section 4.3. It is also the only model in the study permitted to use clear-sky physics, because it is the physics reference.

Hourly climatology. The mean normalised power for each month, hour and minute bin, computed on training plants. It carries no information about the current day at all, so it detects two pathologies: a model that fails to beat it is not using its inputs, and a suspiciously strong climatology result indicates that the evaluation windows are dominated by average conditions rather than by the transitions that matter.

Seasonal naive. The value observed at the same clock time on the previous day. This is the standard reference in general-purpose forecasting benchmarks and is included so that results here can be positioned against that literature.

5.3 Tier 1 — classical machine learning

LightGBM. Gradient-boosted decision trees [32] over a flattened feature vector of lagged power and covariates. A separate model is fitted per quantile with the pinball objective, which yields a genuine probabilistic forecast rather than a point estimate with error bars bolted on. Gradient boosting is the honest strong baseline for tabular problems, and a deep model that cannot beat it has not justified its cost. In the cross-plant setting it has a specific advantage: trees are insensitive to feature scaling and therefore transfer across plants of different capacity without any special handling.

TabPFN-3. A tabular foundation model [24] in its time-series mode: prior-fitted, performing inference in a single forward pass with no gradient-based fitting on the target task. It is the tabular counterpart to the time-series foundation models of Tier 3, and it tests whether the “foundation model” advantage, if any, is specific to sequence architectures or is a property of large-scale prior fitting in general.

5.4 Tier 2 — supervised deep learning

All Tier-2 models are trained from scratch on the training plants and must generalise to disjoint test plants. This is the strongest form of the “do you actually need pretraining” question: these models see the same data as the fine-tuned foundation models and have no pretraining at all.

MLP. A multilayer perceptron over the flattened history window and covariates. The simplest learned baseline, present to bound how much of the achievable skill requires any temporal structure at all.

DLinear. A single linear layer applied to a trend/seasonal decomposition [38]. This is the model that embarrassed a generation of forecasting transformers, and its presence is non-negotiable: reviewers ask for it, and on strongly periodic signals it is frequently competitive. It consumes the target only,

no covariates, which makes it the cleanest measure of how much information is in the power history alone.

PatchTST. A channel-independent patch transformer [35]: the series is divided into patches that act as tokens, and channels are processed independently with shared weights. It is the strongest conventional supervised patch transformer and is architecturally the closest supervised analogue to the Chronos-2 backbone, which makes the comparison between them informative about pretraining specifically rather than about architecture.

iTransformer. Attention is applied across *variables* rather than across time: each channel becomes a token and the attention matrix models inter-variable structure [34]. On a problem where the target is driven by a handful of strongly correlated exogenous quantities — irradiance, cloud cover, temperature (Figure 3.8) — this is an unusually well-matched inductive bias, and Chapter 6 shows the consequence.

TFT. The temporal fusion transformer [33] combines gated residual processing, variable selection and a quantile output head. It is included because it is quantile-native, so it supplies a probabilistic comparison point outside the foundation-model tier. The implementation here is a reduced variant, described in Section 5.10.

5.5 Tier 3 — time-series foundation models

Chronos-2, zero-shot. The backbone of the proposed architecture, evaluated with no adaptation [11]. This row is the anchor for hypothesis H0 and the honest starting point of the whole thesis: whatever MMTSFM achieves must be measured against what its own backbone does untouched.

Chronos-2, fine-tuned. The same model fine-tuned on the training plants. This is the single most important comparison row for the proposed architecture, because it holds the backbone, the data and the protocol fixed and varies only the adaptation strategy. A multimodal architecture that does not beat its own fine-tuned backbone has not demonstrated anything about multimodality.

Chronos-2 oracle variants. Two additional rows in which the model receives *observed* future weather over the forecast horizon rather than only the deterministic covariates known in advance. These are not competitors and are labelled as such throughout; they quantify the headroom available from perfect conditioning, which turns out to be the largest single measurement in the study (Section 6.12).

TimesFM 2.5. A decoder-only foundation model from an independent lineage [16], present so that any conclusion about zero-shot transfer is not an artefact of one pretraining corpus.

TiRex. Built on the xLSTM family rather than on attention [17], and among the strongest zero-shot performers on general-purpose leaderboards. It is the third distinct architecture family, which is the threshold at which a negative zero-shot result becomes a statement about the task rather than about a model.

TTM-R2, zero-shot and fine-tuned. A compact mixer-based model [9] of a few million parameters. It probes the scale axis directly: if zero-shot performance is a function of capacity, a tiny model should fail and a large one should not. The fine-tuned variant additionally shows how much of a weak zero-shot result adaptation can recover.

5.6 Tier 4 — frozen-backbone adaptation

This tier contains the methods that compete with the thesis directly. Each keeps a pretrained backbone frozen and adds an external mechanism for supplying task-specific information — exactly the strategy of the proposed architecture, differing only in what is supplied and how deeply it is coupled.

TS-RAG. Retrieval-augmented forecasting on a frozen backbone [18]: windows similar to the current context are retrieved from a datastore and fused into the forecast. Per Section 4.6 the datastore contains training-plant windows only, and the mixing weight is tuned on validation plants. This is the “you did not need vision, you needed precedent” objection in runnable form.

Cross-RAG. A stronger retrieval variant [21] with clear-sky-aware retrieval keys and per-step mixing weights, retrieving across series rather than only within one. It is included so that the retrieval tier is represented by a tuned modern method rather than by a strawman.

CoRA. Covariate-aware adaptation of a frozen backbone through a zero-initialised residual adapter [12]. This is the closest published competitor to the proposed architecture: same frozen-backbone premise, same idea of injecting external information through a learned interface, but coupling the modalities at a single insertion point rather than throughout the temporal operator. If deep token-level fusion cannot beat CoRA, the fusion-depth claim does not survive.

5.7 Tier 5 — endogenous multimodal

Models whose second modality is synthesised from the numerical series or its covariates. No external sensor is consumed. Their presence is what allows the thesis to separate visual *inductive bias* from visual *information*.

Time-VLM. The series is rendered as an image and passed, with a textual description, to pretrained vision-language backbones [15]. Nothing enters the model that was not already in the numerical track; what is exploited is the prior structure of a visual encoder — locality, periodicity, multi-scale texture — applied to a re-rendering of the input.

VisionTS++. A related approach mapping series to images to exploit a frozen masked autoencoder backbone [20], included to show that the endogenous strategy is not a single lucky implementation.

Aurora. A multimodal foundation model evaluated zero-shot [10], with weather text templated from the covariate track. It answers the question “why not simply use an off-the-shelf multimodal foundation model”.

UniCast. Vision and text embeddings injected as soft prompts into a frozen time-series foundation model [19]. Architecturally this is the shallowest point on the fusion-depth axis while still being a genuine multimodal method, which makes it the natural contrast for the interleaving hypothesis. It is placed in this tier because in the configuration run here no external text encoder is available and the visual path operates on generic features.

5.8 Tier 6 — exogenous multimodal, photovoltaic domain

Models that consume genuine satellite imagery, all of them from the photovoltaic forecasting literature.

Solar-VLM. The primary domain state of the art [23]: satellite imagery, covariates and generated weather text combined through a vision-language backbone. Per the parity rules its text is generated only from covariates available to every other model; no external weather service is consulted.

CrossViVit. The reference deep satellite-plus-series architecture, cross-attending video tokens into a temporal transformer [26]. It is the strongest non-foundation multimodal competitor and the closest published relative of the proposed fusion mechanism — both couple modalities inside the encoder rather than at a readout, differing in whether the coupling is cross-attention or token interleaving.

SUNSET. The canonical convolutional solar baseline [5], a hybrid of an image branch and a power branch. Nearly every paper in the domain compares against it, so its presence anchors this benchmark to that literature.

5.9 What was deliberately excluded

A benchmark’s exclusions are as much a design decision as its inclusions, and leaving them unstated invites the suspicion that they were chosen after the results were known. The following were considered and left out, each for a stated reason.

SPIRIT and PV-VLM. Both are photovoltaic-domain multimodal methods that would have strengthened Tier 6. They were excluded for effort budget rather than on principle: Tier 6 already contains three domain models spanning convolutional, cross-attention and vision-language designs, and a fourth would have added breadth without opening a new axis. They remain the first additions a follow-up should make.

FusionSF and M3S-Net. Non-foundation photovoltaic fusion architectures. Both are released against corpora structured for different tasks (hourly day-ahead windows), and porting them to this protocol would have required changes substantial enough that the resulting row would measure the port rather than the method.

MEMTS and parametric-memory adapters. These occupy the same niche as the retrieval baselines — external information supplied to a frozen backbone — which is already represented by two tuned methods. The marginal information from a third mechanism in the same family was judged low.

Additional foundation-model families. Toto and Moirai were available and were not run. The methodological requirement was three distinct architectural families evaluated zero-shot, which Chronos-2, TimesFM, TiRex and TTM already satisfy; a fifth transformer would not have changed the conclusion of Section 6.5.

Few-shot in-context adaptation curves. Accuracy as a function of K support days per test plant is a different protocol — context matching rather than zero-shot transfer — and mixing the two in one table would obscure both. The harness supports it, and it is available as an appendix experiment on request, with $K \in \{0, 1, 3, 7\}$ days used either as a retrieval datastore or as a linear-probe update.

Post-hoc recalibration. Several models exhibit systematic scale bias (Section 6.8) that an affine correction would substantially repair. It was not applied, because the correction would have to be fitted somewhere — on validation plants at best — and the resulting table would compare forecasters-plus-post-processing rather than forecasters. The bias is reported instead.

5.10 Implementation and deviations

Third-party models were vendored and adapted to the tensor contract of Section 3.10 rather than reimplemented from their papers. The deviations below are recorded because they bound how strongly the corresponding rows may be read; a benchmark that hides its integration compromises is not reproducible.

- **Tier 2** is served by a single port of a common deep time-series library [6], so PatchTST, iTransformer, DLinear and MLP share one trainer and differ only in architecture. This is a deliberate fairness choice: differences between those four rows cannot be attributed to training-loop details.
- **TFT** is a reduced variant retaining the quantile head and gated residual processing but not the full variable-selection stack. Its row should be read as a quantile-native supervised reference rather than as a faithful reproduction.

- **TTM** uses the R2 checkpoint, not R3. The R3 release is a trend-plus-residual decomposition model whose parameter keys do not load into the available library version; loading it silently produces a randomly initialised network that trains and evaluates without error. Histories shorter than the model’s context are edge-padded with an observed mask rather than zero-padded, since zero is a meaningful power value; fine-tuned variants receive a frequency token.
- **CrossViVit** runs with single-channel imagery and synthesised coordinate encodings, because the `uk_pv` product is greyscale and the original model expects multi-channel input with precise geospatial coordinates. Its row is a conservative estimate of the published architecture.
- **Time-VLM** runs on evaluation windows not bit-aligned with the rest of the suite. The number is reported and flagged rather than omitted; an aligned re-run is required before it is used as a precise comparison point.
- **Retrieval baselines** populate datastores from training plants only, with mixing weights tuned on validation plants.
- **Foundation models with fixed maximum context** consume their most recent supported window rather than the full fourteen days. This is stated rather than silently tolerated, because context length is a fairness variable (Section 4.1).

5.11 On tuning budgets

The hardest fairness question in a benchmark of this kind is not which inputs a model receives — that is settled by the parity matrix — but how much effort went into making each one work. A model can lose because its architecture is unsuited to the task or because nobody tuned it, and no protocol fully separates the two.

Three practices bound the problem here without solving it. Within Tier 2, four models share a single trainer and a single hyperparameter search procedure, so differences *within* that tier are architectural rather than effort-driven — which is what makes the iTransformer against PatchTST comparison in Chapter 6 interpretable. Across tiers, vendored models retain their authors’ published defaults wherever those defaults are applicable, on the principle that a method’s own authors are better placed to configure it than a third party with a different dataset. And the methods most threatening to the thesis — the retrieval baselines and the covariate adapter — were tuned rather than accepted at defaults, because a benchmark whose strongest competitors are strawmen proves nothing.

The residual risk runs in a specific direction, and it should be stated: the proposed architecture received far more attention from its author than any baseline did. Where that asymmetry would matter most — the comparison against the same backbone fine-tuned, and against the vision-free arm of the same model — the comparisons are internal, sharing the recipe, the schedule and the data, which is precisely why those two rows and not an external baseline are treated as the decisive ones in Chapter 8.

A correction that changed a conclusion. One row in an earlier revision of this benchmark reported a catastrophic score for Time-VLM. The cause was a stale artefact: the prediction file had been re-saved after inverse scaling, but the metric file was never regenerated, so the table showed an error computed on standardised rather than physical units. Re-scoring from the corrected predictions — with no retraining — moved the model from last place to second. All other externally integrated models re-score bit-identically to their recorded rows, so the defect was isolated; but the episode is recorded here because it is the reason every row in Chapter 6 was re-derived from saved predictions

rather than trusted from its original metric file.

6 Benchmark Results

This chapter reports what the twenty-eight forecasters of Chapter 5 scored under the protocol of Chapter 4, and draws four conclusions from the result. All numbers are cross-plant: fourteen disjoint test plants, fourteen days of history, a six-hour horizon, capacity-normalised metrics macro-averaged over plants. No test-plant data entered any fitted component of any model.

6.1 Leaderboard

#	Tier	Model	NMAE ↓	NRMSE ↓	SS ↑	R^2 ↑
1	T2	iTransformer	0.0699	0.1032	0.552	0.765
2	T5	Time-VLM[†]	0.0692	0.1061	0.540	0.487
3	T3	Chronos-2 oracle FT [‡]	0.0808	0.1142	0.504	0.706
4	T4	TS-RAG	0.0705	0.1203	0.478	0.331
5	T4	Cross-RAG	0.0726	0.1206	0.477	0.375
6	T3	Chronos-2 oracle [‡]	0.0817	0.1213	0.474	0.679
7	T2	PatchTST	0.0886	0.1249	0.458	0.651
8	T6	Solar-VLM	0.0955	0.1283	0.443	0.660
9	T2	TFT	0.0889	0.1330	0.423	0.626
10	T2	MLP	0.0958	0.1352	0.413	0.600
11	T1	LightGBM	0.1000	0.1419	0.384	0.555
12	T4	CoRA	0.1025	0.1444	0.374	0.550
13	T3	TTM-R2 fine-tuned	0.1029	0.1465	0.364	0.520
14	T6	CrossViVit	0.1112	0.1500	0.349	0.565
15	T1	TabPFN-3	0.1076	0.1524	0.339	0.499
16	T3	Chronos-2 fine-tuned	0.1120	0.1543	0.331	0.475
17	T2	DLinear	0.1131	0.1556	0.325	0.462
18	T3	TiRex zero-shot	0.1145	0.1642	0.287	0.440
19	T3	TimesFM 2.5 zero-shot	0.1172	0.1680	0.271	0.429
20	T0	Hourly climatology	0.1353	0.1766	0.234	0.314
21	T5	Aurora	0.1280	0.1769	0.232	0.440
22	T6	SUNSET	0.1384	0.1806	0.216	0.359
23	T3	Chronos-2 zero-shot	0.1376	0.1873	0.187	0.277
24	T5	UniCast	0.1433	0.2025	0.121	0.371
25	T0	Seasonal naive	0.1419	0.2058	0.107	0.276
26	T5	VisionTS++	0.1690	0.2266	0.017	0.397
27	T0	Persistence	0.1643	0.2272	0.014	0.171
—	T0	<i>Smart persistence</i> (reference)	0.1593	0.2304	0.000	0.210
28	T3	TTM-R2 zero-shot	0.1704	0.2490	−0.081	0.155

Table 6.1: Cross-plant leaderboard on **uk_pv** (scenario S2). [†] non-aligned evaluation windows, see Section 5.10. [‡] oracle variants consume observed future weather and are upper bounds, not competitors.

Probabilistic scores. Continuous ranked probability scores are available only for quantile-native models: Chronos-2 oracle fine-tuned 0.0630, Chronos-2 oracle 0.0635, TFT 0.0689, LightGBM 0.0768, TabPFN-3 0.0815, CoRA 0.0816, Chronos-2 fine-tuned 0.0855, Chronos-2 zero-shot 0.1072. The ordering broadly tracks the point metrics with one instructive exception: TFT ranks ninth on skill but third on calibration. A model can be mediocre at predicting the conditional mean and excellent at describing its own uncertainty, and for a grid operator sizing reserve capacity the second property is not a consolation prize.

Metric disagreement. NMAE and NRMSE do not induce the same ordering, and the disagreements are diagnostic. TS-RAG has the second-best NMAE of the whole suite (0.0705) but only the fourth-best NRMSE, and an R^2 of 0.331 — far below models with similar skill. It is very accurate on typical steps and poor on the large deviations that squared error punishes. Solar-VLM shows the opposite profile: a mediocre NMAE with a high R^2 of 0.660, meaning it follows the shape of the day closely while being systematically mis-scaled. The headline skill score, being NRMSE-based, rewards the first pattern less than a mean-absolute framing would and the second more.

Summary of what follows. Table 6.2 states the four conclusions before they are argued, so that a reader can check each against the leaderboard directly.

Finding	Evidence
A well-tuned supervised transformer is the model to beat	iTransformer 0.552, ahead of every foundation model, retrieval scheme and multimodal system
Zero-shot foundation models under-deliver	0.187, 0.271, 0.287 and -0.081 across three architecture families; three of four below climatology
Retrieval beats fine-tuning on a frozen backbone	0.478 against 0.331 for the same backbone, with no weight update
Genuine exogenous imagery has not paid off	Best multimodal model (0.540) uses rendered images; every satellite-consuming model ranks eighth or below

Table 6.2: The four findings and the evidence for each, stated in advance of the argument.

6.2 Reading by tier

Tier 0. The reference tier spans a wide range, which is itself informative. Hourly climatology reaches 0.234 — a model with no knowledge of the current day beats naive persistence (0.014) by a wide margin, and beats three of the four zero-shot foundation models. The gap between persistence and smart persistence is instructive in the other direction: they score 0.014 and 0.000 on skill respectively, yet smart persistence is the harder reference in every regime that matters, because the skill score is defined relative to it and persistence’s apparent edge comes from the trivially predictable diurnal component.

Tiers 1 and 2. Supervised models dominate the upper half of the table. iTransformer leads outright; PatchTST is seventh; TFT, MLP and LightGBM occupy ninth through eleventh. DLinear at 0.325 is the informative low end: a single linear layer on a trend/seasonal decomposition, consuming no covariates, essentially matches fine-tuned Chronos-2 (0.331). Whatever the pretrained model contributes on this task, it is worth approximately as much as a linear layer with the right decomposition.

Tier 3. The foundation-model tier is bimodal. Zero-shot rows occupy positions 18, 19, 23 and 28; adapted rows occupy 13 and 16; oracle rows sit at 3 and 6. The spread within a single backbone — Chronos-2 ranges from 0.187 zero-shot to 0.504 with oracle covariates — is larger than the spread across the entire supervised tier. On this task, what a foundation model is *conditioned on* matters more than which foundation model it is.

Tier 4. The two retrieval methods are nearly tied at 0.478 and 0.477 and are the best non-oracle results built on a foundation model. CoRA trails at 0.374. The ordering within the tier is the interesting part and is developed in Section 6.6.

Tiers 5 and 6. The multimodal tiers straddle the whole table. Time-VLM is second overall; every other multimodal model is eighth or below, and four of the seven fall in the bottom third. The split does not follow tier boundaries — it follows the endogenous/exogenous boundary, which is the subject of Section 6.7.

6.3 Tier-level summary

Table 6.3 aggregates the leaderboard by tier. With four to eight models per tier these are small samples and the spread matters more than the mean, but the pattern is stable enough to be worth stating.

Tier	Models	Best SS	Median SS	Worst SS
T0 Reference	4	0.234	0.061	0.000
T1 Classical ML	2	0.384	0.362	0.339
T2 Supervised DL	5	0.552	0.423	0.325
T3 Foundation (honest)	6	0.364	0.301	−0.081
T3 Foundation (oracle)	2	0.504	0.489	0.474
T4 Frozen adaptation	3	0.478	0.477	0.374
T5 Endogenous multimodal	4	0.540	0.177	0.017
T6 Exogenous multimodal	3	0.443	0.349	0.216

Table 6.3: Leaderboard aggregated by tier. Median is over the models within the tier. Oracle rows are separated from the honest foundation-model rows because they are upper bounds.

Three readings. The supervised tier has both the best model and a high median, so its strength is not one lucky architecture. The honest foundation-model tier has the worst median of any learned tier and by far the widest spread, which is the bimodality of Section 6.2 showing up as a statistic. And the two multimodal tiers differ in a revealing way: the endogenous tier has the higher ceiling (0.540 against 0.443) and the lower median (0.177 against 0.349), meaning it contains both the best multimodal model and most of the worst ones. Rendering a series as an image is a high-variance strategy; consuming satellite frames is a consistently mediocre one.

6.4 Finding 1: a well-tuned supervised transformer is the model to beat

iTransformer tops every metric — skill 0.552, R^2 0.765, best NRMSE — ahead of every foundation model, every retrieval scheme and every multimodal system in the suite. The lead is not marginal: 0.074 skill clear of the best non-oracle foundation-model result and 0.221 clear of honest Chronos-2 fine-tuning.

This inverts the usual framing and deserves to be stated without softening. On this task, architecture plus in-domain supervised training on 69 plants beats large-scale pretraining. The comparison is fair in the strongest sense: the supervised models also see only training plants, are also evaluated on disjoint test plants, and receive the same windows and the same covariates. They have no pretraining and no privileged information.

Why iTransformer specifically? Its inductive bias matches the structure of the problem unusually well. It applies attention across *variates* rather than across time, treating each channel as a token, and Figure 3.8 shows why that pays: normalised power correlates at 0.74 with shortwave radiation, 0.69 with direct radiation and 0.60 with direct normal irradiance, while the covariates are heavily correlated with each other (0.93 between shortwave and direct radiation). The forecasting problem is substantially a problem of resolving a small set of strongly inter-correlated exogenous drivers, and variate-attention models exactly that. PatchTST, which is channel-independent, ranks seventh; the difference between the two is a clean demonstration that on this task the cross-channel structure is where the signal is.

The finding does not show that pretraining is worthless. It shows that on a task with 69 training plants and a strongly structured covariate track, the value of pretraining is smaller than the value of a well-matched inductive bias — and that a paper reporting only foundation-model comparisons would have reached the opposite conclusion by omission.

6.5 Finding 2: zero-shot foundation models underdeliver

The zero-shot tier is the weakest part of the leaderboard. Chronos-2 scores 0.187, TimesFM 0.271, TiRex 0.287, and TTM-R2 reaches −0.081 — worse than assuming cloud conditions persist. All four

trail LightGBM; three of the four trail hourly climatology, a model that has never seen the current day.

Because the result replicates across an encoder transformer, a decoder transformer and a mixer architecture, it is a property of the task rather than of any one pretraining corpus. The natural explanation is distributional. Photovoltaic power is bounded below by zero and above by capacity, is gated by a deterministic astronomical cycle, is exactly zero for a large and predictable fraction of every day, and transitions abruptly when a cloud edge crosses the array (Figure 3.2). General-purpose pretraining corpora are dominated by smooth economic, demand and sensor series with none of those properties. A model whose prior is “series are smooth and roughly stationary in the short run” is systematically wrong here in a way that no amount of context length repairs.

Two secondary observations support the reading. First, the failure is worst for the smallest model: TTM-R2 zero-shot is the only entry below the reference floor, and fine-tuning the same architecture recovers it to 0.364 — a swing of 0.445 skill from adaptation alone. The zero-shot deficit is therefore not a capacity ceiling but a prior mismatch. Second, the ordering among zero-shot models does not track their general-purpose leaderboard ordering, which is what one expects if the limiting factor is domain mismatch rather than model quality.

Adaptation is not optional on this task. That conclusion is what makes Tier 4 the interesting tier rather than a completeness exercise.

6.6 Finding 3: retrieval beats fine-tuning on a frozen backbone

Every adaptation step helps — Chronos-2 moves from 0.187 to 0.331, TTM-R2 from -0.081 to 0.364 — but the ordering *among* adaptation strategies is the result worth carrying forward. Retrieval over a frozen backbone reaches 0.478 (TS-RAG) and 0.477 (Cross-RAG) against 0.331 for fine-tuning the same backbone: 0.147 skill, obtained without a single gradient update to the model.

The covariate adapter sits between them at 0.374, which sharpens the interpretation. What retrieval supplies is not simply “more input” — CoRA also supplies more input, in the form of the full covariate track, and gains less. What retrieval supplies is *relevant precedent*: windows whose dynamics resemble the current one. For a problem whose central difficulty is regime identification — is this a clear day, a broken-cloud day, or a front passing? — an example of how a similar situation evolved is worth more than either a re-fitted weight matrix or an additional covariate channel.

There is a caveat that keeps this from being a stronger claim than it is. The two retrieval models have low R^2 (0.331 and 0.375) relative to their skill, and Section 6.1 noted the same pattern in their metric disagreement. They are excellent on typical steps and comparatively weak at tracking the day’s shape, which suggests they are exploiting the datastore for level calibration at least as much as for dynamics. Whether the gain would survive a harder ramp-weighted evaluation is visible in Appendix D, where both sit mid-table on ramp NRMSE despite their aggregate rank.

6.7 Finding 4: genuine exogenous imagery has not paid off

The best multimodal model in the suite is Time-VLM at 0.540 — second overall, statistically level with iTransformer, and with the lowest ramp NRMSE of the entire suite (0.172, Appendix D), which is precisely where a visual model is supposed to win. And Time-VLM consumes *no external imagery at all*: it renders the numerical series as an image and passes it to pretrained vision-language backbones.

Every model that ingests real satellite frames ranks eighth or below: Solar-VLM 0.443, CrossViVit 0.349, SUNSET 0.216, UniCast 0.121. The domain state of the art, with genuine satellite input and a vision-language backbone, trails a model that only ever saw a plot of its own input by 0.097 skill.

Three explanations are available, and they have very different consequences for the design of a multimodal forecaster.

1. **Redundancy.** The information in a satellite crop is already present in the covariate track. Cloud cover, shortwave radiation and direct normal irradiance are supplied numerically to every Tier-2 and Tier-4 model, so the frames may add nothing that the reanalysis has not already summarised.

2. **Interface loss.** The information is present in the frames but does not survive the encoder-to-forecaster interface. A visual encoder trained on natural video produces representations optimised for object and motion semantics, not for the quantity a forecaster needs, and the learned adapter may be too weak a bridge.
3. **Inductive bias, not information.** What converts into skill in Time-VLM is the *prior structure* of a visual backbone applied to data the model already had — and that resource is available without any external sensor.

Figure 3.10 is evidence against the first explanation, and it is worth dwelling on. It bins daytime observations by shortwave radiation and, within each bin, by cloud cover. If the reanalysis irradiance already captured everything, the three cloud-cover boxes within an irradiance bin would coincide. They do not: in the 250–450 W/m² bin the median normalised power falls from 0.33 under clear skies to 0.27 under mixed conditions to 0.24 under overcast, a monotone residual effect that persists across every irradiance bin. There is genuine cloud-state information beyond what the numerical irradiance channel carries. That is exactly the information a satellite frame observes directly, and exactly the information the exogenous models are failing to convert.

The remaining explanations are therefore the second and third, and they are distinguishable only by controlled experiment — specifically by the shuffled-frames and modality-grid controls specified in Chapter 8. This is the central open question the benchmark leaves behind, and it reframes the thesis: the interesting question is no longer “does deep fusion beat late fusion” but “does any current visual interface transmit what the frames contain”.

6.8 Qualitative behaviour

Aggregate metrics hide failure modes that a plot exposes immediately. Figure 6.1 shows one forecast window on a representative test plant; Figure 6.2 contrasts four models’ calibration pooled over all of that plant’s windows.



Figure 6.1: One forecast window on test site 10793: the best model of each cluster against the truth. Time-VLM is the only model that anticipates elevated output at the start of the horizon; the others decay smoothly from the forecast origin in the manner of persistence, and all of them miss the sharp drop at step 2. Legend metrics are window-local and are not the pooled numbers of Table 6.1.

The trace exhibits the characteristic failure of the whole suite. The truth rises above the forecast

origin, holds, then falls by three quarters within two steps. Every model except Time-VLM produces a monotone decay from the origin — the shape a model outputs when it has no information about what is about to happen and falls back on the conditional mean. Even Time-VLM, which correctly holds output high for two steps, misses the drop entirely and ends the horizon far above the truth. The models are not disagreeing about the ramp; they are collectively blind to it.

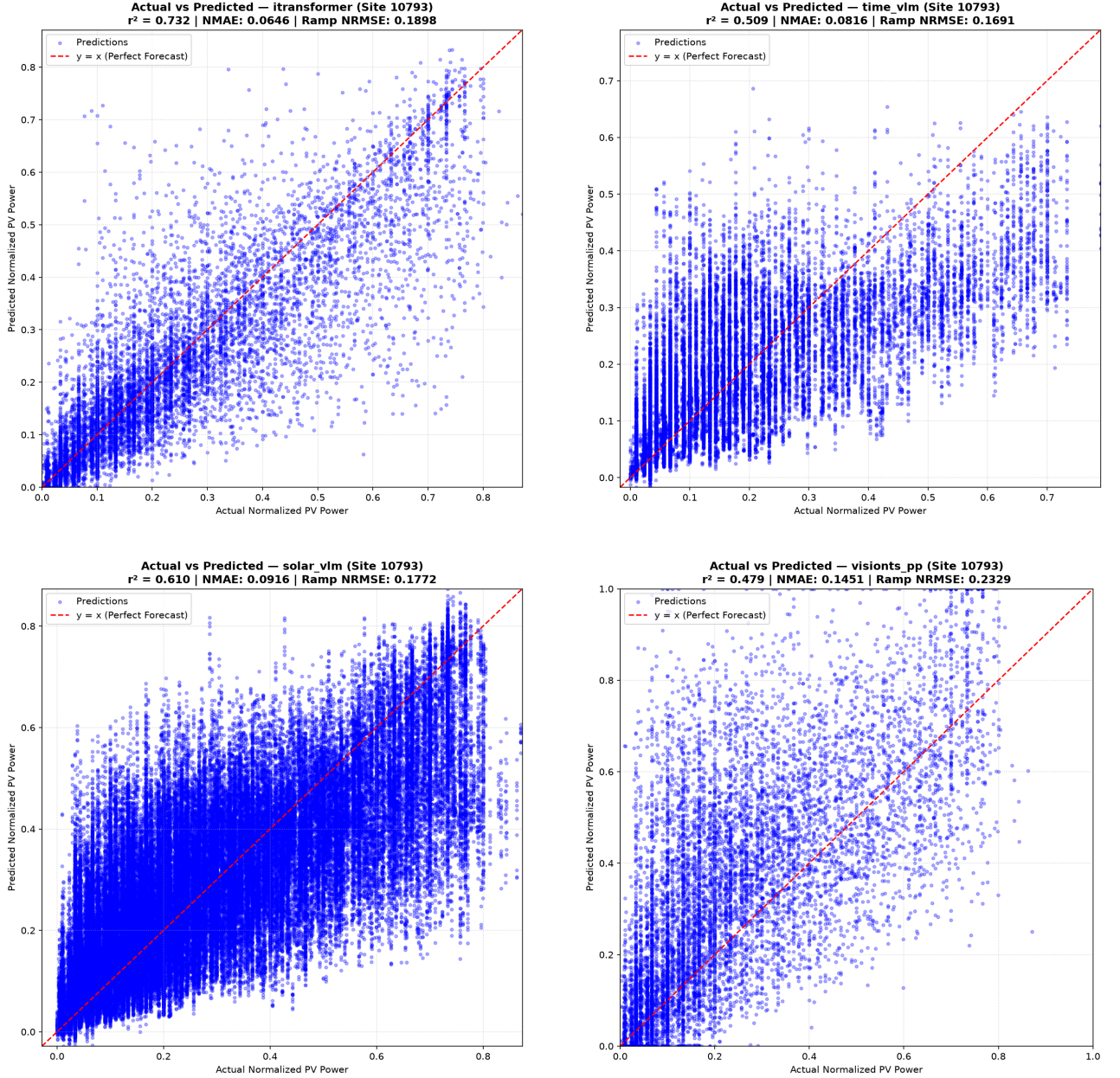


Figure 6.2: Actual against predicted power on test site 10793, pooled over all windows. Top: iTransformer and Time-VLM, the two leaders, both tightly distributed about the diagonal. Bottom: Solar-VLM, which tracks shape but sits below the diagonal at high output, and VisionTS++, whose cloud is over-dispersed and biased upward at low power. Per-panel metrics are site-local and are *not* the pooled fourteen-plant numbers of Table 6.1.

Two distinct pathologies appear in the scatter plots, and neither is visible in a skill column. Several exogenous multimodal models — Solar-VLM, SUNSET, CrossViVit — *systematically under-predict*: they track the shape of the day well (R^2 between 0.36 and 0.66) yet sit below the diagonal at high output. VisionTS++ fails differently, with an over-dispersed cloud biased upward at low power: respectable variance tracking (R^2 0.397) and almost no skill (0.017), because it is wrong about magnitude in both directions rather than consistently in one. UniCast is the cleanest instance of the first pathology, with predictions of approximately $0.61y + 0.063$ at correlation 0.61, compounded by

a late-day skew in its native evaluation windows.

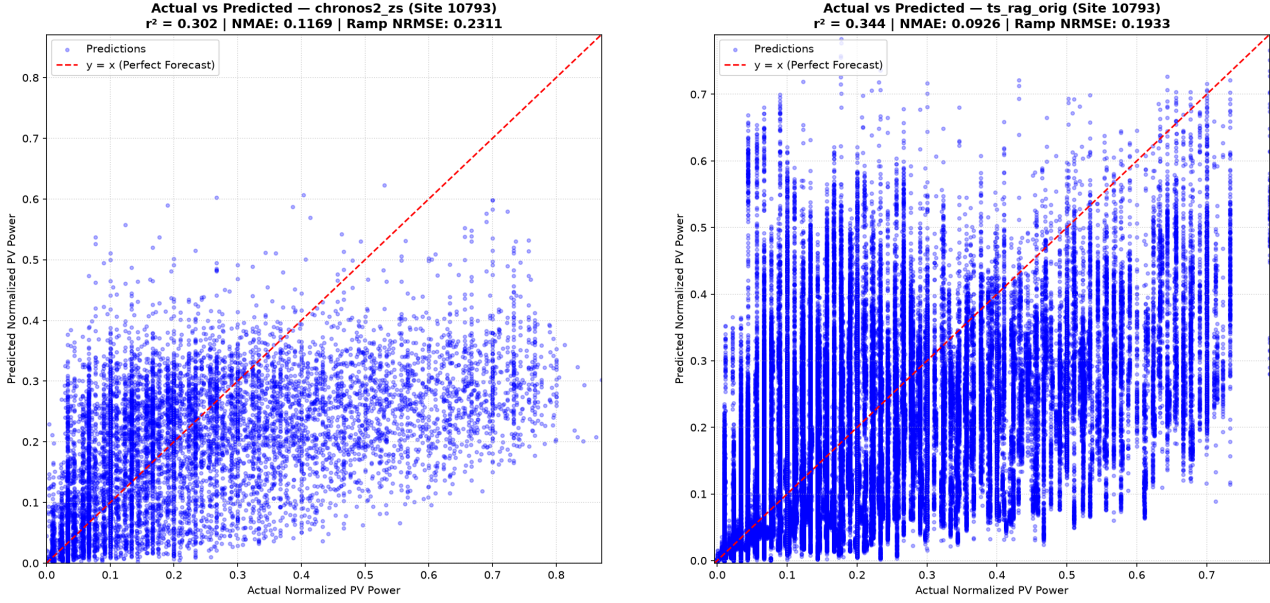


Figure 6.3: The effect of retrieval on a frozen backbone, same test site. Left: Chronos-2 zero-shot. Right: TS-RAG over the same frozen backbone. Retrieval tightens the distribution substantially without any weight update.

Both pathologies are *bias and scale* failures rather than timing failures, and both are in principle recoverable by recalibration. Recalibration was not attempted, for two reasons. It would have to be fitted on validation plants only, to avoid the leakage the protocol forbids; and more fundamentally it would change what the tier measures, converting a comparison of forecasters into a comparison of forecasters-plus-post-processing. The honest reading is that several published multimodal architectures transfer their *shape* knowledge across plants but not their *scale* knowledge — which is itself a finding about cross-plant generalisation, and one that capacity normalisation was supposed to prevent.

Figure 6.5 is worth more attention than its aggregate row. The spread among multimodal models on a single window exceeds the spread of the entire leaderboard: at the final horizon step, predictions range from 0.00 (VisionTS++) to 0.26 (Time-VLM) against a truth of 0.11. Two models overshoot by more than a factor of two, two undershoot to nothing, and one — SUNSET — ignores the horizon entirely and outputs a near-constant. A visual modality is evidently not a stabilising influence; it is a source of variance that each architecture converts differently, which is consistent with the interface-loss explanation of Section 6.7.

6.9 The ramp regime

Appendix D reports every model on the top-decile $|\Delta y|$ subset. Three observations follow, and they matter more for the thesis than the aggregate table does, because the ramp regime is where the multimodal hypothesis was supposed to pay.

First, the ordering is broadly preserved but compressed. Time-VLM leads at 0.172 ramp NRMSE, iTransformer follows at 0.177, and the reference floor sits at 0.281 — a range of 0.109 against a range of 0.147 in the aggregate NRMSE column. Ramps are hard for everyone, and the spread between a good model and a bad one narrows exactly where the value of forecasting is highest.

Second, the retrieval models lose their advantage. TS-RAG and Cross-RAG rank fourth and fifth overall but seventh and eighth on ramp NRMSE, behind Solar-VLM and PatchTST. This supports the reading of Section 6.6: retrieval supplies level calibration and typical-case accuracy more than it supplies transition dynamics. Retrieving a window that looks like the current one is most useful when the current one is unremarkable.

Third — and this is the observation that shapes Chapter 8 — the model with the best ramp accuracy in the entire suite consumes no external imagery. If satellite frames were being exploited for

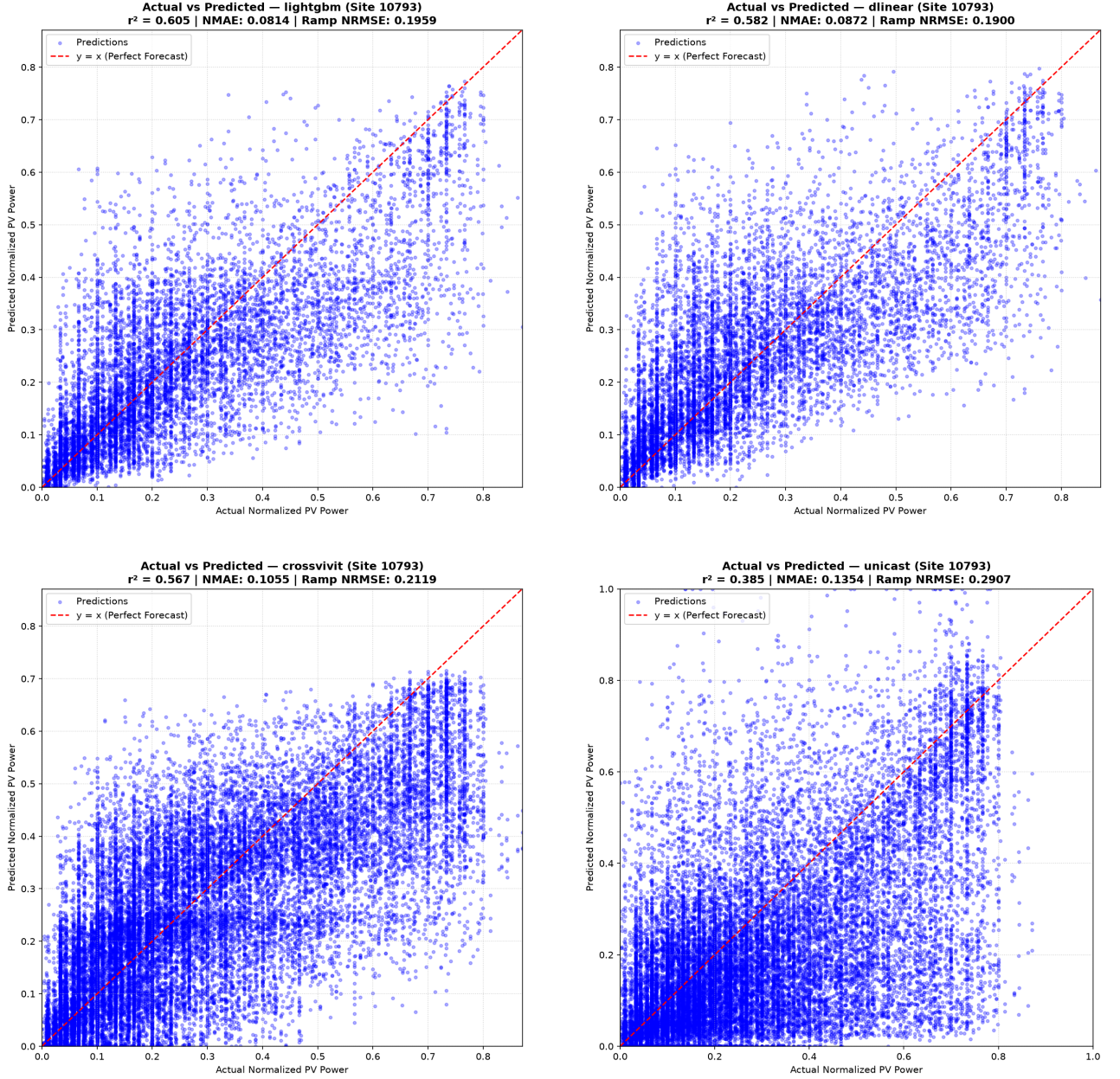


Figure 6.4: Calibration across tiers, same test site. Top: LightGBM and DLinear, the classical and linear references — both well centred, DLinear visibly more dispersed. Bottom: CrossViVit and UniCast, two exogenous multimodal models, both showing the compression toward the mean that produces high R^2 with low skill.

advection, the exogenous tier should own this column. It does not: Solar-VLM is fourth, CrossViVit sixteenth, SUNSET nineteenth. Whatever the cloud-state information demonstrated in Figure 3.10 is worth, no current architecture in this suite is extracting it in the regime where it should matter most.

6.10 Practical considerations

Accuracy is not the only axis on which these models differ, and a benchmark that ignores cost gives misleading guidance to anyone who has to deploy something.

Cost of adaptation. The three adaptation strategies differ by orders of magnitude in what they require. Retrieval needs an index over training-plant windows and a nearest-neighbour lookup at inference; no gradients, no accelerator, and a new plant can be served the moment its history arrives. Adapter tuning needs a short training run over a small parameter set. Full fine-tuning needs a training

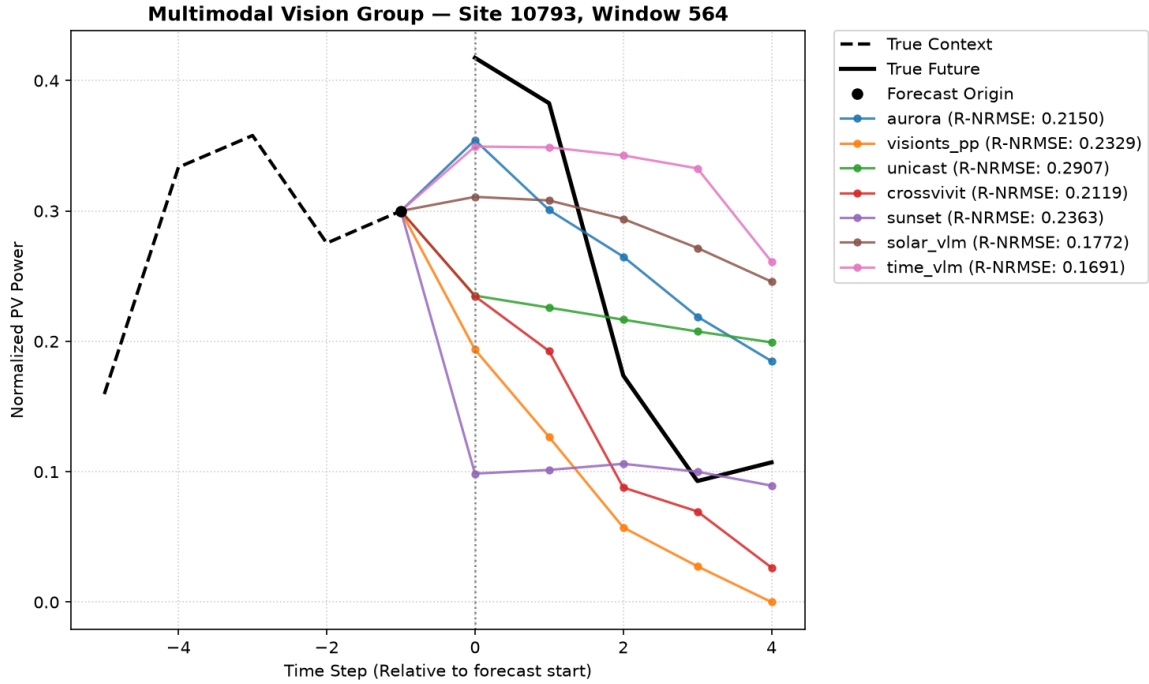


Figure 6.5: The multimodal tier alone, same window as Figure 6.1. The seven models disagree by more than the whole range of the truth. Time-VLM and Solar-VLM hold output high; SUNSET collapses immediately to 0.10 and stays there; VisionTS++ and CrossViVit decay to near zero. All of them consume — or in Time-VLM’s case simulate — a visual modality, and no two of them read this window the same way.

run over the whole backbone and produces a task-specific copy of a large model that must then be stored and served. That the cheapest of the three is also the most accurate on this task (Section 6.6) is the most practically consequential finding in the chapter.

Cost of the visual branch. Every exogenous multimodal model in the suite requires a satellite ingest pipeline: reprojection, per-site cropping, storage of terabytes of frames, and a vision encoder forward pass per forecast. In this study that pipeline is the single largest engineering component of the dataset (Section 3.2) and the vision encoder dominates inference cost. The endogenous alternative — rendering the series as an image — requires none of it. Against that background, Time-VLM outranking every satellite-consuming model is not merely a scientific result but an economic one.

Serving a growing fleet. The deployment scenario that motivated this thesis — new plants arriving continuously — favours methods whose marginal cost per plant is near zero. Supervised in-domain models, including the leader, must be retrained as the fleet grows if they are to exploit new plants; zero-shot and retrieval methods absorb a new plant by adding rows to an index. The leaderboard ordering and the operational ordering are therefore not the same, and a practitioner reading only Table 6.1 would draw the wrong conclusion.

6.11 What the benchmark cannot answer

Four questions are outside what these numbers support, and stating them prevents over-reading.

Whether the results transfer to another fleet. Everything here is `uk_pv`: one country, one satellite product, one capacity class, two years. The `goes_pvdaq` track exists in the dataset and differs in sensor, scale, latitude and cadence, but was not run. Cross-dataset generalisation is unproven — not disproven.

Whether the ordering is statistically significant. Each model was run once, and no significance testing accompanies the table. Differences of a few thousandths of skill — TS-RAG at 0.478 against Cross-RAG at 0.477, for instance — are ties, and are read as such throughout.

Whether the models were equally well tuned. Hyperparameter budgets were comparable but not identical, and vendored implementations carry their authors’ defaults. A model can lose a benchmark because its architecture is unsuited or because nobody tuned it, and this study cannot fully separate the two. The tier structure mitigates rather than eliminates the concern: within Tier 2, four models share one trainer, so differences there are architectural.

Whether the models would rank the same at another horizon. Every number here is a six-hour forecast. At one hour persistence-like methods are far stronger and the spread compresses; at twenty-four hours the observed cloud field is irrelevant and numerical weather dominates, which would penalise every imagery-consuming model and favour those that ingest the covariate track well. The skill-decay scenario that would settle this was only partially run, so the ordering reported should be read as an ordering *at six hours*.

Whether better conditioning would change the ordering. The oracle rows show that the same backbone gains 0.173 skill from privileged future weather. It is entirely possible that the ranking of architectures would reorder under better — but still legitimate — conditioning, and nothing here rules that out.

6.12 What the benchmark establishes

Three numbers define the target for any proposed architecture on this task.

- **0.552** — the overall bar (iTransformer). A model that does not approach it has not beaten supervised in-domain training.
- **0.540** — the multimodal bar (Time-VLM), achieved with no external imagery.
- **0.331** — the honest same-backbone reference (Chronos-2 fine-tuned). Any architecture built on Chronos-2 must clear this to show that its additions, rather than its backbone, are doing the work.

A fourth number is not a bar but a measurement of available headroom. The oracle variants, which see observed future weather, reach 0.474 and 0.504 against 0.331 for the honest fine-tuned model — a gap of **0.173 skill** attributable purely to better conditioning over the horizon. That is larger than the difference between any two architectures in the entire suite. It is the single strongest indication in this work of where effort should be directed, and Chapter 9 returns to it.

7 MMTSFM: Deep Multimodal Fusion on Frozen Backbones

This chapter presents the proposed architecture. MMTSFM couples a largely fixed time-series foundation model with a frozen video foundation model through two mechanisms that, to the best of this work’s knowledge, have not previously been combined: *selective temporal interleaving*, which places visual tokens inside the time-series sequence but only within a short refinement window, and *causal Grassmann mixing*, an $O(L)$ temporal operator that encodes the transition between hidden states as a subspace on a Grassmann manifold.

The chapter describes the system as built and measured. A subsequent redesign exists on paper and is discussed as future work in Section 9.5; nothing in this chapter depends on it.

7.1 Overview

The model consumes, per entity row, a 672-step normalised power history, fourteen covariate channels over history and horizon, and eight satellite frames covering a short recent window. It emits the full six-hour horizon as nine quantiles in a single forward pass.

The data flow is:

1. The power history is patch-tokenised by the *pretrained* Chronos-2 patch embedding into $T_{\text{ctx}} = \lceil 672/16 \rceil = 42$ context tokens, plus $T_{\text{fut}} = \lceil 12/16 \rceil = 1$ future token. Each covariate channel becomes its own token row on the batch axis.
2. The eight frames are encoded as a *clip* by frozen V-JEPA 2.1 into $[B, 4, 196, 1024]$ — four latent time steps, 196 spatial patches — and compressed by a learned latent summariser into n_{vis} tokens of width 768 ($n_{\text{vis}} = 1$ for `uk_pv`, 2 for `goes_pvdaq`).
3. A multimodal embedding layer adds modality, segment, token-type, entity and rotary positional information.
4. The fusion mode determines where visual tokens enter: as additional batch rows (late fusion) or woven into the temporal sequence (interleaved).
5. Twelve Chronos-2 encoder blocks process the result, each applying temporal mixing, group self-attention across the batch axis, and a feed-forward layer.
6. The output head decodes the final future positions into nine quantiles per horizon step and inverts the normalisation.

Figure 7.1 shows the assembled system.

7.1.1 Shapes, worked through

Table 7.1 traces one `uk_pv` sample from input to output. Tracing shapes explicitly is worth a page here because two of the design’s least obvious properties — the aggressive patch compression of the numerical stream, and the extreme compression of the visual stream — are invisible in prose and immediate in a shape table.

Stage	Shape	Note
Power history	$[B, 672]$	14 days at 30-min cadence
After patch embedding	$[B, 42, 768]$	$\lceil 672/16 \rceil$ tokens; one token ≈ 8 h
Future positions	$[B, 1, 768]$	$\lceil 12/16 \rceil$
Covariates	$[B, 684, 14]$	history + horizon
Covariate token rows	$14 \times [B, 1, 768]$	one batch row per channel
Frames	$[B, 3, 8, 224, 224]$	8 frames over a 6-hour window
After V-JEPA	$[B, 4, 196, 1024]$	tubelet stride 2; 4 latent steps
After latent summariser	$[B, 1, 768]$	$n_{\text{vis}} = 1$
Interleaved sequence	$[B, 44, 768]$	$42 + n_{\text{vis}} + T_{\text{fut}}$
After 12 encoder blocks	$[B, 44, 768]$	shape preserved
Output head	$[B, 12, 9]$	horizon $\times$ quantiles

Table 7.1: One `uk_pv` sample end to end. Two compressions dominate: 672 timesteps become 42 tokens, and $4 \times 196 \times 1024$ visual activations become a single 768-dimensional vector.

The second compression is worth dwelling on, because it is a candidate explanation for any weak visual contribution. The V-JEPA output contains $4 \times 196 = 784$ patch embeddings of width 1024; the summariser reduces this to one vector of width 768, a reduction of roughly three orders of magnitude. Whatever spatial structure the cloud field has — a front approaching from the west, a break opening to the south — must survive that bottleneck to influence the forecast. The visual token budget sweep specified in Section 8.8 exists precisely to test whether it does.

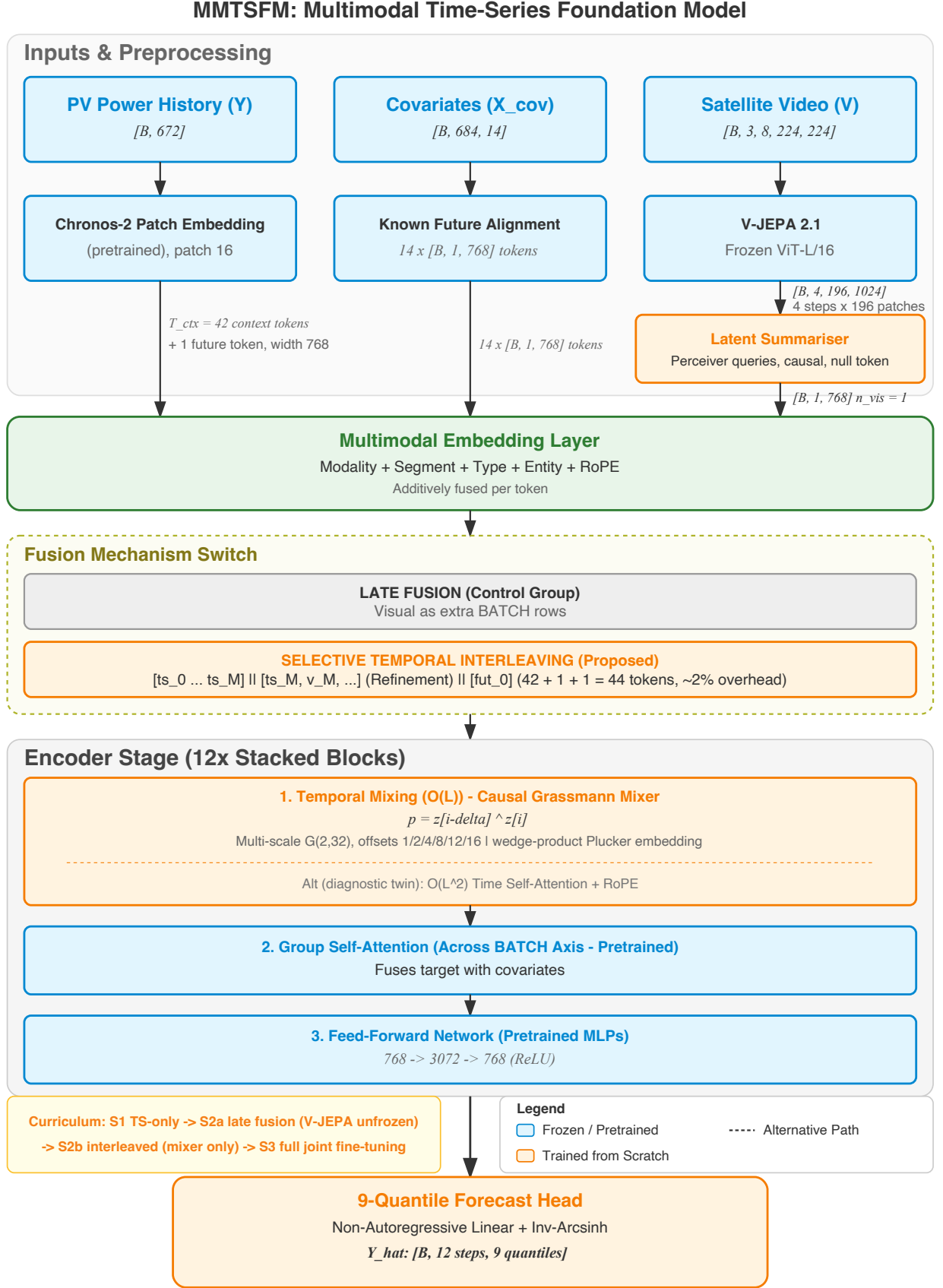


Figure 7.1: MMTSFM as assembled, for `uk.pv`. Cool boxes are pretrained or frozen; warm boxes are trained from scratch — the latent summariser, the fusion interface, the Grassmann mixer and the quantile head are the only newly initialised components. Tensor shapes are annotated on the edges.

7.2 Design principle: decoupled resolution

The two modalities are kept at *different temporal resolutions on purpose*. The numerical stream carries a long macro-context — two weeks, spanning many diurnal cycles, enough to expose seasonality and regime structure. The visual stream carries a short micro-context at high cadence, covering only the last few hours.

Both alternatives are worse. Resampling a fortnight of satellite frames onto the numerical grid is computationally prohibitive and mostly redundant, since frames more than a few hours old carry no information about the coming six hours: cloud fields advect out of the crop. Resampling the numerical series up to the frame cadence discards the macro-context that makes the foundation-model backbone useful in the first place. Decoupling keeps each modality at the resolution at which it is informative, and the cost is that fusion must handle sequences in which the two streams have incommensurable time bases — which is exactly what Section 7.6 addresses.

7.3 Numerical branch

The backbone is Chronos-2 [11] at its native checkpoint size: d_{model} 768, twelve layers, twelve heads with head dimension 64, feed-forward width 3072, input patch size 16, nine protocol quantiles, and an inverse-hyperbolic-sine normalisation that makes series of arbitrary magnitude comparable without per-series statistics.

Because the configured input patch size matches the checkpoint, the patch embedding loads *pre-trained* and transfers; only the nine-quantile output head is reinitialised. This is a deliberate design constraint rather than a happy accident — a mismatched patch size would discard the tokeniser and with it most of the value of the pretrained backbone.

A reproducibility trap worth recording. The configuration files in this project nominally specify a smaller model (d_{model} 512, six layers, feed-forward 2048). These values are *silently ignored*: loading from a pretrained checkpoint preserves the checkpoint’s architecture, and only the Grassmann-specific fields and the tokeniser configuration propagate. Any report of this system that quotes the configured dimensions rather than the checkpoint’s is wrong. The failure is silent — no warning, no shape error — and it is the kind of discrepancy that survives a long time in a research codebase.

Covariates. The fourteen covariate channels enter as separate token rows on the batch axis, sharing a group identifier with the target row, with their masks set so that values survive to the encoder. That they genuinely reach the model in both fusion modes was verified by perturbation: changing a covariate changes the forecast. This is worth stating because a covariate path that is silently masked out is a common and invisible failure, and because it is what makes the honest-versus-oracle distinction of Section 4.6 meaningful.

7.4 Visual branch

The visual encoder is a frozen V-JEPA 2.1 ViT-L/16 [25], applied to the entire eight-frame clip rather than frame by frame. Tubelet embedding with temporal stride two yields four latent time steps of 196 spatial patches each, at width 1024. Motion is therefore encoded *inside* the backbone, and the downstream module only has to perform spatial compression.

That module is a learned **latent summariser**: a Perceiver-style cross-attention block [31] with a small set of learned queries, applied causally across latent time, and equipped with a *null token* so that the summariser can represent “no useful visual evidence” rather than being forced to emit a confident summary of an uninformative sky. It maps $[B, 4, 196, 1024]$ to $[B, n_{\text{vis}}, 768]$.

All domain specialisation lives in this summariser. There is no per-sensor input projection: the data loader delivers uniformly shaped frames for every plant and source, so the queries — and only the queries — learn what to extract from a satellite crop for the purpose of forecasting power. This mirrors the adapter-recycling strategy used to specialise frozen vision-language backbones, and it keeps the number of newly initialised parameters small, which matters for the curriculum of Section 7.11.

7.5 Multimodal embedding

Before fusion, every token receives an additive composition of five embeddings: a *modality* embedding (numeric or visual), a *segment* embedding (context or future), a *token-type* embedding (target, covariate or visual), an *entity* embedding, and rotary positional encoding [36] over the temporal axis. Additive composition keeps the embedding table small and lets the encoder disentangle the factors linearly.

7.6 Fusion

Two fusion modes are implemented, and the difference between them is the architectural hypothesis under test.

Late fusion. The visual summary is passed through a cross-modal adapter and appended as $N_{\text{soft}} = 1$ additional *batch rows* sharing the target’s group identifier. The two modalities therefore meet only in the group-attention operator, once per block, and never interact along the time axis. This is the standard design and serves as the control.

Selective temporal interleaving. Let $T_M = T_{\text{ctx}} - n_{\text{vis}}$. The encoder input sequence is

$$\mathbf{S} = \underbrace{[\text{ts}_0, \dots, \text{ts}_{T_M-1}]}_{\text{macro: pure time series}} \parallel \underbrace{[\text{ts}_{T_M}, v_{T_M}, \dots, \text{ts}_{T_{\text{ctx}}-1}, v_{T_{\text{ctx}}-1}]}_{\text{refinement: interleaved}} \parallel \underbrace{[\text{fut}_0, \dots]}_{\text{future}}.$$

Visual tokens are inserted *only* inside the refinement window. The macro region stays purely numerical, so its temporal geometry — the structure the pretrained backbone knows how to exploit — is left undisturbed; in the refinement region the two modalities alternate, so that the temporal operator computes genuinely cross-modal pairs.

The causal access pattern follows from the ordering. A time-series token at position $T_M + 2k$ attends to the full macro history and all previous refinement pairs, but not to the visual token that follows it; the visual token at $T_M + 2k + 1$ attends to everything above plus its own time-series partner. Causality is preserved by construction and requires no masking special case.

Cost. Total sequence length is $T_{\text{ctx}} + n_{\text{vis}}$: the overhead is $n_{\text{vis}}/T_{\text{ctx}}$, proportional to the number of visual tokens rather than to context length. For `uk_pv` this is $42 + 1 + 1 = 44$ tokens, roughly two percent. Longer numerical contexts make the ratio smaller, not larger — which is the opposite of the scaling behaviour of full cross-attention between two long sequences, and is the practical argument for the design.

7.7 Temporal mixing: causal Grassmann flow

Each encoder block begins with a temporal mixing operator. Two are implemented and are selected by a single configuration switch, with everything else — including the training schedule — held identical, so that any measured difference is attributable to the operator alone.

7.7.1 The operator

Ordinary self-attention scores a pair of positions by the dot product of their representations: a measure of *similarity*. Grassmann mixing instead encodes the *transition* between two positions as an oriented plane. For each position i and each offset δ in a fixed set:

1. **Reduction.** $\mathbf{z}_i = W_{\text{red}} \mathbf{h}_i \in \mathbb{R}^r$ with r even (default 32), so that rotary embedding applies.
2. **Phase injection.** Rotary positional encoding is applied to $\mathbf{z}$.
3. **Plücker encoding.** The wedge product of the two reduced states, normalised,

$$\mathbf{p}_{i,\delta} = \frac{\mathbf{z}_{i-\delta} \wedge \mathbf{z}_i}{\|\mathbf{z}_{i-\delta} \wedge \mathbf{z}_i\| + \varepsilon} \in G(2, r), \quad \dim(\mathbf{p}) = \binom{r}{2},$$

is a point on the Grassmann manifold of two-dimensional subspaces: the plane spanned by the two states, independent of their magnitudes.

4. **Projection.** $\mathbf{g}_{i,\delta} = W_{\text{plu}} \mathbf{p}_{i,\delta} \in \mathbb{R}^d$.
5. **Multi-scale aggregation.** A softmax over the offsets $\delta \in \{1, 2, 4, 8, 12, 16\}$ mixes the projections.
6. **Gated fusion.** $\mathbf{h}'_i = \alpha_i \odot \mathbf{h}_i + (1 - \alpha_i) \odot \mathbf{g}_i$ with $\alpha_i = \sigma(W_{\text{gate}}[\mathbf{h}_i \parallel \mathbf{g}_i])$, so the operator can learn to pass the residual stream through unchanged.

7.7.2 Why this inductive bias

Three arguments motivate the construction. *Geometry over magnitude:* the Plücker normalisation makes the representation scale-invariant, so what is encoded is the *direction* of state evolution. In a physical system the direction of change — “rapidly clouding” as opposed to “stable” — is frequently more predictive than the level, and photovoltaic power is such a system. *Efficiency:* the operator is $O(L)$ in sequence length, since each position pairs with a fixed number of offsets rather than with all predecessors, which admits contexts of a thousand or more steps within a single device’s memory where quadratic attention does not. *Multi-scale by construction:* the offset set spans half an hour to eight hours at the `uk_pv` cadence, so several temporal frequencies are tracked simultaneously without stacking layers.

7.7.3 Modality-aware offset gating

Interleaving interacts with the offset structure in a way that requires a correction. Inside the refinement window, with the pattern $[\text{ts}_k, v_k, \text{ts}_{k+1}, v_{k+1}, \dots]$, the modality composition of a pair depends on the parity of δ : offset 1 produces genuinely cross-modal pairs (time series with visual), while offsets 2, 4 and beyond produce same-modality pairs (time series with time series, or visual with visual), and large offsets reach back across the boundary into the pure-macro region and mix interleaved with non-interleaved cadences.

Table 7.2 works this out explicitly for the interleaved pattern.

δ	Pair at a time-series query (position $T_M + 2k$)	Pair at a visual query (po- sition $T_M + 2k + 1$)	Class
1	$(v_{T_M+k-1}, \text{ts}_{T_M+k})$	$(\text{ts}_{T_M+k}, v_{T_M+k})$	cross-modal
2	$(\text{ts}_{T_M+k-1}, \text{ts}_{T_M+k})$	(v_{T_M+k-1}, v_{T_M+k})	unimodal
4	time series with time series, stride 2	visual with visual, stride 2	unimodal
8, 12, 16	reaches back into the pure- macro region	likewise	unimodal / boundary

Table 7.2: Modality composition of Grassmann pairs inside the refinement window. Offset 1 is the *only* offset producing genuinely cross-modal subspaces; even offsets produce same-modality pairs, and large offsets cross the boundary between interleaved and pure cadence.

These pair classes live on different statistical manifolds and should not share a single projection unmodified. The mitigation is a set of four learned scalar biases, one per modality-pair class — time/time, time/visual, visual/time, visual/visual — added to the offset logit before the aggregation softmax:

$$\ell_{i,\delta} = \langle \mathbf{q}_i, \mathbf{k}_\delta \rangle + b_{m_{\delta,i}}.$$

The model can thereby down-weight offsets that produce semantically incoherent pairs inside the refinement window without disturbing behaviour in the macro region. The cost is four parameters and no change in asymptotic complexity. Whether the correction is necessary is an ablation in its own right (Section 8.4).

7.7.4 The diagnostic twin

The alternative operator is ordinary causal self-attention over the time axis with rotary embeddings — $O(L^2)$, no geometric structure. It is retained not as a weaker baseline but as a *matched diagnostic*:

identical stage schedule, identical data, identical everything except the mixer. Without it, any advantage of the full system could be attributed to the training recipe rather than to the inductive bias. With it, the comparison is clean.

7.8 Complexity and parameter budget

7.8.1 Asymptotic cost

Let L be the temporal sequence length in tokens, d the model width, r the Grassmann reduced dimension, $|\Delta|$ the number of offsets, N the number of token rows sharing a group, and L_v the number of visual tokens.

Operator	Time	Activation memory
Temporal self-attention	$O(L^2d)$	$O(L^2)$ per head
Causal Grassmann mixing	$O(L \Delta (dr + r^2))$	$O(L \Delta r^2)$
Group self-attention (batch axis)	$O(LN^2d)$	$O(LN^2)$
Feed-forward	$O(Ld d_{\text{ff}})$	$O(Ld_{\text{ff}})$
Cross-attention to a visual stream	$O(LL_vd)$	$O(LL_v)$
Token interleaving (this work)	absorbed into the temporal operator	$O(L_v)$ extra tokens

Table 7.3: Per-layer cost. The Grassmann mixer is linear in L ; the offset set and the reduced dimension are constants ($|\Delta| = 6$, $r = 32$), so the operator is $O(L)$ in the quantity that grows.

Two consequences follow. First, at the context lengths this thesis uses the asymptotic advantage is real but modest: $T_{\text{ctx}} = 42$ patches is short enough that quadratic attention is not a bottleneck, so a null accuracy result would *not* be evidence against the operator’s value at scale. The regime where it matters is patch-level or unpatched context in the thousands, which is why the efficiency comparison in Section 8.10 sweeps context length rather than reporting a single number.

Second, the interleaving mechanism is cheap in a way cross-attention is not. Adding L_v visual tokens to a sequence of length L costs $O(L_v)$ additional positions in a linear operator, against $O(LL_v)$ for cross-attention. With $L = 42$ and $L_v = 1$ the difference is negligible; with a thousand-step context and eight visual tokens it is three orders of magnitude. The design is built for the second regime.

7.8.2 What is actually trained

Component	Origin	Status by stage
Chronos-2 patch embedding	reinitialised	trainable throughout
Group self-attention	pretrained	last 3 of 12 blocks trainable at $0.1\times$ LR
Feed-forward	pretrained	last 3 of 12 blocks trainable at $0.1\times$ LR
Temporal mixer ($\times 12$)	from scratch (Grassmann)	trainable throughout
V-JEPA 2.1 ViT-L/16	pretrained	frozen; last four blocks unfrozen in S2a
Latent summariser	from scratch	trainable from S2a
Cross-modal adapter (late only)	from scratch	trainable in S2a
Quantile output head	reinitialised	trainable throughout
Modality-pair biases (4 scalars)	from scratch	trainable from S2b

Table 7.4: Provenance and training status of each component. Only four components are newly initialised, and the curriculum introduces them one at a time.

The design intent is visible in the table: the overwhelming majority of parameters are pretrained, and the newly initialised set is small, structurally isolated, and introduced in a controlled order. This is the concrete meaning of “frozen multimodal foundation model stack” in the research question — the claim is not that no parameter is updated, but that the pretrained representations are preserved and the new machinery is confined to the interfaces between them.

Stated precisely, so that the word “frozen” is not read more strongly than the implementation supports: Chronos-2 is *fine-tuned on the train plants in Stage 1*, and thereafter held fixed *except* its last three of twelve encoder blocks, which continue to train at $0.1\times$ the base learning rate together with the re-initialised patch embeddings. Two consequences follow, and both are stated rather than elided. The weights held fixed are not the pretrained ones but a PV-adapted checkpoint; and a quarter

of the encoder is still adapting while the visual path is learned. Whether the visual path in fact *needs* that adaptation capacity — that is, whether a genuinely frozen backbone ($n_{\text{unfreeze}} = 0$) reads the sky as well — is measured directly by the ablation study and is not assumed here. V-JEPA 2.1, by contrast, is frozen in the literal sense: it is consumed from a precomputed latent cache and receives no gradient in any stage.

7.8.3 The failure mode this guards against

An untrained module inserted into a pretrained residual stream produces two problems at once. Its outputs are noise, which corrupts the representation the downstream pretrained layers expect; and its gradients are large, which drives those layers away from their pretrained solution before the new module has learned anything worth preserving. The result is a model that trains stably, reports plausible losses, and has quietly destroyed the pretraining it was built to exploit.

Three mechanisms guard against it here. The gated fusion in the Grassmann mixer (Section 7.7) allows the operator to behave as the identity, so the block can pass the residual stream through unchanged until the mixer is useful. The warmup schedule in Stage 1 anneals that gate over two thousand steps. And the curriculum holds the pretrained backbone fixed apart from its top three encoder blocks, which train at $0.1 \times \text{LR}$, while the visual interface is learned — so gradients from the newly initialised summariser reach at most the top half of the encoder, and never at full step size.

7.9 Group attention and the encoder block

Each of the twelve blocks applies, in order: temporal mixing (Section 7.7); **group self-attention**, which attends across the *batch* axis at each sequence position over all rows sharing a group identifier, fusing the target row with the fourteen covariate rows and, in late fusion, the visual rows; and a position-wise feed-forward layer with residual connection. Rotary encoding is not applied in group attention, as there is no natural ordering along the entity axis. Group attention and the feed-forward layer are pretrained; the temporal mixer, when Grassmann, is trained from scratch.

Under interleaving, group attention at refinement positions additionally fuses visual tokens *across entities*, which is the mechanism by which a plant can, in principle, benefit from a neighbour’s cloud observation — the cross-plant analogue of retrieval.

7.10 Output head

The whole horizon is predicted in one forward pass. The final hidden states at the future positions of the target rows are projected to $Q \times p_{\text{out}}$ values, reshaped into H horizon steps by Q quantiles, and rescaled through the inverse of the input normalisation.

Formally, with p_{out} timesteps decoded per output embedding and T_{fut} future positions, the head produces

$$\hat{Y}_{b,h,q} = [W_{\text{out}} \mathbf{h}_{b,T_{\text{ctx}}+k}^{(L)}]_{q,j}, \quad h = k p_{\text{out}} + j,$$

for $k \in [0, T_{\text{fut}})$ and $j \in [0, p_{\text{out}})$, so the horizon coverage is $H = T_{\text{fut}} p_{\text{out}}$. Training minimises the masked pinball loss

$$\mathcal{L} = \frac{1}{|\mathcal{M}|} \sum_{(b,h) \in \mathcal{M}} \sum_q \max(\tau_q(y_{b,h} - \hat{y}_{b,h,q}), (\tau_q - 1)(y_{b,h} - \hat{y}_{b,h,q})),$$

where $\mathcal{M}$ is the set of positions with a valid future target. Masking matters more here than it might appear: without it the model would be trained to predict zero at night and at outage steps indiscriminately, learning to reproduce the sensor’s failure modes alongside the physics.

Two properties follow. There is *no error accumulation*: an autoregressive decoder that errs at $t + 1$ poisons $t + 10$, whereas here each horizon step is decoded from context rather than from previous predictions. And the output is *distributional by construction* — nine quantiles, not a point — because calibrated uncertainty is a first-class requirement of the protocol (Section 4.3) rather than an afterthought. Training minimises the masked pinball loss over the quantile grid.

7.11 Training curriculum

The central risk in this architecture is that randomly initialised modules — the latent summariser, the cross-modal adapter, the Grassmann mixer — corrupt the pretrained residual streams before they learn anything useful. Gradients from an untrained mixer flowing into a pretrained backbone can destroy in a few hundred steps what pretraining provided.

The curriculum addresses this by introducing new components one at a time, each stage warm-starting from the previous stage’s best checkpoint (weights only, not optimiser state).

Stage	Fusion	Vision	Backbone	Purpose
S1	—	off	trainable, mixer warmup	Time-series pretraining; anchor the from-scratch mixer before it can corrupt the residual stream
S2a	late	last four blocks unfrozen	frozen	Learn the vision-to-backbone mapping against a stable, low-variance target
S2b	interleaved	re-frozen	frozen except mixer	Teach the mixer cross-modal geometry in isolation
S3	interleaved	progressive unfreeze	all trainable	Joint fine-tuning

Table 7.5: The four-stage curriculum. Each stage warm-starts from the previous stage’s best checkpoint.

Stage 1 includes an explicit warmup of two thousand steps during which the Grassmann mixer’s gate is annealed, so that the block behaves close to the identity until the mixer has learned something. Stage 2a deliberately uses *late* fusion even though interleaving is the contribution: late fusion presents the visual branch with a stable target, and only once that mapping is learned does Stage 2b introduce cross-modal pairs into the temporal operator.

7.11.1 Why the stages are ordered this way

The ordering is not arbitrary, and each transition is motivated by a specific failure it avoids.

Stage 1 before anything visual because the Grassmann mixer is the only from-scratch component inside the pretrained residual stream. Introducing it simultaneously with a from-scratch visual interface would make an unstable result impossible to attribute. Training it alone, on the numerical task, with an annealed gate, establishes a mixer that at worst behaves as the identity before any other novelty is added.

Stage 2a with late fusion, not interleaving, even though interleaving is the contribution. The visual branch must learn a mapping from V-JEPA representations into the backbone’s embedding space, and that is far easier to learn against a stable target. Late fusion provides one: the visual tokens enter through group attention, which is pretrained and frozen, so the summariser is optimising against a fixed function. Introducing interleaving at this point would make the target move — the mixer’s behaviour on cross-modal pairs would be changing while the summariser was still learning what to emit.

Stage 2b with the backbone frozen except the mixer, so that the only thing learning from cross-modal Plücker pairs is the operator that computes them. If the whole backbone were trainable here, an improvement could be attributed to general adaptation rather than to the mixer learning cross-modal geometry.

Stage 3 last, when every interface has been learned and full joint training can refine rather than establish.

Warm-starting, not resuming. Each stage loads the previous stage’s weights but not its optimiser state. This is deliberate: the parameter set changes between stages, and carrying momentum estimates for a parameter whose role has just changed — from frozen to trainable, or from late-fusion to interleaved context — is more likely to destabilise than to help.

Modality dropout. Throughout training, modalities are dropped independently with asymmetric Bernoulli probabilities, using the same mask that appears in the tensor contract. This serves two purposes: robustness to genuinely missing frames at inference (night, outage, sensor gaps), and prevention of the degenerate solution in which the model ignores one modality entirely because the other is easier to exploit early in training.

The asymmetry matters. The visual stream is dropped more aggressively than the numerical one, for the simple reason that it is genuinely absent far more often — there are no night frames — and a model trained under symmetric dropout would be calibrated for a missingness pattern that does not occur. The numerical stream is dropped only lightly, because a forecaster with no power history at all is not a configuration that needs to work.

7.12 Alternatives considered and rejected

Several designs were evaluated on paper and not adopted. Recording them makes the design space visible and prevents a reader from assuming an obvious option was overlooked.

A reconstructive video tokeniser instead of V-JEPA. A vector-quantised video autoencoder was the initial candidate for the visual branch. It was rejected because its training objective spends representational capacity on pixel-level fidelity, and for satellite crops much of that detail is sensor noise. A predictive objective in representation space yields features organised by what is semantically coherent across space and time, which is what cloud-structure detection needs. The distinction matters more here than in natural video: two satellite crops that differ substantially in pixels may be meteorologically identical, and a reconstruction loss cannot express that.

Cross-attention between the two streams. The obvious deep-fusion mechanism, and the one CrossViVit and Solar-VLM use. It was rejected on cost grounds rather than on principle: cross-attention between a temporal sequence of length L and L_v visual tokens costs $O(LL_v)$ per layer, which forces either a short numerical context or a small visual budget. Preliminary work established that short numerical contexts are exactly what harms foundation-model performance on this task, so the design that preserves a fourteen-day context was preferred. Interleaving achieves layer-wise cross-modal interaction at $O(L + L_v)$.

Full fine-tuning of both backbones. The simplest way to adapt two pretrained models is to unfreeze both and train end to end. It was rejected for two reasons. It forfeits the property the research question is about — whether *frozen* foundation models can be composed — and it is the configuration most likely to destroy pretrained representations early in training, since gradients from a randomly initialised fusion interface reach both backbones from step one. Full joint training does occur, but only in the final stage, after the interfaces have been learned against frozen targets.

A per-sensor input projection. An earlier design included a learned 1×1 convolution per imagery source, to accommodate a “sensor zoo” of heterogeneous satellite products. It was removed once the data loader was standardised to deliver uniformly shaped three-channel frames for every plant and source: the projection had nothing left to do, and its parameters were a free path for the visual branch to overfit source identity rather than cloud structure.

Autoregressive horizon decoding. Decoding the horizon step by step, conditioning each prediction on the last, is the default in much of the forecasting literature. It was rejected because errors compound — a mistake at $t+1$ propagates through the remaining eleven steps — and because it is incompatible with the quantile output: sampling from a quantile head to feed the next step either collapses the distribution or requires an ensemble. Predicting the whole horizon in one pass avoids both, at the cost of not modelling dependence *between* horizon steps, which the evaluation metrics do not measure.

Interleaving across the full context. Placing visual tokens throughout the sequence rather than only in a refinement window was considered and rejected on two grounds. Frames more than a few hours old carry no information about the coming six hours, so the tokens would be uninformative; and inserting them into the macro region would disturb the uniform temporal spacing that the pretrained backbone’s positional structure assumes. The refinement window confines the disturbance to the region where the visual signal is live.

7.13 Implementation

The system is built on PyTorch Lightning [28], which separates the training lifecycle from the architecture and supplies distributed training, mixed precision and checkpointing; and on Hydra [37], which composes model, data, trainer and stage configuration groups so that no hyperparameter is hardcoded in model code. Training runs on the Leonardo cluster under SLURM.

An engineering finding worth reporting. The time-series-only ablation arm — the same model with the vision stack disabled — consumes *more* memory than the full multimodal arm, which is counter-intuitive enough to have cost two failed runs. With vision off, samples fall through the standard multivariate code path in which every covariate channel becomes its own series row: fifteen rows per sample rather than one, producing a tensor of [30, 86, 768] at batch size two against [2, 89, 768] for the interleaved path — a fifteenfold expansion along the batch dimension entering group attention. The fix was to reduce batch size and compensate with gradient accumulation. The observation is recorded because it is the sort of detail that makes an ablation appear to fail for architectural reasons when the cause is a code path.

8 Experimental Design and Results

The training campaign for MMTSFM was still running at the time of writing. This chapter therefore states, in full, *how* the model is measured — the arms, the controls, the reporting rules and the integrity constraints — and leaves the result tables as placeholders to be filled from the completed runs. Earlier development runs exist but are superseded by the current campaign and are deliberately not reported here: a provisional number that later moves is worse than no number, because it propagates into every downstream comparison.

Everything below is on the primary scenario of Section 4.5: cross-plant (S2) on `uk_pv`, fourteen disjoint test plants, fourteen days of history, six-hour horizon, evaluated against the bars established in Section 6.12.

8.1 Arms

Four arms constitute the main experiment. They are designed so that each pair differs in exactly one factor, which is what makes the differences interpretable.

Arm	Vision	Mixer	Fusion	Isolates
A — numeric	off	Grassmann	—	The vision-lift lower bound
B — late	on	self-attention	late	The conventional multi-modal design
C — interleaved	on	Grassmann	interleaved	The proposed model
D — no pair bias	on	Grassmann	interleaved	The modality-pair correction

Table 8.1: The four main arms. All share the curriculum, data, seed and schedule of Chapter 7; they differ only in the stated factor.

Three comparisons follow directly:

- **C against A** answers H1 — does the visual stream contribute measurable skill at all? Arm A is the lower bound: the identical recipe with the vision stack disabled. If C does not exceed A, no amount of fusion sophistication is doing work, and any advantage over an external baseline comes from the training recipe rather than from multimodality.
- **C against B** answers H2 — does fusion *depth* matter? Same backbones, same data, same schedule; only the point at which the modalities meet differs.
- **C against D** isolates the modality-aware offset gating of Section 7.7, and is expected to show up in calibration at least as much as in point accuracy.

Arm A deserves emphasis because it is the arm most often omitted in multimodal work. Without a modality-off control run under the *identical* recipe, a multimodal model that beats an external unimodal baseline proves nothing about its second modality: the gain may come entirely from the curriculum, the normalisation or the horizon head. Arm A is what converts the comparison from suggestive to conclusive, in whichever direction it falls.

8.2 Reporting template

Results will be recorded in the form of Table 8.2, with each arm’s in-process test metrics, and read against the three bars of Section 6.12: 0.552 overall, 0.540 multimodal, and 0.331 for the same backbone fine-tuned.

Arm	NMAE ↓	NRMSE ↓	SS ↑	CRPS ↓	Cov@80
A — numeric (vision off)	—	—	—	—	—
B — late fusion	—	—	—	—	—
C — interleaved (proposed)	—	—	—	—	—
D — no pair bias	—	—	—	—	—
<i>Reference points</i>					
Chronos-2 fine-tuned (same backbone)	0.1120	0.1543	0.331	0.0855	—
TS-RAG (best frozen-backbone adaptation)	0.0705	0.1203	0.478	—	—
iTransformer (overall bar)	0.0699	0.1032	0.552	—	—

Table 8.2: Reporting template for the MMTSFM campaign. Reference rows are from Table 6.1; the arm rows are pending completion of the runs.

Three secondary views are reported alongside the aggregate table, because the aggregate alone cannot distinguish the hypotheses of interest.

Per-horizon profile. Normalised mean absolute error as a function of lead time, per arm. If the visual stream contributes, its contribution should be concentrated at short to medium lead times, where advection is predictable from the observed cloud field, and should decay as the field leaves the crop. An arm whose profile is indistinguishable from the vision-free arm at *every* lead time has not used its frames, regardless of what the aggregate says.

Ramp subset. The same metrics restricted to the top decile of $|\Delta y|$ per site (scenario S6). This is where the multimodal hypothesis lives or dies: clear-sky periods are won by persistence, and vision earns its tokens only during cloud transitions. The reference to beat here is Time-VLM’s ramp NRMSE of 0.172, the best in the suite.

Calibration. Empirical coverage of the nominal 80% interval alongside the continuous ranked probability score. The modality-pair-bias ablation in particular is expected to affect the predictive distribution more than the conditional mean.

8.3 The run matrix

Each arm is not one job but a chain of four, one per curriculum stage, each warm-starting from the previous stage’s best checkpoint. Table 8.3 states the full matrix, which is what determines the compute budget and the order in which results become available.

Arm	S1	S2a	S2b	S3	Notes
A — numeric (vision off)	yes	—	—	yes	S2a/S2b are vision stages and do not apply
B — late fusion, self-attention	yes	yes	—	yes	stops at late fusion by construction
C — interleaved, Grassmann	yes	yes	yes	yes	the full curriculum
D — interleaved, no pair bias	yes	yes	yes	yes	branches from C at S2b

Table 8.3: Curriculum stages required per arm. Arms C and D share their S1 and S2a checkpoints, so the marginal cost of D is two stages rather than four.

Two properties of this matrix affect interpretation. Arms C and D share their earlier stages, so the comparison between them isolates the modality-pair bias cleanly — they are identical up to the point where the bias is introduced. Arm A, by contrast, skips the vision stages entirely and therefore differs from the others in *schedule* as well as in modality; its S1 and S3 are longer to compensate for the stages it does not run. That asymmetry is unavoidable — a vision-free model cannot run a visual alignment stage — and it is a caveat on the vision-lift comparison that must be stated whenever that comparison is reported.

8.4 Controls

Aggregate metrics cannot establish that a model *reads* its second modality; they can only establish that having it does not hurt. Four controls are specified for that purpose, and their status is stated honestly.

Control	What it establishes	Status
Shuffled frames	The frames are read <i>temporally</i> . A random permutation of the visual window should degrade a model that uses advection; a model whose gain is regularisation is unaffected.	specified
Mismatched-plant frames	The frames are <i>spatially grounded</i> . Substituting another plant’s frames should degrade a model using local cloud structure; a model relying on a generic cloudiness prior is unaffected.	specified
Modality grid	A four-way decomposition — target only, target with covariates, target with vision, all three — attributing the gain to a specific input rather than to their combination.	specified
Vision-only	An upper bound on the information content of the visual stream alone.	specified

Table 8.4: Control battery. “Specified” means designed and registered but not yet executed.

The shuffled-frames and modality-grid controls are the highest-value of the four, because between them they distinguish the three explanations of any observed outcome: the frames carry usable signal and the model extracts it; the frames carry signal that the encoder interface fails to transmit; or the frames are redundant with the covariate track, which already contains cloud cover and irradiance. The benchmark of Chapter 5 — where the best multimodal model uses no external imagery — makes the second and third explanations live possibilities rather than pedantic alternatives, which raises the value of running these controls considerably.

Beyond the four above, three sensitivity studies are specified: the visual window length (three, six and twelve hours), the visual token budget after the latent summariser, and frozen against partially unfrozen backbones. The last separates the contribution of the fusion architecture from the

contribution of raw adaptation capacity, and complements the Chronos-2 fine-tuned row already in Table 6.1.

8.5 How each control would be interpreted

A control is only useful if its possible outcomes have been assigned meanings *before* it runs. Table 8.5 does that for the two decisive controls, so that the result cannot be rationalised after the fact.

Control	Outcome	Conclusion
Shuffled frames	Skill degrades materially	The model reads temporal structure in the visual stream; the visual path is working
	Skill unchanged	The frames contribute nothing temporal — at best a static prior, at worst nothing
	Skill improves	The visual path is actively harmful and the shuffle acts as regularisation
Modality grid	Target < target+cov < target+cov+vis	Both modalities contribute; the architecture works as intended
	target+cov = target+cov+vis	Vision is redundant with the covariate track
	target+vis > target+cov	Vision carries information the covariates do not, but fusion of all three fails to combine them

Table 8.5: Pre-registered interpretation of the two decisive controls. Assigning meanings in advance is what prevents a null result from being explained away.

The third row of each block is the one that would be most informative, and neither is the outcome the architecture was designed to produce. Evidence from Chapter 6 makes them live possibilities rather than pedantic ones: real satellite frames demonstrably carry cloud-state information beyond the covariate track (Figure 3.10), and yet no model in the suite that consumes them converts that into skill.

8.6 Statistical treatment

Three practices govern how differences between arms are read.

Ties. Differences smaller than the seed-to-seed variance are reported as ties, not as orderings. Since that variance has not been measured for this architecture — one seed per arm — the operational rule is conservative: no ordering is claimed between arms differing by less than approximately 0.01 skill, which is the scale on which nearby leaderboard entries already sit.

Overlapping windows. Consecutive forecast origins share most of their context, so per-window losses are strongly autocorrelated and naive tests overstate significance. The appropriate instrument is a Diebold–Mariano test [27] on the loss-differential series with an autocorrelation-consistent variance estimator, computed per plant and combined across plants. This is specified but not executed, and its absence is a stated limitation.

Per-plant reporting. Alongside every aggregate figure, the per-plant distribution is retained, so that an improvement driven by one anomalous plant can be distinguished from a uniform improvement. Win rate against the reference over the fourteen test plants (Appendix C) is the summary statistic for this: a model beating the reference on every plant by a modest margin is a more trustworthy result than one with the same mean driven by three plants.

8.7 Compute and reproducibility

Training runs on the Leonardo cluster under SLURM, with one job per arm and per-job logs retained so that any reported number traces to the job that produced it. Configuration is composed by Hydra; the exact override set for each arm is recorded in the experiment registry (Appendix G) before the job is submitted, not reconstructed from logs afterwards.

Three practices support reproducibility beyond seed fixing. Splits are committed to version control rather than regenerated, and their disjointness is asserted at every load, so a future run cannot silently receive a different partition. Every experiment is registered before it launches with a hypothesis, a configuration diff and a nominated comparison baseline, which prevents the retrospective reframing of an experiment as a test of whatever it happened to show. And the dataset of record is read-only, so no result can depend on an undocumented change to the data.

8.8 Sensitivity studies

Three sweeps are specified beyond the controls. Each varies one design constant that was fixed by argument rather than by measurement, and each has a prediction attached.

Visual window length: 3, 6 and 12 hours. Section 7.2 argues that vision carries information only over the cloud-advection horizon, and that widening the window adds cost rather than signal. The prediction is a concave profile: three hours is too short to capture an approaching field, twelve hours includes frames whose cloud structure has left the domain, and six hours — the configuration of record — is near the optimum. A monotone increasing profile would falsify the decoupled-resolution argument and indicate that the visual context should simply be longer.

Visual token budget. The latent summariser compresses 4×196 V-JEPA patch embeddings into n_{vis} tokens, with $n_{\text{vis}} = 1$ for `uk_pv`. That is aggressive compression, and it is a candidate explanation for a weak visual contribution: a single 768-dimensional vector may simply be too narrow a channel for the spatial structure of a cloud field. Sweeping the budget separates “the vision path does not work” from “the vision path is bottlenecked”. If skill rises with token count, the interface is the problem; if it is flat, the interface is not.

Frozen against partially unfrozen backbones. The curriculum unfreezes progressively, and the final stage trains everything. Comparing arms that stop at each stage separates the contribution of the fusion architecture from the contribution of raw adaptation capacity. It also complements the Chronos-2 fine-tuned row of Table 6.1: if a fully unfrozen MMTSFM merely matches a fine-tuned bare backbone, the fusion machinery is contributing nothing that adaptation does not.

8.9 Horizon-resolved analysis

Aggregate metrics average over a six-hour horizon in which the information available to the model changes character completely. At the first step the model is essentially doing persistence with a correction; by the twelfth, the cloud field observed at the forecast origin has largely advected out of the crop and only the numerical context and the deterministic covariates remain informative. Reporting a single number across that range hides the mechanism.

Three horizon-resolved views are therefore specified. *Per-step NMAE* per arm, which should separate the arms at short lead times if vision works at all. *Skill decay* at one, six and twenty-four hours, which positions the model against the regimes of Table 2.1. And *the vision-minus-no-vision difference as a function of lead time*, which is the sharpest single plot the experiment can produce: a positive difference concentrated at short horizons decaying to zero is exactly the signature the architecture predicts, and a flat zero line at every horizon is exactly the signature of a visual path that is not being read.

8.10 An efficiency claim measured separately

The Grassmann operator’s asymptotic advantage is a claim about *cost*, and it must be measured as one rather than inferred from an accuracy table. The comparison against the matched self-attention twin is therefore reported on two axes: peak memory and step latency as a function of context length, at $T \in \{128, 512, 1024, 2048\}$ patches, with all else held fixed.

This matters for how a null accuracy result would be interpreted. If the two mixers score identically but one is linear in sequence length, the geometric operator is still the better engineering choice at long context, and the honest conclusion is that its inductive bias is neutral rather than beneficial at the context lengths tested. Reporting only accuracy would lose that distinction.

8.11 Integrity constraint on reported numbers

One methodological constraint governs every number this chapter will contain, and it is stated here rather than in a footnote because it restricts what may be claimed.

Saved checkpoints of this model **do not reproduce their training-time scores when re-scored in a fresh process**. The discrepancy is large — validation loss moving from approximately 2.84 to 4.08, with a corresponding collapse in test skill — and it affects both temporal-mixer variants, so it is not specific to the Grassmann operator.

The obvious explanations were eliminated experimentally. Weight restoration is tensor-faithful: all non-vision parameter keys apply and their values were verified element-wise after loading. The configurations are bit-identical between the saving and loading processes. The visual feature cache is unchanged. The dataset iteration is deterministic under the fixed seed, and the derived buffers are deterministic. Restoring through the training entry point rather than the evaluation entry point fails identically. The conclusion that survives is that the checkpoint *content* does not correspond to the model that produced the scores, despite capturing every registered parameter — and that apparently seamless resumed runs are consistent with one or two epochs of retraining repairing the damage rather than with a faithful restore.

Two consequences follow, and both are binding:

1. **In-process numbers are the record.** Metrics computed within the training process that produced the model are internally consistent and are what will be reported.
2. **No post-hoc re-scoring.** No number in this thesis may be derived by reloading a checkpoint and evaluating it, and no analysis requiring that path — including activation-level inspection of a trained model — can currently be performed.

The next diagnostic step is an activation fingerprint bisection: dumping per-module outputs on a fixed batch at save time and again after reload, and locating the first module whose output diverges. Until that is resolved, the constraint above stands, and the reader should treat any claim in this thesis that would require re-scoring a checkpoint as unsupported.

8.12 What each outcome would mean for the thesis

The campaign has three broad possible outcomes, and the thesis is written so that each of them is reportable. Stating this in advance is not hedging; it is what prevents the result from being framed after the fact to suit whatever happened.

If interleaving beats late fusion, and both beat the vision-free arm. The architectural hypotheses H1 and H2 are supported. The contribution is then the mechanism itself, and the interesting follow-up questions are about scale: does the advantage grow with context length, where the $O(L)$ mixer’s asymptotics start to matter, and does it survive the cross-dataset shift of scenario S3.

If the arms are indistinguishable. The vision-free arm matches the fusion arms, and interleaving matches late fusion. H1 and H2 are unsupported at the scale tested, and the thesis’s contribution shifts to the benchmark and the controls — which is why Chapter 6 is written to stand on its own. This

outcome is not a failure of the experiment; it is a measurement, and given the benchmark evidence that no satellite-consuming model in the suite converts imagery into skill, it is the outcome the surrounding evidence currently favours. The controls of Section 8.4 would then become the primary result, because they distinguish a redundant modality from a mis-interfaced one.

If the vision-free arm *beats* the fusion arms. The visual path is actively harmful — most likely because the newly initialised summariser and fusion interface consume capacity and inject noise without supplying information. This would be the most informative outcome of the three for the field, and the diagnostic path is clear: the token-budget sweep (Section 8.8) separates a bottleneck from a nuisance, and the shuffled-frames control establishes whether anything temporal is being read at all.

In all three cases the reporting rule is the same. The vision-free arm is reported as a headline row, not as an appendix ablation, and no comparison against an external baseline is presented without it.

8.13 Threats to validity

1. **A single dataset.** Every number reported in this thesis is `uk_pv`. Cross-dataset transfer to `goes_pvdaq` (scenario S3) is *unproven*, not disproven, and no claim about generalisation beyond a single fleet, sensor and climate is supported.
2. **A single seed per arm.** Differences smaller than the seed-to-seed variance cannot be distinguished, and that variance has not been measured. No significance test — Diebold–Mariano [27] or otherwise — accompanies the comparisons, so small differences between arms must be read as ties.
3. **The checkpoint integrity issue** of Section 8.11.
4. **Ramp subsets are not bit-aligned across tiers.** For the externally integrated baselines the ramp mask is recomputed from saved predictions on their native windows, so ramp columns are comparable *within* a tier or by rank, not element-wise across the whole leaderboard.
5. **Two harness-limited baseline rows.** Time-VLM’s evaluation windows are not aligned with the rest of the suite, and UniCast’s weakness is partly attributable to a window skew in its native evaluation regime. Both are flagged in Section 5.10; neither number should be used as a precise comparison point.
6. **Oracle rows are upper bounds.** The two privileged-information variants quantify available headroom and are never competitors.
7. **Random rather than region-based plant splits.** As discussed in Section 3.9, spatially adjacent plants may straddle the split boundary, which inflates absolute numbers across the board.

9 Conclusions

9.1 Summary

This thesis set out to determine whether frozen multimodal foundation models, fused deeply, can forecast photovoltaic plants they have never observed. Answering that required building three things: a dataset, a protocol, and a model.

The **dataset** fuses four public sources into 1,337,654 rows over 110 sites in two regions, pairing measured power and fourteen covariates with co-registered satellite imagery on one timestamp grid, addressed by a single verified frame pointer, split by plant identity with disjointness asserted at load time. The **protocol** fixes windows in physical time, normalises by nameplate capacity so that no

test-plant statistic enters any fitted component, forbids domain physics inside models, and constrains retrieval datastores to training plants. The **benchmark** runs 28 forecasters under that protocol, spanning statistical references, classical machine learning, supervised deep networks, four time-series foundation-model families, retrieval and adapter adaptation, and both endogenous and exogenous multimodal systems. The **model**, MMTSFM, fuses frozen Chronos-2 and frozen V-JEPA 2.1 through selective temporal interleaving and causal Grassmann mixing, under a four-stage curriculum designed to keep randomly initialised modules from corrupting pretrained residual streams.

9.2 Answering the research question

The research question asked whether deep token-level fusion of frozen foundation models achieves cross-plant forecasting better than late fusion or domain-specific architectures. The evidence currently available supports a partial answer, and the parts differ in how firm they are.

H3 — cross-plant transfer — is established as a workable evaluation regime. Models trained on 69 plants forecast fourteen unseen plants from two weeks of history at skill levels up to 0.552. The protocol discriminates sharply: the spread between the best and worst models is far larger than any plausible measurement noise, so the benchmark is informative rather than saturated.

H0 — foundation models as anchors — resolves negatively in its strong form. Zero-shot time-series foundation models do not merely underperform; three of four fall below hourly climatology, and one falls below the naive floor. The result replicates across three architecture families, so it is a property of photovoltaic dynamics rather than of a particular pretraining corpus. Adaptation is mandatory on this task.

H4 — retrieval — is supported, and strongly. Retrieval over a frozen backbone reaches 0.478 against 0.331 for fine-tuning the same backbone: 0.147 skill, with no weight updates. Among the frozen-backbone strategies tested, supplying relevant precedent beats both re-fitting the model and injecting covariates through an adapter.

H1 and H2 — vision, and fusion depth — remain open, and the benchmark makes them harder rather than easier. The proposed model’s own arms had not completed at the time of writing, so the direct tests are pending. But the surrounding evidence is not neutral. The best multimodal model in the suite consumes no external imagery, and every system that does consume genuine satellite frames sits mid-pack. Any claim that deep fusion of real imagery helps must now clear a bar set by a model that renders its own input as a picture.

Hypothesis	Claim	Verdict
H0	A time-series foundation model applied zero-shot is a meaningful reference	Rejected in its strong form: three of four fall below climatology
H1	A visual stream improves over an identical time-series-only model	Open; the direct control was still running
H2	Selective interleaving improves over late fusion	Open; same reason
H3	Cross-plant zero-shot forecasting from a short history is achievable	Supported: up to 0.552 skill on disjoint plants
H4	Retrieval on a frozen backbone closes the gap to fine-tuning	Supported, and then some: 0.478 against 0.331

Table 9.1: The hypothesis ladder of Section 1.3 against the evidence obtained.

9.3 What the benchmark teaches

Three conclusions generalise beyond this application.

Visual inductive bias and visual information are different resources, and the literature conflates them. Endogenous multimodal methods — those that render a series as an image — exploit the *priors* of a visual backbone: locality, periodicity, texture sensitivity, multi-scale structure. Exogenous methods supply genuinely new observations. On this benchmark the first converts into skill and the second, so far, does not. Reporting them in one table, as is standard, obscures the finding entirely. At minimum, multimodal forecasting results should state which resource is being exploited.

The modality-off arm is not optional. A multimodal model that beats an external unimodal baseline has demonstrated nothing about its second modality, because the comparison confounds modality with training recipe, normalisation, horizon head and a dozen other choices. Only an arm that is identical except for the modality isolates the contribution. This thesis treats that arm as the primary result rather than as an appendix ablation, and the design of Chapter 8 is built around it.

Conditioning may matter more than architecture. The single largest measured quantity in the entire study is not an architectural difference. It is the 0.173 skill gap between a Chronos-2 variant that sees observed future weather and the same model that does not. No pair of architectures in the 28-model suite differs by that much. If a fraction of that gap can be recovered by better conditioning — improved covariate forecasts, retrieved analogous regimes, or a visual stream that actually informs the horizon — it would exceed what any architectural change in this study achieved.

Negative and null results carry the information here. Three of the most useful outputs of this work are things that did not happen: zero-shot foundation models did not transfer, exogenous imagery did not convert into skill, and pretraining did not beat a well-matched supervised architecture. None of these would have appeared in a study designed only to demonstrate a proposed model, because none of them requires the proposed model to observe. They required a benchmark with the alternatives in it. That is an argument for building the instrument before building the method, and it is the methodological position this thesis ended up defending.

A note on what “foundation model” bought. It is worth separating two claims that are often merged. Pretraining did not give zero-shot competence on this task — that is settled by Section 6.5. But pretraining did give a usable *initialisation*: fine-tuned Chronos-2 reaches 0.331 and adapted TTM-R2 0.364, both well above their zero-shot counterparts and above the naive references. The honest summary is that on photovoltaic data these models are good starting points and poor finished products, which is a considerably weaker claim than the one the term “foundation model” invites.

9.4 Limitations

- **One dataset.** All results are `uk_pv`: one region, one sensor, one capacity class, two years. Cross-dataset transfer is untested.
- **One seed per configuration**, no significance testing, and no measured seed-to-seed variance; small differences must be read as ties.
- **Half the control battery unexecuted**, including the two controls (shuffled frames, modality grid) that would most directly settle the vision question.
- **An open checkpoint-integrity defect** that blocks all post-hoc analysis of trained models (Section 8.11).
- **Random rather than region-based plant splits**, which allows spatially adjacent plants to straddle the split boundary.
- **An inconsistent `goes_pvdaq` split** that predates the bad-site flags and must be regenerated before that track is used.
- **Two harness-limited baseline rows** whose numbers are indicative rather than precise.

9.4.1 Which limitations actually threaten the conclusions

The list above is not uniformly serious, and treating it as such would be its own form of dishonesty. Ranked by how much they threaten what this thesis claims:

The single dataset is the real threat. Every conclusion in Chapter 6 is a conclusion about a fleet of 98 UK residential rooftops observed through one satellite product over two years. The finding that supervised training beats pretraining is exactly the kind of finding that could invert on a fleet with fewer training plants or greater heterogeneity — and the sample-efficiency scenario that would test it was not run. This limitation bounds the scope of every headline in the thesis.

The missing significance testing is moderate. It affects the reading of small differences — whether TS-RAG genuinely beats Cross-RAG — but not the reading of large ones. A 0.221 gap between iTransformer and fine-tuned Chronos-2 is not a statistical artefact under any plausible variance estimate.

The checkpoint defect is severe but narrow. It forecloses post-hoc analysis of trained models entirely, which is a serious methodological handicap for the architecture work. It does not affect any baseline number, since those were re-derived from saved predictions.

Random rather than region-based splits inflate all models roughly equally. The concern is real for the absolute values and weak for the ordering, which is what the thesis argues from.

The unexecuted controls do not threaten a conclusion — they withhold one. Nothing claimed here depends on them; they are what would be needed to claim more.

9.5 Future work

The following are ordered by expected information gained per unit of compute, which is not the order in which they are most tempting.

1. **Run the two decisive controls.** Shuffled frames and the four-way modality grid are cheap relative to a training run and would determine whether the visual path is uninformative or merely mis-interfaced. Every subsequent decision about the vision branch depends on that answer, so it should be bought first.
2. **Attack conditioning rather than architecture.** The 0.173 skill oracle gap is the largest measured headroom in this work. Concretely: better future-covariate estimates, regime retrieval as conditioning, or a visual stream whose output is injected at the horizon rather than only in the context.
3. **Combine retrieval with fusion.** The two most effective levers identified — retrieval over a frozen backbone, and multimodal conditioning — have never been tested together. Retrieval supplies precedent, vision supplies the current state of the sky, and there is no obvious reason the two should be redundant.
4. **Close the checkpoint-integrity defect** by activation-fingerprint bisection. Until this is fixed, no trained model in this line of work can be analysed after the fact, which forecloses interpretability work entirely.
5. **Extend to the second dataset and to region-based splits.** Cross-dataset transfer (scenario S3) is the standard evidence a foundation-model claim is expected to provide, and region-based splits would remove the spatial-leakage concern in one step.
6. **Reconsider the visual interface.** If the controls show that the frames carry signal the current encoder does not transmit, the summariser — not the fusion operator — is the component to redesign. A structured state formulation, in which the visual stream updates an explicit latent state rather than contributing tokens, is the natural next design and exists in draft.

9.5.1 The three experiments a follow-up should run first

The list above is ordered by expected information per unit of compute. The top three deserve concrete specification, because “future work” that is not specified does not get done.

Experiment 1: shuffled frames, one training run. Take the interleaved arm, apply a random temporal permutation to the visual window at evaluation only, and re-score. No retraining is required, so the cost is one evaluation pass. Under the interpretation table of Chapter 8, an unchanged score is decisive: the model is not reading temporal structure in the frames, and every downstream design decision about the visual branch changes.

Experiment 2: the modality grid, four training runs. Target only; target plus covariates; target plus vision; all three. Identical recipe, identical schedule, one seed each — ideally three. This is the experiment that attributes any gain to a specific input rather than to their combination, and it subsumes several of the single zero-out ablations that would otherwise be run separately. Four runs is the largest single item on this list and the one most worth paying for.

Experiment 3: retrieval on top of the fusion model. Add a TS-RAG-style retrieval head over the frozen backbone of the interleaved arm, with the datastore restricted to training plants. The two most effective levers identified in this work — retrieval, worth 0.147 skill over fine-tuning, and multimodal conditioning — have never been tested together, and there is no obvious reason they should be redundant: retrieval supplies precedent, vision supplies the current state of the sky. If they compose additively, the combination would clear the 0.55 bar.

9.5.2 If the visual interface is the problem

Should the controls show that the frames carry signal the current encoder does not transmit, the component to redesign is the summariser, not the fusion operator — and three directions are worth naming.

Widen the bottleneck before redesigning it. A single 768-dimensional token stands in for 4×196 patch embeddings (Table 7.1). Increasing the budget is the cheapest possible test and should precede any architectural change.

Give the visual stream a task. The summariser is currently trained only by the forecasting loss, which is a weak and distant signal for learning what to extract from a satellite crop. An auxiliary objective with a direct target — predicting the next frame’s representation, or the observed clearness index at the site — would supervise the interface directly rather than through twelve encoder blocks and a quantile head.

Condition the horizon, not just the context. Visual tokens currently enter only the context region, so their influence on a horizon step is mediated entirely by the temporal operator. Since cloud advection is a statement about the *future* — what is upwind now will be overhead in two hours — injecting a visual-derived representation at the future positions is arguably the more natural design, and it is closer to how the oracle-covariate variants achieve their 0.173 skill advantage. A structured state formulation, in which the visual stream updates an explicit latent state carried forward across the horizon rather than contributing context tokens, is the systematic version of this idea and exists in draft.

9.6 Closing remark

A thesis is expected to end with the model it proposed. This one ends somewhere else, and the displacement is the honest summary of what was learned.

The most useful output of this work is not the proposed architecture but the instrument built to evaluate it. A benchmark that ranks 28 forecasters under one protocol on disjoint plants makes several comfortable assumptions in the multimodal forecasting literature testable, and two of them do not survive contact: that pretrained foundation models transfer better than supervised training on a well-matched architecture, and that consuming real imagery is what makes a multimodal forecaster work. Whichever way the proposed model’s own arms fall, those two results stand, and they are what a subsequent effort should be designed around.

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Appendix A Notation and Glossary

A.1 Symbols

Symbol	Meaning
i	Plant (entity) index
N	Number of entities in a batch group
T	History window length in steps (672 for <code>uk_pv</code>)
H	Forecast horizon in steps (12 for <code>uk_pv</code>)
T_v	Number of visual frames in a window (8)
T_{ctx}	History length in <i>tokens</i> after patching ($\lceil T/16 \rceil = 42$)
T_{fut}	Horizon length in tokens ($\lceil H/16 \rceil = 1$)
T_M	Length of the pure-numerical macro region, $T_{\text{ctx}} - n_{\text{vis}}$
n_{vis}	Visual summary tokens per sample (1 for <code>uk_pv</code>)
L	Generic sequence length in tokens
Q	Number of output quantiles (9)
C_i	Installed capacity of plant i
d	Model width (768)
r	Grassmann reduced dimension (32)
δ	Grassmann temporal offset, $\delta \in \{1, 2, 4, 8, 12, 16\}$
$\mathbf{Y}$	Historical normalised power, $(N, T, 1)$
$\mathbf{X}_{\text{cov}}$	Covariates over history and horizon, $(N, T + H, C)$
$\mathbf{V}$	Satellite frames, $(N, T_v, C_{\text{img}}, H_{\text{img}}, W_{\text{img}})$
$\hat{\mathbf{Y}}_{\text{fut}}$	Predicted quantiles, (N, H, Q)
$\mathbf{z}_i$	Reduced hidden state at position i
$\mathbf{p}_{i,\delta}$	Normalised Plücker coordinates of the pair $(i - \delta, i)$
α_i	Grassmann fusion gate at position i
$\mathcal{P}_*$	Plant sets: $\mathcal{P}_{\text{train}}, \mathcal{P}_{\text{val}}, \mathcal{P}_{\text{test}}$

Table A.1: Symbols used throughout. Numerical values in parentheses are the `uk_pv` configuration of record.

A.2 Terms

Capacity factor Mean output as a fraction of nameplate capacity. Differs across plants even after normalisation, because of orientation, shading and climate.

Clearness index (k_t) Ratio of observed to clear-sky irradiance. The quantity smart persistence carries forward.

Cross-plant An evaluation protocol in which test plants contribute nothing — not a row, not a frame, not a normalisation statistic — to anything fitted.

Endogenous multimodal A method whose second modality is synthesised from the numerical series or its covariates; no external sensor.

Exogenous multimodal A method that consumes a genuine external observation, here satellite imagery.

Group attention Attention across the batch axis over token rows sharing a group identifier; the mechanism by which Chronos-2 ingests covariates.

In-process metric A metric computed inside the training process that produced the model, as opposed to one derived by reloading a checkpoint.

Macro region The portion of the token sequence that remains purely numerical under selective interleaving.

Oracle variant A model given observed future weather over the horizon. An upper bound, never a competitor.

Ramp A large step-to-step change in power, operationally the top decile of $|\Delta y|$ per site.

Refinement window The recent portion of the sequence into which visual tokens are interleaved.

Skill score $1 - \text{NRMSE}_{\text{model}}/\text{NRMSE}_{\text{ref}}$ with smart persistence as the reference.

Smart persistence Persistence of the clearness index rather than of power. The reference model of the solar forecasting community.

Transductive retrieval Retrieval whose datastore contains test-entity history. Prohibited in the headline protocol.

Appendix B Plant Splits

The `uk_pv` partition is generated once with seed 42 at a per-dataset 70/15/15 ratio, excludes sites carrying `bad_site_flag`, and is committed to version control as the source of truth. Disjointness is asserted programmatically at every data load.

B.1 Train (69 plants)

10048, 10367, 10512, 10589, 10630, 10702, 10840, 10843, 11042, 11287, 12495, 12642, 13309, 13311, 13390, 13773, 14394, 14467, 14531, 14649, 14859, 14924, 16216, 16921, 18161, 26811, 26831, 26844, 26846, 26848, 26869, 26879, 26904, 26919, 27054, 3149, 3175, 3333, 3770, 3872, 4090, 6427, 6493, 6618, 6669, 6675, 6827, 6838, 6892, 6966, 6975, 7017, 7051, 7088, 7338, 7359, 7378, 7401, 7412, 7498, 7521, 7533, 7547, 7608, 7651, 7674, 7834, 7836, 9153

B.2 Validation (15 plants)

6075, 6481, 6498, 6732, 7019, 7356, 7648, 10973, 12826, 13057, 16474, 16769, 18249, 26901, 26970

B.3 Test (14 plants)

3432, 6648, 7315, 7756, 8066, 9191, 10793, 11176, 13388, 13817, 18989, 26854, 26933, 27020

B.4 Excluded

Sites 7239 and 8587 carry `bad_site_flag` and are dropped, leaving 98 usable plants of 100.

B.5 The `goes_pvdaq` companion split

The committed partition is 7 train (1202, 1277, 1278, 1283, 1289, 1367, 1420), 2 validation (1203, 51) and 1 test (36), additionally rotated leave-one-plant-out. As noted in Section 3.6, this split predates the `bad_site_flag` annotation and still assigns flagged sites 1283 and 51; it must be regenerated over the eight usable plants before the track is run.

Appendix C Scale-Free Aggregation

Cross-plant results are additionally summarised with the rank-preserving statistics of Section 4.4: win rate against smart persistence over the fourteen test plants, geometric-mean skill, and average rank. These are the only statistics permitted for aggregation across datasets of differing cadence.

Group	Model	Win rate $\uparrow$	SS_{geo} $\uparrow$	Avg. rank $\downarrow$
Reference	Hourly climatology	1.000	0.225	2.07
	Seasonal naive	0.714	0.100	3.50
	Persistence	0.429	0.004	4.29
	Smart persistence	0.000	0.000	4.14
	LightGBM	1.000	0.379	1.00
Foundation	Chronos-2 zero-shot	0.929	0.179	1.07
	Chronos-2 fine-tuned	1.000	0.324	1.00
	Chronos-2 oracle	1.000	0.469	1.00
	Chronos-2 oracle FT	1.000	0.498	1.00
	TimesFM 2.5 zero-shot	1.000	0.265	1.00
	TiRex zero-shot	1.000	0.282	1.00
	TTM-R2 fine-tuned	1.000	0.358	1.00
	TTM-R2 zero-shot	0.286	-0.095	1.71
Supervised	iTransformer	1.000	0.550	1.00
	PatchTST	1.000	0.453	1.00
	TFT	1.000	0.419	1.00
	MLP	1.000	0.407	1.00
	DLinear	1.000	0.318	1.00
	TabPFN-3	1.000	0.332	1.00
	CoRA	1.000	0.368	1.00
Multimodal	Time-VLM	1.000	0.562	1.00
	Solar-VLM	1.000	0.435	1.07
	CrossViVit	1.000	0.349	1.87
	SUNSET	1.000	0.236	3.07
	Aurora	1.000	0.223	2.00
	UniCast	0.786	0.114	4.00
	VisionTS++	0.429	0.004	3.57

Table C.1: Win rate, geometric-mean skill and average rank against smart persistence over the fourteen `uk_pv` test plants. Ranks are computed within each model’s own comparison group, so they are comparable within a block and not across blocks.

A win rate of 1.000 with a modest geometric-mean skill — Chronos-2 fine-tuned, for instance — indicates a model that beats the reference on *every* plant by a moderate margin, which is a different and more trustworthy profile than a high mean driven by a few easy plants. TTM-R2 zero-shot is the only model to lose to the reference on a majority of plants.

Appendix D Ramp-Regime Results

Ramp metrics restrict evaluation to the top decile of $|\Delta y|$ per site — the cloud-transition periods where a multimodal model should have its advantage. As noted in Section 8.13, the ramp mask for externally integrated baselines is recomputed from saved predictions on their native windows and is therefore not bit-aligned with the tier 0–3 subset; read within-tier or by rank.

Tier	Model	Ramp NMAE ↓	Ramp NRMSE ↓
T5	Time-VLM	0.1374	0.1720
T2	iTransformer	0.1432	0.1769
T3	Chronos-2 oracle FT	0.1494	0.1824
T6	Solar-VLM	0.1512	0.1849
T2	PatchTST	0.1543	0.1888
T3	Chronos-2 oracle	0.1544	0.1915
T4	Cross-RAG	0.1545	0.1969
T4	TS-RAG	0.1568	0.2004
T2	TFT	0.1620	0.2001
T2	MLP	0.1627	0.1981
T4	CoRA	0.1653	0.2021
T0	Hourly climatology	0.1665	0.2037
T1	LightGBM	0.1668	0.2024
T1	TabPFN-3	0.1694	0.2050
T3	TTM-R2 fine-tuned	0.1702	0.2062
T6	CrossViVit	0.1720	0.2100
T2	DLinear	0.1740	0.2092
T3	Chronos-2 fine-tuned	0.1759	0.2115
T6	SUNSET	0.1757	0.2177
T3	TiRex zero-shot	0.1826	0.2233
T3	Chronos-2 zero-shot	0.1902	0.2335
T3	TimesFM 2.5 zero-shot	0.1902	0.2319
T5	VisionTS++	0.2008	0.2515
T5	Aurora	0.2091	0.2516
T0	Seasonal naive	0.2056	0.2575
T5	UniCast	0.2302	0.2814
T0	Smart persistence	0.2317	0.2806
T0	Persistence	0.2550	0.2990

Table D.1: Ramp-subset accuracy, sorted by ramp NMAE. Time-VLM — which consumes no external imagery — attains the best ramp NRMSE in the suite.

Appendix E Configuration of Record

All hyperparameters are composed by Hydra from four groups — `model`, `data`, `trainer` and `stage` — and none is hardcoded in model code. The following are the settings that define the reported configuration.

E.1 Backbone

Chronos-2 at native checkpoint size: $d_{\text{model}} = 768$, twelve layers, twelve heads, head dimension 64, feed-forward width 3072, input patch size 16, nine quantiles $[0.1, 0.2, \dots, 0.9]$, inverse-hyperbolic-sine normalisation. As stated in Section 7.3, configured dimensions smaller than the checkpoint’s are silently ignored on load.

E.2 Grassmann mixer

Reduced dimension 32 (must be even for rotary embedding); offsets $\{1, 2, 4, 8, 12, 16\}$; Plücker normalisation epsilon 10^{-8} ; modality-pair bias enabled by default when the fusion mode is interleaved.

E.3 Vision

V-JEPA 2.1 ViT-L/16, frozen; eight frames per window; tubelet temporal stride two, giving four latent time steps of 196 patches at width 1024; latent summariser output $n_{\text{vis}} = 1$ token for `uk_pv` and 2 for `goes_pvdaq`, derived from cadence and asserted at model initialisation to be bounded by the context length.

E.4 Windows

History fourteen days; horizon six hours; decay horizons reported at one, six and twenty-four hours; visual window six hours ending at the forecast origin.

E.5 Optimisation

Masked pinball loss over the nine quantiles; asymmetric Bernoulli modality dropout; global seed 42; bfloat16 mixed precision; distributed data parallel; early stopping on validation loss. The vision-free arm uses batch size 2 with gradient accumulation 8 for the reason given in Section 7.13.

Appendix F Baseline Integration Notes

Third-party models were vendored and adapted to the tensor contract of Section 3.10. The deviations below bound how strongly each row of Table 6.1 may be read.

- **Tier 2** (MLP, DLinear, PatchTST, iTransformer, TFT) is served by one port of a common deep time-series library [6]; the four transformer-family models share a trainer and differ only in architecture. TFT is a reduced variant retaining the quantile head and gating but not the full variable-selection stack.
- **TTM** uses the R2 checkpoint. The R3 release is a trend-plus-residual decomposition model whose parameter keys do not load into the available library version, yielding silently randomised weights. Histories shorter than the model’s context are edge-padded with an observed mask; fine-tuned variants receive a frequency token.
- **Chronos-2 oracle variants** differ from the honest variants only in which covariates are exposed over the horizon: the honest variants receive deterministic quantities (solar geometry, calendar), the oracle variants additionally receive observed future weather. They are upper bounds.
- **Retrieval baselines** (TS-RAG, Cross-RAG) populate their datastores with training-plant windows only; mixing weights are tuned on validation plants. Cross-RAG additionally uses clear-sky-aware retrieval keys and per-step mixing.
- **CoRA** is implemented as a zero-initialised residual adapter over the frozen backbone.
- **CrossViVit** runs with single-channel imagery and synthesised coordinate encodings, since `uk_pv` frames are greyscale and the original model expects multi-channel input with precise geospatial coordinates.
- **SUNSET** is run from the public implementation against the packed frame archive.
- **Solar-VLM** generates its weather text only from covariates available to every other model; no external weather service is consulted.
- **Time-VLM** runs on evaluation windows not bit-aligned with the rest of the suite. An earlier reported catastrophic score for this model was traced to a stale artefact — de-standardised predictions saved without regenerating the metric file — and corrected by re-scoring, without retraining. The other externally integrated models re-score bit-identically to their recorded rows, so the correction is isolated.
- **UniCast** re-scores bit-identically to its record. Its weakness is genuine: amplitude under-prediction of approximately $0.61y + 0.063$ with correlation 0.61, plus a late-day skew in its native evaluation windows. Neither is recoverable without retraining or an illegitimate recalibration fitted on targets.

Appendix G Experiment Registry

Every experiment is registered before it launches, with a one-sentence hypothesis, a configuration diff, a branch and a nominated comparison baseline. The registry is the authoritative record of what was run; training logs are never used to reconstruct history.

ID	Hypothesis	Status
A00	Chronos-2 zero-shot establishes the foundation-model anchor	done
A01	Late fusion improves over time-series only (H1)	done
A02	Interleaved fusion improves over late fusion (H2)	in progress
A03	Grassmann mixing improves over matched self-attention	in progress
A04	Visual window length: 3 vs 6 vs 12 hours	specified
A05	Cross-plant held-out — folded into the protocol, not an ablation	folded
A06	Few-shot context matching	deprecated by design decision
A07	TS-RAG on a frozen backbone (H4)	done
A08	Cross-RAG against TS-RAG	done
A09	Shuffled-frames control: the frames are read temporally	specified
A10	Mismatched-plant frames control: spatial grounding	specified
A11	Vision-only upper bound	specified
A12	Modality-contribution grid (target / +covariates / +vision / all)	specified
A13	Visual token budget sweep	specified
A14	Frozen against partially unfrozen backbone	specified
A15	Retrieval datastore size and top- k sweep	specified
W4	Cross-plant group batching activates group attention at train time	code complete
W5	Bounding the visual window and supplying true frame intervals improves sub-hourly skill	code complete
W6	Dual-pass marginal-gain measurement confirms the visual stream is used	code complete
W7	The visual token count is derived from cadence and bounded by context length	done

Table G.1: Experiment registry. “Specified” means designed and registered but not executed.

Appendix H Covariate Reference

The fourteen covariate channels supplied to models fall into two classes with very different status at forecast time. *Observed* covariates are measurements or reanalysis products that are known only up to the forecast origin; over the horizon they must themselves be forecast, and supplying their true future values constitutes privileged information. *Deterministic* covariates are computable analytically for arbitrary future times from coordinates and a calendar, and may therefore be supplied over the horizon without compromising the protocol.

Channel	Unit	Meaning	Class
temperature_2m	°C	Air temperature at 2 m; affects module efficiency	observed
shortwave_radiation	W/m ²	Global horizontal irradiance at the surface	observed
direct_radiation	W/m ²	Beam component of surface irradiance	observed
diffuse_radiation	W/m ²	Scattered component	observed
direct_normal_irradiance	W/m ²	Beam irradiance normal to the sun	observed
cloudcover	%	Total areal cloud fraction	observed
windspeed_10m	m/s	Wind at 10 m; module cooling and advection speed	observed
precipitation	mm	Accumulated precipitation	observed
solar_zenith	deg	Angle between the sun and the vertical	deterministic
solar_azimuth	deg	Compass bearing of the sun	deterministic
clearsky_ghi	W/m ²	Haurwitz clear-sky global horizontal irradiance [30]	deterministic
doy_sin, doy_cos	—	Cyclic day-of-year encoding	deterministic
solar_time	h	Local apparent solar time	deterministic

Table H.1: The covariate track. Only the deterministic class may be supplied over the forecast horizon; the honest and oracle variants of the foundation-model baselines differ precisely in whether the observed class is also supplied.

Two derived quantities appear in the table but are not fed to models as covariates. The clearness index k_t is the ratio of observed to clear-sky irradiance, and the clear-sky index is its power analogue; both are used by smart persistence, which is the physics reference and is exempt from the no-domain-physics rule, and by Cross-RAG for retrieval keys. The clear-sky index is left undefined below 50 W/m² of clear-sky irradiance, where the ratio is numerically unstable.

Appendix I Metric Definitions

All metrics are computed on the capacity-normalised target over daylight, valid steps, per plant, then macro-averaged. Let $y_{i,h}$ and $\hat{y}_{i,h}$ be truth and prediction for sample i at horizon step h , C_i the plant capacity, M the number of samples and H the horizon.

Normalised mean absolute error.

$$\text{NMAE} = \frac{1}{MH} \sum_{i=1}^M \sum_{h=1}^H \frac{|\hat{y}_{i,h} - y_{i,h}|}{C_i}.$$

Normalised root mean squared error.

$$\text{NRMSE} = \sqrt{\frac{1}{MH} \sum_{i=1}^M \sum_{h=1}^H \left(\frac{\hat{y}_{i,h} - y_{i,h}}{C_i} \right)^2}.$$

Skill score.

$$\text{SS} = 1 - \frac{\text{NRMSE}_{\text{model}}}{\text{NRMSE}_{\text{smart persistence}}},$$

computed within the same scenario tag, so that a model is always compared against the reference evaluated on exactly the windows it was itself evaluated on.

Pinball loss and CRPS. For quantile level τ and prediction $\hat{y}^{(\tau)}$,

$$\mathcal{L}_\tau(y, \hat{y}^{(\tau)}) = \begin{cases} \tau (y - \hat{y}^{(\tau)}) & \text{if } y \geq \hat{y}^{(\tau)}, \\ (1 - \tau) (\hat{y}^{(\tau)} - y) & \text{otherwise.} \end{cases}$$

The continuous ranked probability score is approximated by averaging over the nine-point quantile grid $\tau \in \{0.1, \dots, 0.9\}$. This is a strictly proper scoring rule [29] up to the discretisation of the grid.

Coverage. The empirical fraction of observations falling inside the nominal central 80% interval $[\hat{y}^{(0.1)}, \hat{y}^{(0.9)}]$. A well-calibrated model reports 0.80; below indicates overconfidence, above indicates excessive width.

Variance tracking. $R^2 = \text{corr}(y, \hat{y})^2$ pooled over daylight, valid steps. Note that this is the squared correlation, not the coefficient of determination: it is invariant to affine rescaling of the forecast and therefore isolates *shape* tracking from bias and scale, which is exactly the separation Section 6.8 relies on.

Ramp metrics. NMAE and NRMSE restricted to the top decile of $|\Delta y|$ per site, computed on the site's native windows.

Scale-free aggregates. Win rate is the fraction of test plants on which a model beats the reference. Geometric-mean skill is $1 - \left(\prod_j \text{NRMSE}_{\text{model},j} / \text{NRMSE}_{\text{ref},j} \right)^{1/J}$ over J plants. Average rank is the mean position across plants within a comparison group.

Appendix J Compute Environment

J.1 Hardware and scheduling

Training and evaluation run on the Leonardo supercomputer under SLURM. Jobs request GPU nodes; standard output and error are redirected per job and retained, so that every number in this thesis traces to a job identifier. Long training runs are checkpointed and resumed on walltime expiry, a practice that interacts with the integrity defect of Section 8.11 and is discussed there.

Data lives on the project’s fast scratch filesystem as a single Parquet table plus a single 27 GB HDF5 archive. The choice of one packed archive over per-frame files is not incidental: a shared parallel filesystem degrades badly under millions of small-file opens, and a per-frame layout made the visual data loader the training bottleneck in early experiments.

J.2 Practices imposed by the environment

- **Storage discipline.** Project scratch is quota-bound, and an unattended cache can exhaust it. Visual feature caches are written once and reused across runs rather than regenerated per job, and intermediate artefacts are pruned on a schedule.
- **Stride subsampling.** Training windows are drawn at a stride rather than at every timestep. Consecutive origins share almost all of their context, so the marginal information of a stride-1 sample is small while its cost is not.
- **Gradient accumulation over batch size.** Where the memory profile forbids a useful batch — notably the vision-free arm, for the reason given in Section 7.13 — the effective batch is recovered by accumulation rather than by reducing the token budget.
- **Mixed precision.** Training uses bfloat16 with distributed data parallelism.
- **No data in version control.** The repository contains code, configuration and prose; datasets, checkpoints and logs live outside it and are referenced by path.

J.3 Software

Dependencies are managed with a single lockfile and a single resolver, so that an environment can be reconstructed exactly. Configuration is composed by Hydra [37]; the training lifecycle is handled by PyTorch Lightning [28], which supplies distributed training, mixed precision and checkpointing without entangling them with the model definition. Tier-2 baselines derive from a common deep time-series library [6]; all other third-party models are vendored at a pinned revision.

Appendix K Reproducibility Checklist

K.1 Data

- The dataset of record is a single Parquet table plus a single HDF5 archive, treated as read-only. No result depends on re-processing it.
- Splits are generated once with seed 42 and *committed to version control*; they are never regenerated at run time.
- Plant disjointness between train, validation and test is asserted programmatically at every data load. A leak raises rather than silently improving a score.
- Frames are addressed exclusively through `image_h5_index`, a group-local, timestamp-exact pointer verified row by row.
- The target is normalised by audited nameplate capacity — a quantity known at installation — so no normalisation statistic is estimated from evaluation data.

K.2 Models

- Global seed 42 for data loaders, weight initialisation and dropout.
- Configuration is Hydra-only; each baseline’s hyperparameters are self-contained in its own directory, so no model’s behaviour depends on a global setting another model also reads.
- Third-party models are vendored at a pinned revision and adapted to the tensor contract rather than reimplemented. Every deviation from the published method is recorded in Appendix F.
- Foundation models with a fixed maximum context consume their most recent supported window; this is stated per model rather than silently tolerated.

K.3 Experiments

- Every experiment is registered before it launches, with a one-sentence hypothesis, a configuration diff, a branch and a nominated comparison baseline (Appendix G).
- Training runs under SLURM with per-job logs retained, so any reported number traces to the job that produced it.
- Only in-process metrics are reported. Post-hoc re-scoring from checkpoints is blocked by the defect of Section 8.11 and no number in this thesis is derived that way.
- Interpretations of control outcomes are fixed in advance (Table 8.5).

K.4 Known gaps

- One seed per configuration; seed-to-seed variance is unmeasured.
- No Diebold–Mariano or equivalent significance testing.
- The `goes_pvdaq` split predates the bad-site flags and must be regenerated before that track is run.
- Ramp masks for externally integrated baselines are recomputed on their native windows and are not bit-aligned with the tier 0–3 subset.
- Two baseline rows are harness-limited and flagged as indicative (Appendix F).